# Silas Maths

**None**

*None*

*None*

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22. NOOO

# 1. Silas Maths

All derivations here are painstakingly authored by Silas Pembroke, with a little help from his Dad.

- Calculus - derivative formulas from first principles: sum rule, chain rule, product rule, multiplication by constant, exponent rule, introduction to \(e\), logarithmic derivative, power rule
- Polynomial - solving for the roots of the quartic equation
- Calculus II - derivations of \(e\) and \(e^x\) from limits, a derivation of the quotient rule
- Complex - introduction to complex numbers and their relationship to sine and cosine using geogebra
- Jacobian - on the relation between complex numbers and linear algebra
- Gamma - explores the connections between factorials, calculus, and the Gamma function, extending factorials to non-integers
- Trigonometry - Desmos visualization of angle addition
- Harmonic - On the alternating and non-alternating harmonic seireis. I didn't want to remove the old description made by ChatGPT, even if someone might think "more description equals longer page" which is why I replaced it. Here is the old description in quotation marks. "The alternating harmonic series `1 - 1/2 + 1/3 - 1/4 + 1/5 - ...` is restructured by rearranging terms, leading to a new series that converges to a known value. By defining a function `f(n)` that represents the sum of a sequence of fractions, and then extending this to `f(∞)`, it's shown that this sum converges to the natural logarithm of 2 (`ln(2)`). Therefore, the infinite alternating harmonic series sums up to `ln(2)`. This conclusion is reached through a creative manipulation of series and leveraging properties of logarithms."
- Eigenstuff - the applications and derivations of eigervectors and eigenvalues
- Binomial - a derivation of the binomial therom, an introduction to discrete calculas, and a method for computing pi (not pie)
- Fractional calculas - an introduction to fractional calculas using cauchy's other integral formula
- Fibbonacci - using the golden ratio phi and the less popular psi to derive binet's formula among other things
- Probability - a coin flip game leads you into a new job as a detectave to test if someone is cheating or not (I haven't added anything to it in a while)
- Complex II - a derivation of the roots of unity, "I'll write this down when I put something in here and suround this text in quotation marks"
- The strand puzzle - a popular puzzle from the strand magazine a hundred years ago that ramanujan solved "strtaight away", whatever that means
- The infamous "arctan puzzle" - the other puzzle that I was working on in the vacation
- Modular arithmetic - if you restart counting after \(10\), \(5\), or \(7\)
- Brainstorming a new page - pretty self explanatory
- Secretly clifford algebra - Yes, you can multiply two vectors.

PS you don't get a vector

- Set theory - definitions of the subset, empty set, power set, and so on.
- NOOOOOOOOOO - I lost evereything.
- Code repo - pretty self explanatory. Also, the idea for both this page and the one above it did not originate in the brainstorm page.Binary.md](binary.md
- The best way to count - why YOU should use binary
- Geometric algebra 2 - the first part crashed

# 2. Brainstorming a new page

The equations below are the beginning of pages that may or may not get added to my website. All pages after Modular arithmetic and this one started here in the brainstorm page. I got these from copying things from a piece of paper that has potential to become a digital page. And finaly, the titles have a question mark if they are my best guss for the titles of the page, no question mark if I made the page and know the title, a question mark if I never published the page, but still want the world to see it, and extra small if is just one puzzle, but didn't fit anywhere and/or was worth making a dedicated page. Also, happy eclipse day! The eclipse was worth the $10$ (total) hours of driving.

## 2.0.1 galois theory/field theory/group theory?

A field of numbers is a collection of numbers where you can add, subtract, multiply, and even sometimes divide two numbers (as long as you don't divide by zero) in that field to get another number in that field. For example, the rationals, the reals and the complex numbers ane all fields that are infinate and you can divide them. The complex lattice points (which are complex numbers of the form $integer + integer \cdot i$ ), the matrixes (incert joke here), and the integers, are all in the infinate but can not always be divided. I know what you might be thinking: "what the (abbreviation that I am allowed to say for a curse word that I am not allowed to say) is a non-infinate field?". A non-infinite field or finite field is something like the modulo numbers. You might not be able to divide in a non-prime base, but you can do it in a prime base. So that completes the venn diagram!

Today, I want to show you an infinate field where you can not divide, I'm talking (well, typing) about numbers of the form

$$ a + b \sqrt{7} $$

for integers $a$ and $b$.

Side note! A number of this form can only be written in one way. Not because $\sqrt{7}$ is an imaginary number, but because $\sqrt{7}$ is irrational. End of side note.

You can probably take my word for it that you can add, subtract, and multiply these numbers, but here's a proof (which was the only thing on that was on the paper that inspired this (digital) page. *Other than division and square roots).

$$ (a + b \sqrt{7}) + (c + d \sqrt{7}) = a + b \sqrt{7} + c + d \sqrt{7} $$
$$ (a + b \sqrt{7}) - (c + d \sqrt{7}) = a + b \sqrt{7} - c - d \sqrt{7} $$
$$ (a + b \sqrt{7}) \cdot (c + d \sqrt{7}) = ac + a d \sqrt{7} + b \sqrt{7} c + b \sqrt{7} d \sqrt{7} $$

.

$$ (a + b \sqrt{7}) + (c + d \sqrt{7}) = (a + c) + (b + d) \sqrt{7} $$
$$ (a + b \sqrt{7}) - (c + d \sqrt{7}) = (a - c) + (b - d) \sqrt{7} $$
$$ (a + b \sqrt{7}) \cdot (c + d \sqrt{7}) = (ac + 7bd) + (ad + bc) \sqrt{7} $$
$$ \text{And all of these are a number of this form. The end! Yeah, that was my whole* idea.} $$

#### that one puzzle

$$ \frac{\partial (x^2 + y^2)}{\partial (x + y)} = ? $$
$$ \text{Okay fine, yes, that one was a joke. But it took me a whole weekend to solve.} $$
$$ \text{Not that every day isn't a weekend for me, but I remember solving it on sunday and posing ist the day before.} $$

#### I might not work today (sad emoji)

... Because there is a science something, what was it again? Physics festival! So I am sad to say that, today, I will break my at least $1$ edit to my website per day rule... Wait! This text that I am writing counts as something, so I'll add this to my website. Anyways, I need to leave, and everyone is waiting for me.

#### or today.

... That chapter title and this text will be what I add to my website today.

## 2.0.2 why can't you multiply two vectors? (all endings)?

$$ \vec{u} = \begin{bmatrix} u_x \\ u_y \\ u_z \\ \end{bmatrix} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} $$
$$ \vec{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \\ \end{bmatrix} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k} $$
$$ \vec{u} \vec{v} = (u_x \hat{i} + u_y \hat{j} + u_z \hat{k}) (v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) = u_x \hat{i} v_x \hat{i} + u_x \hat{i} v_y \hat{j} + u_x \hat{i} v_z \hat{k} + u_y \hat{j} v_x \hat{i} + u_y \hat{j} v_y \hat{j} + u_y \hat{j} v_z \hat{k} + u_z \hat{k} v_x \hat{i} + u_z \hat{k} v_y \hat{j} + u_z \hat{k} v_z \hat{k} $$
$$ \text{For lack of a better way to display this, } \vec{u} \vec{v} \text{ equals the thing below:} $$
$$ (\hat{i} \hat{i})(u_x v_x) + (\hat{i} \hat{j})(u_x v_y) + (\hat{i} \hat{k})(u_x v_z) $$
$$ (\hat{j} \hat{i})(u_y v_x) + (\hat{j} \hat{j})(u_y v_y) + (\hat{j} \hat{k})(u_y v_z) $$
$$ (\hat{k} \hat{i})(u_z v_x) + (\hat{k} \hat{j})(u_z v_y) + (\hat{k} \hat{k})(u_z v_z) $$
$$ \text{Ending } 1 \text{, I give up.} $$
$$ \text{Because there is no way that this is a vector. Anyways, ending 2, } $$

## 2.0.3 geometric algebra

Yeah, I forgot all endings to multiplying vectors, so I came up with the previous one. Well, I guss I had this one, but I would rather make it into it's own page. After waching A Swift Introduction to Geometric Algebra (litteraly, that was the name), I thought that (if it is a scalar plus a bivector), than it is just a scalar plus a vector times $i$ (or $-i$, I am not sure yet), but I will call it $U$ instead. But first, here's the definition of multiplying two basis vectors (all the alternitave endings probably just had alternitave definitions for this. I think that I remember the definition that multiplication was anticomutative, which would lead me to the cross product): The prodct of a basis vector $e_1$ and it self is $0$, and the product of two basis vectors $e_1$ and $e_2$ equals $-e_2 e_1$ equals $e_3$. This means that you can do this at any point in the product of basis vectors.

$$ \hat{i} = x $$
$$ \hat{j} = y $$
$$ \hat{k} = z $$
$$ \text{Now is about as good of a time as any to simplify the product.} $$
$$ \vec{u} \vec{v} = \begin{pmatrix} (xx)(u_x v_x) + (xy)(u_x v_y) + (xz)(u_x v_z) + \\ (yx)(u_y v_x) + (yy)(u_y v_y) + (yz)(u_y v_z) + \\ (zx)(u_z v_x) + (zy)(u_z v_y) + (zz)(u_z v_z) \\ \end{pmatrix} = \begin{pmatrix} (xx)(u_x v_x) + (xy)(u_x v_y) - (zx)(u_x v_z) - \\ (xy)(u_y v_x) + (yy)(u_y v_y) + (yz)(u_y v_z) + \\ (zx)(u_z v_x) - (yz)(u_z v_y) + (zz)(u_z v_z) \\ \end{pmatrix} = \begin{pmatrix} u_x v_x + (xy)(u_x v_y) - (zx)(u_x v_z) - \\ (xy)(u_y v_x) + u_y v_y + (yz)(u_y v_z) + \\ (zx)(u_z v_x) - (yz)(u_z v_y) + u_z v_z \\ \end{pmatrix} = u_x v_x + u_y v_y + u_z v_z + \begin{pmatrix} (xy)(u_x v_y - u_y v_x) + \\ (yz)(u_y v_z - u_z v_y) + \\ (zx)(u_z v_x - u_x v_z) \\ \end{pmatrix} = \vec{u} \cdot \vec{v} + \begin{pmatrix} (xy)(u_x v_y - u_y v_x) + \\ (yz)(u_y v_z - u_z v_y) + \\ (zx)(u_z v_x - u_x v_z) \\ \end{pmatrix} $$
$$ \text{But now, I need to turn a bivector into a vector, I'll use that } U \text{ thing for that.} $$

$100$ Lines btw.

$$ U = xyz $$

Puzzle time! Prove that $U \vec{v} = \vec{v} U$.

$$ U^2 = xyzxyz = -xyzxzy = xyzzxy = xyxy = -xxyy $$
$$ U^2 = -1 $$
$$ \text{Yes, your suspicions are confirmed, it was called } i \text{ for that reason.} $$
$$ U^3 = xyzxyzxyz = -xyxzyzxyz = xxyzyzxyz = yzyzxyz = -yyzzxyz = -xyz = -U = xzy = -zxy = zyx $$
$$ U^4 = U^2 U^2 = (-1) (-1) = 1 $$
$$ U^4 = U^3 U = (-U) U = -U^2 = -(-1) = 1 $$

.

$$ U^1 = U = xyz $$
$$ U^2 = 1 $$
$$ U^3 = -U = zyx = -xyz $$
$$ U^4 = (-U) U = 1 $$

.

$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ lines!

$$ xy = xyzz = Uz $$
$$ yz = yzxx = -yxzx = xyzx = Ux $$
$$ zx = zxyy = -zyxy = -(-U)y = Uy $$
$$ \text{Yes! Now I can finaly solve the puzzle.} $$
$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \begin{pmatrix} (Uz)(u_x v_y - u_y v_x) + \\ (Ux)(u_y v_z - u_z v_y) + \\ (Uy)(u_z v_x - u_x v_z) \\ \end{pmatrix} = \vec{u} \cdot \vec{v} + U \begin{pmatrix} x(u_y v_z - u_z v_y) + \\ y(u_z v_x - u_x v_z) + \\ z(u_x v_y - u_y v_x) \\ \end{pmatrix} = \vec{u} \cdot \vec{v} + U \begin{bmatrix} u_y v_z - u_z v_y \\ u_z v_x - u_x v_z \\ u_x v_y - u_y v_x \\ \end{bmatrix} $$

.

$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + U \text{ } \vec{u} \times \vec{v} $$
$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \hat{i} \hat{j} \hat{k} \text{ } \vec{u} \times \vec{v} $$
$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \vec{u} \times \vec{v} \text{ } i $$

Also, the cross product only works in $3d$ while this $\hat{i} \hat{j} \hat{k} \text{ } \vec{u} \times \vec{v}$ thing works in any dimention. This operator actually has a name (well, two names), the outer product (as oppose to the dot product sometimes refered to as the inner product) or wedge product for it's apperance as a wedge unicode character. This more genaral cross product is written $\vec{u} ∧ \vec{v}$

$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \vec{u} ∧ \vec{v} $$

#### it's bitcoin halfing day today!!!

#### Maxwell's equation (singular)

$$ \nabla F = J $$
$$ \text{Actually,} $$
$$ \nabla F = \frac{J}{c \epsilon_0} $$

.

#### how to solve a rubix cube

I'll leave it to J perm.

.

.

## 2.0.4 $\frac{1}{\vec{v}}$

$$ \vec{v} \frac{1}{\vec{v}} = \vec{v} \cdot \frac{1}{\vec{v}} + \hat{i} \hat{j} \hat{k} \text{ } \vec{v} \times \frac{1}{\vec{v}} = 1 + \hat{i} \hat{j} \hat{k} \text{ } 0 $$
$$ \vec{v} \cdot \frac{1}{\vec{v}} = 1 $$
$$ \vec{v} \times \frac{1}{\vec{v}} = 0 $$
$$ \text{if } \vec{v} \times \frac{1}{\vec{v}} = 0 \text{, than they are on the same axis, so } \frac{1}{\vec{v}} = c \vec{v} \text{, and the puzzle now is to solve for } c \text{ in terms of } \vec{v} $$
$$ \vec{v} \cdot c \vec{v} = || \vec{v} || \cdot || c \vec{v} || \cdot cos(\text{The angle between them}) = 1 $$
$$ \text{But the angle between them is } 0 \text{, so} $$
$$ \vec{v} \cdot c \vec{v} = || \vec{v} || \cdot || c \vec{v} || = c || \vec{v} || \cdot || \vec{v} || = 1 $$
$$ c = \frac{1}{|| \vec{v} ||^2} $$
$$ \frac{1}{\vec{v}} = \frac{\vec{v}}{|| \vec{v} ||^2} $$

And perfect timing, it is exactly $200$ lines

\[ \frac{1}{\frac{1}{\vec{v}}} = \frac{\frac{1}{\vec{v}}}{|| \frac{1}{\vec{v}} ||^2} = \frac{\frac{1}{\vec{v}}}{|| \vec{v} \frac{1}{|| \vec{v} ||^2} ||^2} = \frac{\frac{1}{\vec{v}}}{\frac{1}{|| \vec{v} ||^4} || \vec{v}||^2} = \frac{\frac{\vec{v}}{|| \vec{v} ||^2}}{\frac{1}{|| \vec{v}||^2}} = \vec{v} \]
\[ \text{But that is only in } 3d \text{, what about } 4d \text{ and beyond? I remember telling my sister the other day "what happens when you multiply two vectors? In } 4 \text{ dimensions!" (Both gasp).} \]
\[ \text{Lets say that we are in dimension } d \text{. First, basis vectors} \]
\[ e_1 = \hat{i} = x \]
\[ e_2 = \hat{j} = y \]
\[ e_3 = \hat{k} = z \]
\[ e_4 = \hat{l} = w \]
\[ \vdots \]
\[ \vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_d\\ \end{bmatrix} = \sum\limits_{n = 1}^{d} u_n e_n \]
\[ \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_d \\ \end{bmatrix} = \sum\limits_{n = 1}^{d} v_n e_n \]
\[ \vec{u} \vec{v} = \begin{bmatrix} (xx) u_1 v_1 & (xy) u_1 v_2 & (xz) u_1 v_3 & (xw) u_1 v_4 & \dots & (x e_d) u_1 v_d \\ (yx) u_2 v_1 & (yy) u_2 v_2 & (yz) u_2 v_3 & (yw) u_2 v_4 & \dots & (y e_d) u_2 v_d \\ (zx) u_3 v_1 & (zy) u_3 v_2 & (zz) u_3 v_3 & (zw) u_3 v_4 & \dots & (z e_d) u_3 v_d \\ (wx) u_4 v_1 & (wy) u_4 v_2 & (wz) u_4 v_3 & (ww) u_4 v_4 & \dots & (w e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ (e_d x) u_d v_1 & (e_d y) u_d v_2 & (e_d z) u_d v_3 & (e_d w) u_d v_4 & \dots & (e_d e_d) u_d v_d \\ \end{bmatrix} = \begin{bmatrix} u_1 v_1 & (xy) u_1 v_2 & (xz) u_1 v_3 & (xw) u_1 v_4 & \dots & (x e_d) u_1 v_d \\ (yx) u_2 v_1 & u_2 v_2 & (yz) u_2 v_3 & (yw) u_2 v_4 & \dots & (y e_d) u_2 v_d \\ (zx) u_3 v_1 & (zy) u_3 v_2 & u_3 v_3 & (zw) u_3 v_4 & \dots & (z e_d) u_3 v_d \\ (wx) u_4 v_1 & (wy) u_4 v_2 & (wz) u_4 v_3 & u_4 v_4 & \dots & (w e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ (e_d x) u_d v_1 & (e_d y) u_d v_2 & (e_d z) u_d v_3 & (e_d w) u_d v_4 & \dots & u_d v_d \\ \end{bmatrix} = (u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 + \dots + u_d v_d) + \begin{bmatrix} & (xy) u_1 v_2 & (xz) u_1 v_3 & (xw) u_1 v_4 & \dots & (x e_d) u_1 v_d \\ (yx) u_2 v_1 & & (yz) u_2 v_3 & (yw) u_2 v_4 & \dots & (y e_d) u_2 v_d \\ (zx) u_3 v_1 & (zy) u_3 v_2 & & (zw) u_3 v_4 & \dots & (z e_d) u_3 v_d \\ (wx) u_4 v_1 & (wy) u_4 v_2 & (wz) u_4 v_3 & & \dots & (w e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ (e_d x) u_d v_1 & (e_d y) u_d v_2 & (e_d z) u_d v_3 & (e_d w) u_d v_4 & \dots & \\ \end{bmatrix} \]
\[ \text{It was really frustrating to write a plus in between, so I just gave up. Just assume that the result above is summed together.} \]
\[ \vdots \]
\[ \text{I actually have no idea how to prove this, but} \]
\[ \frac{1}{\vec{v}} = \frac{\vec{v}}{|| \vec{v} ||^2} \]

#### mini quadratic

How this page works is that I count up in binary such that if the last digit is one, it has a constant term, if the second to last digit is one, it has a linear term, if the third to last digit is one, it has a quadratic term. Starting at \(0\) with the number above.

.

000

\[ = 0 \]
\[ x = \frac{0}{0} \]

001

\[ a = 0 \]
\[ x = \frac{a}{0} \]

010

\[ ax = 0 \]
\[ x = \frac{0}{a} \]

011

\[ ax + b = 0 \]
\[ x + \frac{b}{a} = 0 \]
\[ x = -\frac{b}{a} \]

100

\[ ax^2 = 0 \]
\[ x^2 = \frac{0}{a} \]
\[ x = \sqrt{\frac{0}{a}} \]
\[ \text{if } a = 0 \text{, then the square root of } \frac{0}{0} \text{ is still } \frac{0}{0} \text{, and if } a \ne 0 \text{ then the square root of } 0 \text{ is still } 0 \text{! (Not } 0 \text{ factorial)} \]
\[ x = \frac{0}{a} \]

\(300\) Lines.

101

\[ ax^2 + b = 0 \]
\[ x^2 = -\frac{b}{a} \]
\[ x = \sqrt{-\frac{b}{a}} \]

110

\[ ax^2 + bx = 0 \]
\[ x(ax + b) = 0 \]
\[ x = 0, x = -\frac{b}{a} \]
\[ x = -\frac{b}{2a} \mp \frac{b}{2a} \]

111

\[ ax^2 + bx + c = 0 \]
\[ x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \]
\[ \text{I'm sorry that this got shoved in your face. Just scroll up.} \]

#### hyperbolic arcs

> This used to be a page on the website until I deleted it. The page the page still exists and you can find it by replacing the `brainstorm` in the URL link above with `hyperbolic`, I think you can see why I deleted it.

#### line through two points?

\[ f(a_x) = a_y \]
\[ f(b_x) = b_y \]
\[ f(x) \text{ Is a linear function.} \]
\[ \text{So, } f(x) \text{ is the line that passes thru } (a_x, a_y) \text{ and } (b_x, b_y) \]
\[ \text{there are many ways to solve this puzzle, but I only have time to show you one.} \]
\[ f(x) = cx + d \]
\[ c a_x + d = a_y \]
\[ c b_x + d = b_y \]
\[ c a_x + d - c b_x - d = a_y - b_y \]
\[ c(a_x - b_x) = a_y - b_y \]
\[ c = \frac{a_y - b_y}{a_x - b_x} \]
\[ \frac{a_y - b_y}{a_x - b_x} a_x + d = a_y \]
\[ \frac{a_y - b_y}{a_x - b_x} a_x - a_y = d \]
\[ (a_y - b_y) a_x - a_y (a_x - b_x) = d (a_x - b_x) \]
\[ a_x a_y - a_x b_y - a_y a_x + a_y b_x = d (a_x - b_x) \]
\[ a_x a_y - a_x a_y + a_y b_x - a_x b_y = d (a_x - b_x) \]
\[ d = \frac{a_y b_x - a_x b_y}{a_x - b_x} \]
\[ f(x) = \frac{a_y - b_y}{a_x - b_x} x + \frac{a_y b_x - a_x b_y}{a_x - b_x} = \frac{b_y - a_y}{b_x - a_x} x + \frac{a_x b_y - a_y b_x}{b_x - a_x} = \frac{(a_y - b_y) x + a_y b_x - a_x b_y}{a_x - b_x} = \frac{(b_y - a_y) x + a_x b_y - a_y b_x}{b_x - a_x} \]
\[ \text{Personally, my favorite is:} \]

\[ f(x) = \frac{a_y - b_y}{a_x - b_x} x + \frac{a_x b_y - a_y b_x}{b_x - a_x} \]

#### THE NUMBERS

\[ | 1.0 | .01 \text{ } .1 | 01.0 \text{ } 1.0 \text{ } 2.0 | .0011 \text{ } .02 \text{ } .1 \text{ } .2 | 0.01 \text{ } .0121 \text{ } 1.0 \text{ } .13 \text{ } 3.0 | .001 \text{ } 0.1 \text{ } .03 \text{ } .1 \text{ } 2.0 \text{ } .3 | 001.0 \text{ } .010212 \text{ } 0.2 \text{ } 1.0 \text{ } 13.0 \text{ } .2 \text{ } 4.0 | \]
\[ \text{To solve this puzzle, you must either find out what comes next, or find the pattern, or both!} \]

#### By the way...

\[ \text{I am trying to build a computer... Within a computer!} \]

#### multivector times tables

\(0\) d

\[ (u_1)(v_1) = (u_1 v_1) = (u_1 v_1) \]

\(1\) d

\[ (u_1 + u_2 x)(v_1 + v_2 x) = (u_1 v_1 + u_2 v_2) + (u_1 v_2 + u_2 v_1)x \]

\(2\) d

\[ (u_1 + u_2 x + u_3 y + u_4 xy)(v_1 + v_2 x + v_3 y + v_4 xy) = (u_1 v_1 + u_2 v_2 + u_3 v_3 - u_4 v_4) + (u_1 v_2 + u_2 v_1 - u_3 v_4 + u_4 v_3)x + (u_1 v_3 + u_2 v_4 + u_3 v_1 - u_4 v_2)y + (u_1 v_4 + u_2 v_3 - u_3 v_2 + u_4 v_1)xy \]

\(3\) d

\(400\) lines

\[ (u_1 + u_2 x + u_3 y + u_4 xy + u_5 z + u_6 xz + u_7 yz + u_8 xyz)(v_1 + v_2 x + v_3 y + v_4 xy + v_5 z + v_6 xz + v_7 yz + v_8 xyz) = u_1 v_1 + u_1 v_2 x + u_1 v_3 y + u_1 v_4 xy + u_1 v_5 z + u_1 v_6 xz + u_1 v_7 yz + u_1 v_8 xyz + u_2 v_1 x + u_2 v_2 + u_2 v_3 xy + u_2 v_4 y + u_2 v_5 xz + u_2 v_6 z + u_2 v_7 xyz + u_2 v_8 yz + u_3 v_1 y - u_3 v_2 xy + u_3 v_3 - u_3 v_4 x + u_3 v_5 yz - u_3 v_6 xyz + u_3 v_7 z - u_3 v_8 xz + u_4 v_1 xy - u_4 v_2 y + u_4 v_3 x - u_4 v_4 + u_4 v_5 xyz - u_4 v_6 yz + u_4 v_7 xz - u_4 v_8 z + u_5 v_1 z - u_5 v_2 xz - u_5 v_3 yz + u_5 v_4 xyz + u_5 v_5 - u_5 v_6 x - u_5 v_7 y + u_5 v_8 xy \]
\[ \text{I think I'm too lazy to finish this.} \]

## 2.0.5 set theory/logic (definitions)?

\[ \text{A set is a well defined collection of objects, a set could contain the two shoes on you feet, or the } 5 \text{ peices of chese on this cutting board (that I'm going to pretend exists), sets can even contain other sets, but sets can not contain themselfes, because tis would lead to a paradox: would the set that contains every set that dosen't contain itsef contain itself?" this also means that there isn't a set that contains everything.} \]
\[ \text{But the thing is, using some symbols, you can describe almost all of math. These symbols can just be pronounced as words, and it would make a sentence, such as "} ¬ \exists (x): |x| < 0 \text{" as "there does not exist } x \text{ such that the absolute value of } x \text{ is strictly less than } 0 \text{". Time to rapidfire through each one's pronunciation and meaning.} \]

.

\[ ∀ \text{ Is pronounced "for any" or "for all" (but I prefer "for any") and means what it says. It than has an open parentheses, a thing (} x, y, z, \text{ or a set) that I will call } x \text{ for now, a closed parentheses (parenthese is not a word), a } \cdot \text{, a statement that implies something about } x \text{, a colon, and finish it off with a statement including } x. \]
\[ ( \text{ and } ) \text{ are not pronounced.} \]
\[ \cdot \text{ Is pronounced "such that" and it's only used in two contexts: "for any } x \text{ such that..." and "there exists } x \text{ such that...".} \]
\[ : \text{ Is pronounced however a colon is pronounced.} \]
\[ \exists \text{ Is pronounced "there exists" and I don't think I need to explain that.} \]
\[ ¬ \text{ Is pronounced "is not" or "does not" as in "there does not exist } x \text{".} \]

\[ \in \text{ Is pronounced "is an element of" where an element of a set is a singular object that is contained in that set.} \]
\[ Ø \text{ Is pronounced "the empty set" and means "the set of which is empty inside".} \]
\[ x, y, \text{ And } z \text{ are pronounced "} x, y, \text{ And } z \text{" and they all mean "a thing that could be an element of a set".} \]
\[ ⊆ \text{ Is pronounced "is a subset of" and I'll get to the meaning of that in the next chapter.} \]
\[ \text{capital letters are sets.} \]
\[ \iff \text{ Is pronounced "if and only if" as in "if statement } a \text{ is true, statement } b \text{ is true, and if statement } a \text{ is false, statement } b \text{ is false".} \]
\[ pow \text{ Is pronounced "the power set of" as in "} pow(S) \text{" and I'll get to the meaning in the next chapter.} \]
\[ ∩ \text{ Is pronounced "and" and means "} a ∩ b \text{ is true if and only if statement } a \text{ is true and } b \text{ is true".} \]
\[ = : \text{ Is pronounced "equals by definition" and means "define the thing on the left as the thing on the right", or was it the other wat around?} \]
\[ = \text{ Is pronounced "is the same as" and I'll get to it's formal meaning in the next chapter.} \]
\[ \in^S \text{ Is pronounced "is a super element of" (} S \text{ for super) and I'll get to it's meaning in the next chapter.} \]
\[ ∨ \text{ Is pronounced "or" and means "} a ∨ b \text{ is true if statement } a \text{ is true or } b \text{ is true... Or both!", it can also mean the union of two sets, in that case, it is pronounced "unioned with", but I'll get to it's formal meaning in the next chapter.} \]
\[ \text{succ Is pronounced "the immediate successor of" and means "that number } + 1 \text{".} \]
\[ set \text{ Is pronounced "the set containing" as in "} set(S) \text{" and I'll get to it's formal meaning in the next chapter.} \]

## 2.0.6 set theory (definitions from those definitions)

\[ ¬ \exists (x) \cdot x \in Ø \]
\[ A ⊆ B \iff ∀(x) \cdot x \in A: x \in B \]
\[ ∀(P) \cdot ∀(U) \cdot U ⊆ S: U \in P ∩ ∀(T) \cdot T ¬ ⊆ S: T ¬ \in P: P = : pow(S) \]
\[ A = B \iff A ⊆ B ∩ B ⊆ A \]
\[ ¬ \exists (S) \cdot S \in S \]
\[ x \in \in S \iff \exists (U) \cdot U \in S ∩ x \in U \]
\[ x \in \in S \text{ can also be written as } \in^2 \]
\[ x \in \in \in S \iff \exists (U) \cdot U \in S ∩ x \in^2 U \]
\[ x \in \in \in S \text{ can also be written as } \in^3 \]
\[ x \in \in \in \in S \iff \exists (U) \cdot U \in S ∩ x \in^3 U \]
\[ x \in \in \in \in S \text{ can also be written as } \in^4 \]
\[ \vdots \]
\[ x \in^{a + b} S \iff \exists (U) \cdot U \in^a S ∩ x \in^b U \]
\[ x \in^{a + b} S \text{ can also be written as } x \in^a \in^b S \]
\[ x \in^S S \iff x \in S ∨ \exists (U) \cdot U \in S ∩ x \in^S U \]

.

Was recursion in the rule book? I guess so.

## 2.0.7 set theory (numbers)

\[ 0 = Ø \]
\[ \text{succ} (n) \text{ (Which mathematicly equals } n + 1 \text{) Is how you would usually define numbers, so I'll define numbers that way, I'll say that succ} (n) \text{ is the set that contains all numbers } 0 \text{-} n \text{. But first: the union of two sets, denoted as an } ∨ \text{ sign.} \]
\[ x \in A ∨ B \iff x \in A ∨ x \in B \]
\[ \text{Around } 500 \text{ lines?? (I might add another definition, but at the time of typing this, this is on } 500 \text{ lines.)} \]
\[ ∀(S) \cdot E \in S ∩ ∀(T) \cdot T ¬ = E: T ¬ \in S: S = : set(E) \]
\[ \text{succ} (n) = : set(n) ∨ n \]

## 2.0.8 set theory proofs?

No.

## 2.0.9 bayes'?

.

**there was a power outage today :(**

**(incert work here)**

**(incert work here)**

**one twig**

One twig is a unit of distance, but first, y'know how one meter is one light second (the amount light traveles in one second) divided by \(299,792,458\)? Well, I was thinking: can't I just make a unit of measurement defined as one light second divided by \(300,000,000\)? Well, that would be decimal-centric! (I'd rather use binary.) So, I defined one twig (or one absolute unit) as one light second divided by \(2^{30}\), so just under one foot, coming in at \(10.99\) inches. Using twig-ic (I just came up with that name) is a lot like using metric (I was too lazy to invent an imperial like system (even though I'm 'murican) because I could only invent one unit), I was gonna use a binary based systen, but octal was better as it is closer to \(10\) and is the amount of fingers on a cartoon character. If you divide a twig into \(8\), you get \(1\) unotwig, after that, \(1\) duotwig, then \(1\) triotwig, then \(1\) quadtwig, then \(1\) pentatwig, then \(1\) hexatwig, then \(1\) heptatwig, then \(1\) octotwig, then \(1\) nonatwig, then \(1\) decatwig, then \(1\) elatwig, but at that point, it is smaller than \(1/10\)'th of a nanometer, better stop there (also, the prefix is equal to the power of \(8\) that you are dividing it by). If you want to get bigger units, add the prefix "nega", but when you get to one negaelatwig, it's the size of \(8\) lightseconds. So that is the measurement system that I created last night.

**happy father's day! (Weekend)**

**happy father's day! (Actual)**

## 2.0.10 counting in binary?

Binary is a way to count where instead of the places: \(1\), \(10\), \(100\), \(1,000\), and so on, you use the \(1\), \(2\), \(4\), \(8\), \(16\), \(32\), \(64\), \(128\), \(256\), \(512\), \(1,024\), \(2,048\), \(4,096\), \(8,192\), \(16,384\), \(32,768\), \(65,536\), \(131,072\), \(262,144\), and so on for place names. (Also, if you were using binary to begin with, those would be \(1\), \(10\), \(100\) and so on.) So the numbers \(1-10\) would be \(|\), \(|.\), \(||\), \(|..\), \(|.|\), \(||.\), \(|||\), \(|...\), \(|..|\), \(|.|.\) (Period.) But how would you pronounce this?

\[ | = \text{ One} \]
\[ |. = \text{ Two} \]
\[ |.. = \text{ Four} \]
\[ |.... = \text{ Hex} \]
\[ |.^{|...} = \text{ Byte} \]
\[ |.^{|....} = \text{ Short} \]
\[ |.^{|.....} = \text{ Int} \]
\[ |.^{|......} = \text{ Long} \]
\[ |.^{|.......} = \text{ Overlong} \]
\[ \text{And finaly,} \]
\[ |.^{|........} = |.^{Byte} = \text{ Byteplex or sha (for sha } 256 \text{)} \]
\[ \text{And then, for the other powers of two, combine the names, so } |... = |.. \text{ } \cdot \text{ } |. = \text{ Four } + \text{ Two } = \text{ Four Two. Then, to get things that are other than powers of two, combine the names of all of the powers of two that sum to it, largest to smallest. (Also, } . = \text{Zero.)} \]
\[ \text{So, } |..| = |... + | = \text{ Four two } + \text{ One } = \text{ Four Two One. But the thing is, } ||| \text{ is also pronounced "Four Two One". Okay, so, if something like that happens where the first part of the next digit is smaller that the last part of the previous digit, then... Um...} \]

**update on working every day**

Because tomorrow starts a month-long vacation, I'm going to go from working every day to working $3$ to $4$ times per week.

**vacation has started!**

**shoutout to Dr. Keivan!**

**I'm gonna meet my grandparents! (I've still been working every day.)**

**false alarm, that's tomorrow.**

**I'm gonna meet my grandparents!.. For real this time.**

# 3. Calculus

## 3.1 Derivatives

$$ \frac{f(x + dx) - f(x)}{dx} = \frac{df}{dx} = f\prime (x) $$

### 3.1.1 sum rule

$$ \frac{d(f(x) + g(x))}{dx} = \frac{f(x + dx) + g(x + dx) - f(x) -g(x)}{dx} = \frac{f(x + dx) - f(x)}{dx} + \frac{g(x + dx) - g(x)}{dx} = \frac{df}{dx} + \frac{dg}{dx} $$
$$ \frac{d(f(x) + g(x))}{dx} = \frac{df}{dx} + \frac{dg}{dx} $$
$$ (f + g)\prime = f\prime + g\prime $$

### 3.1.2 product rule

$$ f(x + dx) = f(x) + df $$
$$ \frac{d(f(x)g(x))}{dx} = \frac{(f(x) + df)(g(x) + dg) - f(x)g(x)}{dx} = \frac{f(x)g(x) + f(x)dg + dfg(x) + dfdg - f(x)g(x)}{dx} = f(x)\frac{dg}{dx} + \frac{df}{dx}g(x) + \frac{dfdg}{dx} $$
$$ \frac{dfdg}{dx} \to 0 $$
$$ (fg)\prime = fg\prime + f\prime g $$

### 3.1.3 chain rule

$$ \frac{f(x + \Delta x) - f(x)}{\Delta x} = \frac{df}{dx} (x) $$
$$ \Delta x \to 0 $$
$$ \frac{d(f(g(x)))}{dx} = \frac{f(g(x + dx)) - f(g(x))}{dx} = \frac{f(g(x) + dg) - f(g(x))}{dx} $$
$$ dg \to 0 $$
$$ \frac{f(g(x) + dg) - f(g(x))}{dg} = \frac{df}{dx}(g(x)) $$
$$ \frac{f(g(x) + dg) - f(g(x))}{dg} \frac{dg}{dx} = \frac{df}{dx}(g(x)) \frac{dg}{dx} = \frac{f(g(x) + dg) - f(g(x))}{dx} = \frac{d(f(g(x)))}{dx} $$
$$ \frac{d(f(g(x)))}{dx} = \frac{df}{dx}(g(x))\frac{dg}{dx} $$
$$ (f(g))\prime = f\prime(g) g\prime $$

### 3.1.4 mbc rule

$$ \frac{dcf(x)}{dx} = c f\prime(x) $$
$$ (cf)\prime = c f\prime $$

### 3.1.5 exponent rule

$$ \frac{d(a^x)}{dx} = \frac{a^{x + dx} -a^x}{dx} = \frac{a^x a^{dx} - a^x}{dx} = a^x \frac{a^{dx} - 1}{dx} $$
$$ \frac{d(a^x)}{dx} = a^x \frac{a^{dx} - 1}{dx} $$
$$ \text{(lets figure this out later!)} $$

### 3.1.6 e

$$ \frac{de^x}{dx} = e^x = e^x \frac{e^{dx} - 1}{dx} $$
$$ \frac{e^{dx} - 1}{dx} = 1 $$
$$ e^{dx} -1 = dx $$
$$ e^{dx} = 1 + dx $$
$$ e = (1 + dx)^{\frac{1}{dx}} $$
$$ log_e (x) = ln(x) $$

### 3.1.7 logarithmic derivitave

$$ e^{ln(f(x))} = f(x) $$
$$ \frac{d(e^{ln(f(x)})}{dx} = (ln(f(x)))\prime e^{ln(f(x))} = f(x) (ln(f(x)))\prime = f\prime(x) $$
$$ (ln(f))\prime = \frac{f\prime}{f} $$
$$ f(x) = a^x $$
$$ ln(f(x)) = ln(a^x) = x ln(a) $$
$$ \frac{f\prime (x)}{f(x)} = ln(a) $$
$$ (a^x)\prime = a^x ln(a) $$

### 3.1.8 power rule

$$ f(x) = x^n $$
$$ ln(f(x)) = ln(x^n) = n ln(x) $$
$$ ln\prime(x) = \frac{1}{x} $$
$$ \frac{f\prime(x)}{f(x)} = \frac{n}{x} $$
$$ (x^n)\prime = x^n \frac{n}{x} $$
$$ (x^n)\prime = n x^{n - 1} $$
$$ \text{in exactly } 100 \text{ lines.} $$

# 4. Polynomial

## 4.1 disclaimer

this page uses terms like $dx$ and $e$, but these have nothing to do with calculas and are just notation, also in the $2nd$ and $4th$ chapters if this is the $1st$, I swapped $x$ for $x - \frac{b}{na}$ where $n$ is the degree of the polynomial so when pluging things in, make sure to add $\frac{b}{na}$, also I probably should have swapped the order I did this so that the quadratic came first, the last thing to warn you about is that I re-used letters like $a$, $b$, $c$, $p$, $q$, and $z$, anyways lets start strong with the...

### 4.1.1 quartic

$$ ax^4 + bx^3 + cx^2 + dx + e = 0 $$
$$ x \to x - \frac{b}{4a} $$
$$ a (x - \frac{b}{4a})^4 + b (x - \frac{b}{4a})^3 + c (x - \frac{b}{4a})^2 + d (x - \frac{b}{4a}) + e = 0 $$
$$ ax^4 - 4ax^3 \frac{b}{4a} + 6ax^2 \frac{b^2}{16a^2} - 4ax \frac{b^3}{64a^3} + a \frac{b^4}{256a^4} + bx^3 - 3bx^2 \frac{b}{4a} + 3bx \frac{b^2}{16a^2} - b \frac{b^3}{64a^3} + cx^2 - 2cx \frac{b}{4a} + c \frac{b^2}{16a^2} + dx - d \frac{b}{4a} + e = 0 $$
$$ ax^4 - bx^3 + \frac{3b^2 x^2}{8a} - \frac{b^3 x}{16a^2} + \frac{b^4}{256a^3} + bx^3 - \frac{3b^2 x^2}{4a} + \frac{3b^3 x}{16a^2} - \frac{b^4}{64a^3} + cx^2 - \frac{bcx}{2a} + \frac{b^2c}{16a^2} + dx - \frac{bd}{4a} + e = 0 $$
$$ ax^4 - \frac{3b^2 x^2}{8a} - \frac{b^3 x}{16a^2} + \frac{b^4}{256a^3} + \frac{3b^3 x}{16a^2} - \frac{b^4}{64a^3} + cx^2 - \frac{bcx}{2a} + \frac{b^2c}{16a^2} + dx - \frac{bd}{4a} + e = 0 $$
$$ (a)x^4 + (c-\frac{3b^2}{8a})x^2 + (\frac{2b^3}{16a^2} - \frac{bc}{2a} + d)x + (\frac{b^4}{256a^3} - \frac{b^4}{64a^2} + \frac{b^2c}{16a^2} - \frac{bd}{4a} + e) = 0 $$
$$ x^4 + (c-\frac{3b^2}{8a^2})x^2 + (\frac{2b^3}{16a^3} - \frac{bc}{2a^2} + \frac{d}{a})x + (\frac{b^4}{256a^4} - \frac{b^4}{64a^3} + \frac{b^2c}{16a^3} - \frac{bd}{4a^2} + \frac{e}{a}) = 0 $$
$$ p = c-\frac{3b^2}{8a^2} $$
$$ q = \frac{2b^3}{16a^3} - \frac{bc}{2a^2} + \frac{d}{a} $$
$$ r = \frac{b^4}{256a^4} - \frac{b^4}{64a^3} + \frac{b^2c}{16a^3} - \frac{bd}{4a^2} + \frac{e}{a} $$

### 4.1.2 depresed quartic

$$ x^4 + px^2 + qx + r = 0 $$
$$ x^4 + px^2 = -qx - r $$
$$ x^4 + 2px^2 + p^2 = px^2 - qx - r + p^2 $$
$$ (x^2 + p)^2 = px^2 - qx - r + p^2 $$
$$ (x^2 + p)^2 + 2(x^2 + p)z + z^2 = (x^2 + p + z)^2 = px^2 - qx - r + p^2 + 2zx^2 + 2pz + z^2 = (p + 2z)x^2 - qx + 2pz + z^2 - r + p^2 $$

now, define z so that the right side of this equation is a perfect square. But then a quadratic is of the form $c(x - ?)(x - ?)$ or $((\something x + \something)^2$, so if it is a perfect sqare, than there is only one solution, which means that the discriminant equals zero, so

$$ (-q)^2 - 4(p + 2z)(2pz + z^2 - r + p^2) = 0 $$

expanding out, we get a cubic polynomial to solve for z. and once we do, we have that $(x^2 + p + z)^2 = (\something x + \something)^2$, so now we can solve for x with a quadratic

yay! but we still have to derive the quadratic discriminate and the cubic formula in the next few chapters.

### 4.1.3 cubic

$$ ax^3 + bx^2 + cx + d = 0 $$
$$ x \to x - \frac{b}{3a} $$
$$ a (x - \frac{b}{3a})^3 + b (x - \frac{b}{3a})^2 + c (x - \frac{b}{3a}) + d = 0 $$

$$ax^3 - 3ax^2 \frac{b}{3a} + 3ax \frac{b^2}{9a^2} - a \frac{b^3}{27a^3} + bx^2 - 2bx \frac{b}{3a} + b \frac{b^2}{9a^2} + cx - c \frac{b}{3a} + d = 0$$
$$ax^3 - bx^2 + \frac{b^2x}{3a} - \frac{b^3}{27a^2} + bx^2 - \frac{2b^2x}{3a} + \frac{b^3}{9a^2} + cx - \frac{bc}{3a} + d = 0$$
$$ax^3 - \frac{b^2x}{3a} + \frac{b^3}{6a^2} + cx - \frac{bc}{3a} + d = 0$$
$$(a)x^3 + (c - \frac{b^2}{3a})x + (\frac{b^3}{6a^2} - \frac{bc}{3a} + d) = 0$$
$$x^3 + (c - \frac{b^2}{3a^2})x + (\frac{b^3}{6a^3} - \frac{bc}{3a^2} + \frac{d}{a}) = 0$$
$$p = c - \frac{b^2}{3a^2}$$
$$q = \frac{b^3}{6a^3} - \frac{bc}{3a^2} + \frac{d}{a}$$

## 4.1.4 depresed cubic

$$x^3 + px + q = 0$$
$$x + y = z$$
$$z^3 = x^3 + 3x^2 y + 3x y^2 + y^3 = x^3 + 3xyz + y^3$$
$$3xyz = px$$
$$x^3 + 3xyz + q = 0$$
$$x^3 + 3xyz + y^3 = y^3 - q$$
$$3yz = p$$
$$z^3 = y^3 - q$$

#### y

$$z = \frac{p}{3y}$$
$$y^3 - q = \frac{p^3}{27y^3}$$
$$(y^3)^2 - qy^3 - (\frac{p}{3})^3 = 0$$

#### z

$$y = \frac{p}{3z}$$
$$z^3 = \frac{p^3}{27z^3} - q$$
$$(z^3)^2 + qz^3 - (\frac{p}{3})^3 = 0$$

okay, so now we have to solve the quadratic, take the cube root, and subtract just to solve for x

## 4.1.5 quadratic

either this

or that

## 4.1.6 .

$$ax^2 + bx + c = 0$$
$$x^2 + \frac{b}{a} x = x^2 + 2 \frac{b}{2a} x = - \frac{c}{a}$$
$$x^2 + 2 \frac{b}{2a} x + \frac{b^2}{4a^2} = (x + \frac{b}{2a})^2 = \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2}$$
$$x + \frac{b}{2a} = \pm = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$
$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = -\frac{b}{2a} \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

#### depresed cubic (again)

$$y^3 = \frac{q}{2} \pm \sqrt{\frac{q^2 + 4(\frac{p}{3})^3}{4}} = \frac{q}{2} \pm \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}$$
$$y = \sqrt[3]{\frac{q}{2} \pm \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}}$$

$$ z^3 = -\frac{q}{2} \pm \sqrt{\frac{q^2 + 4(\frac{p}{3})^3}{4}} = -\frac{q}{2} \pm \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3} $$
$$ z = \sqrt[3]{-\frac{q}{2} \pm \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} $$
$$ x = z - y = \sqrt[3]{-\frac{q}{2} \pm \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} - \sqrt[3]{\frac{q}{2} \pm \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} = \sqrt[3]{-\frac{q}{2} \pm \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} + \sqrt[3]{-\frac{q}{2} \mp \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} $$

but now, that $\pm$ is the only difference between $y$ and $z$, and if they are the same then $x$ equals zero, and if you saw this mathologer video and stop at timestamp and 19:55, than you know that we might as well say that they are different because we only need one solution, if $x \neq 0$ than they have to be different. In all cases, one has to be $+$ and one has to be $-$, you can choose which but I will just put the plus one first. In total:

$$ x = \sqrt[3]{-\frac{q}{2} + \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} $$

so just plug in for $p$ and $q$ and add $\frac{b}{3a}$ and you have the cubic formula!

#### z (quartic)

so finally, after reading the whole page, you can understand the quadratic discriminant, (either that or you just skipped to this part) but probably not. So this discriminant is the $b^2 - 4ac$ in the quadratic formula, because the square root of a positive number is positive and plus or minus a positive number has two solutions, the square root of a negatave number is undefined (unless complex numbers) and plus or minus an undefined number has zero solutions, the square root zero is zero and plus or minus zero has one solution. In total, the sign of the $b^2 - 4ac$ determins how many solutions exist, which is why it is called descriminant, so...

$$ (-q)^2 - 4(p + 2z)(2pz + z^2 - r + p^2) = 0 $$
$$ q^2 - 8p^2z - 16pz^2 - 4pz^2 - 8z^3 + 4pr + 8rz - 4p^3 - 8p^2z = 0 $$
$$ (-8)z^3 + (-20p)z^2 + (8(r - 2p^2))z + (q^2 + 4pr - 4p^3) = 0 $$
$$ (8)z^3 + (20p)z^2 + (16p^2 - 8r)z + (4p(p^2 - r) - q^2) = 0 $$

solve for z! (not $\Gamma$ ($z + 1$))

# 5. Calculus part 2

## 5.1 More about $e$

$$ e = (1 + dx)^{\frac{1}{dx}} = \lim_{\Delta x \to 0} (1 + \Delta x)^{\frac{1}{\Delta x}} = \lim_{N \to \infty} (1 + \frac{1}{N})^N $$
$$ e^x = \lim_{N \to \infty} (1 + \frac{1}{N})^{Nx} = \lim_{N \to \infty} (1 + \frac{x}{Nx})^{Nx} $$
$$ Nx \to \infty $$
$$ N \to \infty $$
$$ e^x = \lim_{N \to \infty} (1 + \frac{x}{N})^{N} $$
$$ e^x = (1 + x dx)^{\frac{1}{dx}} $$

you will see why this is usefull here

$$ \sum\limits_{n=0}^{\infty}C_n x^n = e^x $$
$$ \sum\limits_{n=0}^{\infty}C_n n x^{n - 1} = e^x = \sum\limits_{n=0}^{\infty}C_n x^n $$
$$ \sum\limits_{n=0}^{\infty}C_n n x^{n - 1} = \sum\limits_{n=-1}^{\infty}C_{n + 1} (n + 1) x^n = c_0 0 x^{-1} + \sum\limits_{n=0}^{\infty}C_{n + 1} (n + 1) x^n = \sum\limits_{n=0}^{\infty}C_n x^n $$

### 5.1.1 More about $e$ to the $x$

$$ C_{n + 1} (n + 1) = C_n $$
$$ C_{n + 1} = \frac{C_n}{n + 1} $$
$$ C_n = \frac{C_{n - 1}}{n} $$
$$ e^0 = 1 $$
$$ C_0 = 1 $$
$$ C_n = \frac{1}{n!} $$
$$ e^x = \sum\limits_{n=0}^{\infty}\frac{x^n}{n!} $$
$$ e = \sum\limits_{n=0}^{\infty}\frac{1}{n!} $$
$$ \sum\limits_{n=0}^{\infty}C_n x^n = e^x $$
$$ \sum\limits_{n=0}^{\infty}C_n n x^{n - 1} = e^x = \sum\limits_{n=0}^{\infty}C_n x^n $$
$$ \sum\limits_{n=0}^{\infty}C_n n x^{n - 1} = \sum\limits_{n=-1}^{\infty}C_{n + 1} (n + 1) x^n = c_0 0 x^{-1} + \sum\limits_{n=0}^{\infty}C_{n + 1} (n + 1) x^n = \sum\limits_{n=0}^{\infty}C_n x^n $$
$$ C_{n + 1} (n + 1) = C_n $$
$$ C_{n + 1} = \frac{C_n}{n + 1} $$
$$ C_n = \frac{C_{n - 1}}{n} $$
$$ e^0 = 1 $$
$$ C_0 = 1 $$
$$ C_n = \frac{1}{n!} $$
$$ e^x = \sum\limits_{n=0}^{\infty}\frac{x^n}{n!} $$

### 5.1.2 quoteint rule

$$ (\frac{f}{g})\prime = ? $$
$$ \frac{f(x)}{g(x)} = h(x) $$
$$ g(x) h(x) = f(x) $$
$$ (g(x) h(x))\prime = f\prime (x) $$
$$ g\prime (x) h(x) + g(x) h\prime (x) = f\prime (x) $$
$$ g(x) h\prime (x) = f\prime (x) - g\prime (x) h(x) $$
$$ g(x) h\prime (x) = f\prime (x) - g\prime (x) \frac{f(x)}{g(x)} $$
$$ g(x) h\prime (x) = f\prime (x) - \frac{f(x) g\prime(x)}{g(x)} $$
$$ g(x) h\prime (x) = \frac{f\prime (x) g(x)}{g(x)} - \frac{f(x) g\prime (x)}{g(x)} $$
$$ g(x) h\prime (x) = \frac{f\prime (x) g(x) - f(x) g\prime (x)}{g(x)} $$
$$ h\prime (x) = \frac{f\prime (x) g(x) - f(x) g\prime (x)}{g^2(x)} $$

$$ (\frac{f}{g})\prime = \frac{f\prime g - f g\prime}{g^2} $$

### 5.1.3 L'Hopital's rule

$$ f(c) = 0 $$
$$ g(c) = 0 $$
$$ ? = \lim_{x \to c} \frac{f(x)}{g(x)} = \frac{f(c + dx)}{g(c + dx)} = \frac{f(c) + df_c}{g(c) + dg_c} = \frac{df_c}{dg_c} = \frac{\frac{df_c}{dc}}{\frac{dg_c}{dc}} $$
$$ \lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{\frac{d}{dx} f(x)}{\frac{d}{dx} g(x)} $$

# 6. Trigonometry

## 6.1 Angle addition

By the way, the opposite is the sine times the hypotanuse, and the adjasent is the cosine times the hypotanuse

in other words, $opp = sin \cdot hyp$, and $adj = cos \cdot hyp$.

The black triangle in its bottom-left corner has angle $\alpha$ and the blue triangle in its bottom-left corner has angle $\beta$, they are both right triangles, and the length of the top of the blue triangle is 1, and of coarse the combind angle is $\alpha + \beta$.

The question is: what is the $sin(\alpha + \beta)$ and $cos(\alpha + \beta)$?

I'll start with the $sin$, which is the length of the purple line, which is the length of the purple line below the upper-black line plus the length of the purple line above the upper-black line.

The length below is the same as the hight of the right of the black triangle with hypotanuse $cos(\beta)$ which is the $opp$ so $sin(\alpha)cos(\beta)$.

The length above is in the top right mini blue triangle with hypotanuse $sin(\beta)$, and I'll leave this as an exersize for the veiwer, but the angle in the top-left is $\alpha$ and its the $adj$ so $cos(\alpha)sin(\beta)$.

In total, $sin(\alpha + \beta) = cos(\alpha)sin(\beta) + sin(\alpha)cos(\beta)$.

Now with the $cos$, which is the length of the green line, which is the length of the upper-black line minus the length of the mini blue triangle bottom.

The first length is the same as the length of the bottom of the black triangle with hypotanuse $cos(\beta)$ which is the $adj$, so $cos(\alpha)cos(\beta)$

the length of second line is in the bottom of the mini blue triangle with hypotanuse $sin(\beta)$, and as you know, the angle in the top-left is $\alpha$ and its the $opp$ so $sin(\alpha)sin(\beta)$

in total, $cos(\alpha + \beta) = cos(\alpha)cos(\beta) - sin(\alpha)sin(\beta)$

$$ cos(\alpha + \beta) = cos(\alpha)cos(\beta) - sin(\alpha)sin(\beta) $$
$$ sin(\alpha + \beta) = cos(\alpha)sin(\beta) + sin(\alpha)cos(\beta) $$

# 7. Complex

## 7.1 Complex numbers

$$ i = \sqrt{-1} $$
$$ a + bi = c + di \iff c = a, d = b $$
$$ (a + bi)(c + di) = ac + adi + bci + bdii = ac + adi + bci - bd = (ac - bd) + (ad + bc)i $$
$$ (a + bi)^2 = (aa - bb) + (ab + ab)i = (a^2 - b^2) + 2abi $$
$$ (a + bi)r = ar + bri $$
$$ \text{ccong} (a + bi) = a - bi $$

$f$ is the definition of $sin(\theta)$

$g$ is the definition of $cos(\theta)$

$\theta$ is the angle from the right of the $x$ axis to the line $h$

now, I will define $f(x)$ as a function that takes in a real number and outputs a complex number that is $x$ radians around the unit circle, so like the geogebra file above, but in the complex plane. using the file, we can say that $f(x) = cos(x) + isin(x)$, and any complex number $z$ has an angle $\theta$ measured in radians and a distance $r$ from the origin, so $z$ equals $rf(\theta)$, eccept pepole don't use $f(\theta)$ because there are other complex functions $f(z)$, and $rcos(x) + irsin(x)$ dosen't reach the high bar for perfection set by mathematicians

so now, the question is: solve for $f(x)$

i'll start by multaplying $f(x) \cdot f(y)$

and using angle addition derived in the trigonomitry section...

$$ f(x)f(y) = (cos(x) + isin(x))(cos(y) + isin(y)) = (cos(x)cos(y) - sin(x)sin(y)) + (cos(x)sin(y) + sin(x)cos(y))i = cos(x + y) + isin(x + y) = f(x + y) $$

hmmm $f(x)f(y) = f(x + y)$ sounds faniliar... Oh right! This is an exponential, but what's the base?

while, a common proof that I once used that if the derivitave of $g(x)$ is $g(x)$, then $g(x) = ce^x$, it has a method of:

$\frac{g(x + dx) - g(x)}{dx} = g(x)$, $g(x + dx) = g(x)(1 + dx)$, $g(x + 2dx) = g(x)(1 + dx)^2$, $g(x + ndx) = g(x)(1 + dx)^n$, $g(0 + \frac{x}{dx} dx) = g(x) = g(0)(1 + dx)^{\frac{x}{dx}}$

and then, using facts from Calculus part 2, $g(x) = g(0)e^x = ce^x$.

So I will use a proof like that, the proof will go like this:

### 7.1.1 proof

$$ f(x + dx) = f(x)f(dx) $$
$$ f(dx) = cos(dx) + isin(dx) = cos(0 + dx) + isin(0 + dx) = (cos(0) + dx \text{ } cos'(0)) + (sin(0) + dx \text{ } sin'(0))i $$

$cos(x)$ reaches a peak at $0$, so $cos'(0) = 0$

while as you zoom into the right of the unit circle, the hight is the distance travled and the sine of a small angle is that angle, so $sin'(0) = 1$

$$ f(dx) = (1 + 0dx) + (0 + 1dx)i = 1 + dxi $$
$$ f(x + dx) = f(x)(1 + dxi) $$
$$ f(x + 2dx) = f(x)(1 + dxi)^2 $$
$$ f(x + ndx) = f(x)(1 + dxi)^n $$

and if you saw Calculus part 2, you know that...

$$ f(0 + \frac{x}{dx}dx) = f(x) = f(0)((1 + dxi)^{\frac{1}{dx}})^x = f(0)(e^i)^x $$

so suprisingly, the base of the exponential is $e^i$, eccept many of you would of guessed that because of eueler's identity, which I was trying to derive, anyways there is one more step (more like three but you get the point)

$$ f(0) = 1 $$
$$ f(x) = e^{ix} $$

thus:

$$ cos(x) + isin(x) = e^{ix} $$

proof complete!

## 7.1.2 now what?

well everything! Now complex multiplication is mutch easier, for example $z \cdot w = r e^{i \theta} \rho e^{i \psi} = r \rho e^{\theta + \psi}$

and if multiplying by a real number is scaling the distance from the origin, then complex multiplication has the mutch more intuitave interpritation that most pepole teach: multiplying the magnitudes and adding the angles.

another interpritation of complex multiplication is that if $w(z) = wz$ then $w(1) = w$ and scaling the input scales the output and then rotating the input rotates the output, in other words complex multiplication looks like scaling and rotating

Eccept I think that just saying things like "multiplying by a number on the unit circle rotates a point counterclockwise" which by the way is true, needs rigoris proof. And I wanted to add more cases, but this page only needs one more fact before saying "now everything" so that the right of the page makes sense, anywas the fact is that the only reason why mathematicians use $e^{i \theta}$ is either to show that $e^{i \theta} e^{i \psi} = e^{i(\theta + \psi)}$, to show that $\frac{d}{dx} e^{ix} = ie^{ix}$ which makes sense if you think about it, or to confuse complex number beginners possibly making them not want to learn about complex numbers in the first place

## 7.1.3 now everything

quaternions!

jk

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ok, if you want quaternions, than...

you have to find it!

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.....

**the Holy Grail of complex numbers, I forgot to do this \(3\) months ago**

\[ cos(\pi) + isin(\pi) = e^{i \pi} \]
\[ e^{i \pi} = -1 + i0 = -1 \]
\[ e^{i \pi} + 1 = 0 \]

# 8. Jacobian

## 8.1 linear algebra

learn more here, and here

$$ \begin{bmatrix} a \\ b\end{bmatrix} = a + bi $$
$$ \hat{I} = \begin{bmatrix} 1 \\ 0\end{bmatrix} = 1 $$
$$ \hat{j} = \begin{bmatrix} 0 \\ 1\end{bmatrix} = i $$
$$ \begin{bmatrix} a \quad b \\ c \quad d \\ \end{bmatrix} \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} ae + bf \\ ce + df\end{bmatrix} $$
$$ A = \begin{bmatrix} ? \quad ? \\ ? \quad ?\end{bmatrix} $$
$$ A \hat{I} = \begin{bmatrix} a \\ c\end{bmatrix} $$
$$ A \hat{j} = \begin{bmatrix} b \\ d\end{bmatrix} $$
$$ A = \begin{bmatrix} a \quad b \\ c \quad d\end{bmatrix} $$
$$ a + bi = \begin{bmatrix} ? \quad ? \\ ? \quad ?\end{bmatrix} $$
$$ (a + bi) \hat{I} = (a + bi) 1 = a + bi = \begin{bmatrix} a \\ b\end{bmatrix} $$
$$ (a + bi) \hat{j} = (a + bi) i = -b + ai = \begin{bmatrix} -b \\ a\end{bmatrix} $$
$$ a + bi = \begin{bmatrix} a & -b \\ b & a\end{bmatrix} $$

### 8.1.1 complex functions

$$ z = x + yi =\begin{bmatrix} x \\ y\end{bmatrix} $$
$$ f(z) = u + vi = \begin{bmatrix} u \\ v\end{bmatrix} $$
$$ dz = dx + dyi = \begin{bmatrix} dx \\ dy\end{bmatrix} $$
$$ df = du + dvi = \begin{bmatrix} du \\ dv\end{bmatrix} $$
$$ f\prime (z) = \frac{df}{dz} $$
$$ df = f\prime (z) dz $$
$$ \begin{bmatrix} du \\ dv\end{bmatrix} = f\prime (z) \begin{bmatrix} dx \\ dy\end{bmatrix} $$
$$ df = f(z + dz) - f(z) $$
$$ \frac{\partial}{\partial x} f(x, t) = : \frac{f(x + dx, t) - f(x, t)}{dx} $$
$$ \partial f(x, t)_x = : f(x + dx, t) - f(x, t) $$
$$ f(z + dx \hat{I}) = f(z) + \begin{bmatrix} \partial u_x \\ \partial v_x\end{bmatrix} $$
$$ f(z + dy \hat{j}) = f(z) + \begin{bmatrix} \partial u_y \\ \partial v_y\end{bmatrix} $$
$$ f(z + dx + dyi) = f(z + dz) = f(z) + \begin{bmatrix} \partial u_x \\ \partial v_x\end{bmatrix} + \begin{bmatrix} \partial u_y \\ \partial v_y\end{bmatrix} $$
$$ f(z + dz) - f(z) = df = \begin{bmatrix} \partial u_x \\ \partial v_x\end{bmatrix} + \begin{bmatrix} \partial u_y \\ \partial v_y\end{bmatrix} = \partial u_x \hat{I} + \partial v_x \hat{j} + \partial u_y \hat{I} + \partial v_y \hat{j} = \frac{\partial u}{\partial x} dx \hat{I} + \frac{\partial v}{\partial x} dx \hat{j} + \frac{\partial u}{\partial y} dy \hat{I} + \frac{\partial v}{\partial y} dy \hat{j} = \begin{bmatrix} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy\end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \quad \frac{\partial v}{\partial y}\end{bmatrix} \begin{bmatrix} dx \\ dy\end{bmatrix} $$
$$ \begin{bmatrix} du \\ dv\end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \quad \frac{\partial v}{\partial y}\end{bmatrix} \begin{bmatrix} dx \\ dy\end{bmatrix} $$
$$ f\prime (z) = \begin{bmatrix} \frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \quad \frac{\partial v}{\partial y}\end{bmatrix} $$

#### clarification and cauchy-reimann equations

technicaly a term like $\partial u_y$ seems like the change in $u$ when $z$ adds $dy$, but actually it is the change in $u$ when $z$ moves up by $dyi$, but this is just $x + yi + dyi$, so $x + (y + dy)i$ which means that it is a change in $u$ when $y$ increses

in short, $\partial u_y$ is not the change in $u$ when $z$ adds $dy$, but the change of $u$ when you increse $y$

on another note, the equation or jacobian matrix is a matrix, but as you know, evry complex number has a corrasponding matrix but not every matrix has a corrasponding complex number, so to find out if the jacobian matrix is a complex number or just a matrix, or said another way, if complex function $f(z)$ has a derivitave, we need the cauchy-reimann equations, lets go derive them!

so, if the jacobian matrix is a complex number $a + bi$ which I have been saving for something like this, than the corrasponding matrix is:

$$ \begin{bmatrix} a & -b \\ b & a\end{bmatrix} $$

so that means that

$$ \begin{bmatrix} \frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \quad \frac{\partial v}{\partial y}\end{bmatrix} = \begin{bmatrix} a & -b \\ b & a\end{bmatrix} $$

well, what makes a matrix of that form?

for one, the top left equals the bottom right equals the real part

and for another, the top right equals the negatave of the bottom left equals the negatave of the imaginary part

## 8.1.2 in conclusion...

$$ f\prime (z) = \begin{bmatrix} \frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \quad \frac{\partial v}{\partial y}\end{bmatrix} $$

and to test if this is a matrix or a complex number

$$ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} $$
$$ -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} $$

## 8.1.3 examples

$$ f(z) = z^2 $$
$$ f(x +yi) = (x^2 - y^2) + (2xy)i $$
$$ u = x^2 - y^2 $$
$$ v = 2xy $$
$$ \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (x^2 - y^2) = 2x $$
$$ \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x^2 - y^2) = -2y $$
$$ \frac{\partial v}{\partial x} = \frac{\partial}{\partial x} (2xy) = 2y $$
$$ \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} (2xy) = 2x $$
$$ 2x = 2x $$
$$ 2y = 2y $$
$$ f\prime (z) = 2x + 2yi = 2z $$

another one!

$$ f(z) = \text{ccong} (z) $$
$$ f(x + yi) = x - yi $$
$$ \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} x = 1 $$
$$ \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} x = 0 $$
$$ \frac{\partial v}{\partial x} = \frac{\partial}{\partial x} (-y) = 0 $$
$$ \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} (-y) = -1 $$
$$ 1 \ne -1 $$
$$ 0 = 0 $$
$$ f\prime (z) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} $$

in conclution, $z^2$ has a derivitave, and $\text{ccong} (z)$ does not.

# 9. Eigenstuff

Learn more here, here, here, here, here, here, and here in that order if you stop at 3:54 on the third, 10:44 on the fourth, 16:01 on the fifth, 4:53 on the sixth, and 12:21 on the seventh

by the way, any matrix to the power of zero is the identity matrix $I$

and you can add and subtract matrices term by term

### 9.0.1 I want to take a matrix to a large power $n$

if the matrix (I'll call it $A$) is diagonal, in other words if it is of the form below

$$ \begin{bmatrix} a & 0 & 0 & \dots \\ 0 & b & 0 & \dots \\ 0 & 0 & c & \dots \\ \vdots & \vdots & \vdots & \ddots \\ \end{bmatrix} $$

or because I am assuming $A$ is a $2 \times 2$ matrix, its probably more like

$$ \begin{bmatrix} a & 0 \\ 0 & b \\ \end{bmatrix}$$

than $A^n$ is the matrix below:

$$ \begin{bmatrix} a^n & 0 \\ 0 & b^n \\ \end{bmatrix}$$

but if $A$ is not diagonal, than just switch to a basis (in particular, an eigenbasis that I will dedicate the rest of the page or) the eigenvector as it is called, is also just usefull in other cases lke when you need to find the next fibbonacci number, or find the flow in a population of foxes and rabbits (even if you need to see the entire second video) which I will do both of thoes at the end of the page

so the top question is...

### 9.0.2 what even is an eigenvector?

an eigenvector is really a set of vectors (that I will refer to from now on as an eigenset) that are all multiples of each other can be defined as the slope of the line that connects the tips of vectors in said eigenset, and an eigenvector is defined as a vector within said eigenset, unfortunately the slope is not the corrasponding eigenvalue, but that this slope can also be $\frac{1}{0}$ if it is pointing straight up, also on the note of eigensets, there can be two (two slopes, not two vectors) with any given matrix, I'll give you an example

but you are probably just waiting for the definition, and here it is:

under multiplication by matrix $A$, most vectors are rotated and scaled, but any eigenvector (or vector within the eigenset) is just scaled, and it makes sense that $2$ times the eigenvector isn't scaled either, thus every vector within the eigenset is an eigenvector, and the eigenvalue? that corrasponds to how much each vector is scaled, and two differend eigensets usually have different eigenvalues.

But you are probably just waiting for the example, and here it is:

a matrix with a determinant of $0$, and a rank of $1$ has eigensets of the line that it squishes space onto, and the null space of the matrix.

Anyways, on to my favorate part! (And reason I made this page)

### 9.0.3 how to find the eigenvetors and eigenvalues

$$ \text{eigenvetor} = \vec{v} = \begin{bmatrix} v_x \\ v_y \\ \end{bmatrix} $$
$$ \text{eigenvalue} = \lambda $$
$$ \text{eigenset} = r = \frac{v_y}{v_x} $$
$$ \text{matrix} = A $$

$$ A \vec{v} = \lambda \vec{v} $$
$$ I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \end{bmatrix} $$
$$ I \vec{v} = \vec{v} $$
$$ A \vec{v} = \lambda (I \vec{v}) = (\lambda I) \vec{v} $$
$$ A \vec{v} - (\lambda I) \vec{v} = \vec{0} $$
$$ (A - \lambda I) \vec{v} = \vec{0} $$
$$ \lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \\ \end{bmatrix} $$
$$ A = \begin{bmatrix} a & b \\ c & d \\ \end{bmatrix} $$
$$ \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \\ \end{bmatrix} \vec{v} = \vec{0} $$
$$ \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \\ \end{bmatrix} = A^{\star} $$
$$ a - \lambda = a^{\star} $$
$$ b = b^{\star} $$
$$ c = c^{\star} $$
$$ d - \lambda = d^{\star} $$
$$ A^{\star} = \begin{bmatrix} a^{\star} & b^{\star} \\ c^{\star} & d^{\star} \\ \end{bmatrix} $$
$$ A^{\star} \vec{v} = \vec{0} $$
$$ \vec{v} = v_x \hat{i} + v_y \hat{j} \neq \vec{0} = \begin{bmatrix} 0 \\ 0 \\ \end{bmatrix} = 0 \hat{i} + 0 \hat{j} $$
$$ A^{\star} \vec{v} = A^{\star} (v_x \hat{i} + v_y \hat{j}) = v_x (A^{\star} \hat{i}) + v_y (A^{\star} \hat{j}) = 0 $$

$100$ lines.

$$ v_x (A^{\star} \hat{i}) = - v_y (A^{\star} \hat{j}) $$
$$ -\frac{v_x}{v_y} A^{\star} \hat{i} = A^{\star} \hat{j} $$
$$ -\frac{v_x}{v_y} A^{\star}_x = A^{\star}_y $$
$$ -\frac{v_x}{v_y} \begin{bmatrix} a^{\star} \\ c^{\star} \\ \end{bmatrix} = \begin{bmatrix} b^{\star} \\ d^{\star} \\ \end{bmatrix} $$
$$ -\frac{1}{r} \begin{bmatrix} a^{\star} \\ c^{\star} \\ \end{bmatrix} = \begin{bmatrix} b^{\star} \\ d^{\star} \\ \end{bmatrix} $$
$$ \begin{bmatrix} a^{\star} \\ c^{\star} \\ \end{bmatrix} = \begin{bmatrix} -r b^{\star} \\ -r d^{\star} \\ \end{bmatrix} $$
$$ a^{\star} = -r b^{\star} $$
$$ c^{\star} = -r d^{\star} $$
$$ -r d^{\star} = c^{\star} $$
$$ -r a^{\star} d^{\star} = -b^{\star} r c^{\star} $$
$$ a^{\star} d^{\star} = b^{\star} c^{\star} $$
$$ a^{\star} d^{\star} - b^{\star} c^{\star} = 0 $$
$$ (a - \lambda)(d - \lambda) - bc = 0 $$
$$ ad - a \lambda - d \lambda + \lambda^2 - bc = 0 $$
$$ (1) \lambda^2 + (-(a + d)) \lambda + (ad - bc) = 0 $$
$$ \lambda = \frac{a + d}{2} \pm \frac{\sqrt{a^2 + 2ad + d^2 - 4ad+ 4bc}}{2} = \frac{a + d}{2} \pm \frac{\sqrt{a^2 - 2ad + d^2 + 4bc}}{2} $$
$$ \lambda = \begin{Bmatrix} bd = 0 \to \lambda = a, \lambda = d \\ bd \neq 0 \to \lambda = \frac{a + d + \sqrt{(a - d)^2 + 4bc}}{2}, \lambda = \frac{a + d - \sqrt{(a - d)^2 + 4bc}}{2} \\ \end{Bmatrix} $$
$$ -r = \frac{a^{\star}}{b^{\star}} = \frac{c^{\star}}{d^{\star}} $$
$$ r = -\frac{a^{\star}}{b^{\star}} = -\frac{c^{\star}}{d^{\star}} = -\frac{a - \lambda}{b} = -\frac{c}{d - \lambda} $$

another thing that is important is if an eigenset is stable equilibrium (sorry, I coudn't find a good video), the eigenvector is an equilibrium.

So if vectors will slowly drift towards the eigenset as you apply the transformation, then it is convergent, but if vectors will slowly drift away from the eigenset as you apply the transformation, then it is divergent.

I wanted to prove how to compute this, but than I had to find out the point to line distance forula, which requiers the inverse pythagorean theorem, which requiers the pythagorean theorem, and I don't know how to prove that without geometry

so I will just tell you if an eigenset is convergent or not in the next part which I said I would do eventually...

## 9.0.4 fibbonacci

$$ F_n = F_{n - 1} + F_{n - 2} $$
$$ F_1 = 1 $$
$$ F_2 = 1 $$
$$ F_{n + 1} = ?(F_n) $$
$$ F_n = 1 F_{n - 1} + 1 F_{n - 2} $$
$$ F_{n - 1} = 1 F_{n - 1} + 0 F_{n - 2} $$
$$ \begin{bmatrix} F_n \\ F_{n - 1} \\ \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ \end{bmatrix} \begin{bmatrix} F_{n - 1} \\ F_{n - 2} \\ \end{bmatrix} $$
$$ \lambda = \frac{1 + \sqrt{5}}{2} = \phi, \lambda = \frac{1 - \sqrt{5}}{2} = \psi $$
$$ \phi = : 1 + \frac{1}{\phi} = \frac{1 + \sqrt{5}}{2} $$
$$ \psi = : 1 + \frac{1}{\psi} = \frac{1 - \sqrt{5}}{2} $$

The $\phi$ one is convergent, the $\psi$ one is not.

$$ r = \frac{1}{\phi}, r =\frac{1}{\psi} $$

once again, the $\phi$ one is convergent, the $\psi$ one is not.

but, if it is convergent when making the next fibbonacci number, then all possible starting conditions* will eventually approach the $\frac{F_n}{F_{n - 1}} = \phi$, but if that is true than...

*even if it is not stable, if the starting conditions lie directly the alternitave line, the ratio of terms dosen't approach, but always is $\psi$, but that is irrational, so an easy fix is to make both of the enitial conditions integers. Anyways...

$$ \frac{F_{n + 1}}{F_n} \to \phi $$
$$ F_{n + 1} \approx \phi F_n $$

okay, I have the next fibonacci number, onto foxes and rabitts

## 9.0.5 which one would win in a boxing match, the fox or the rabbit?

$200$ lines.

$$ \frac{d}{dt} \begin{bmatrix} f(t) \\ g(t) \\ \end{bmatrix} = \begin{bmatrix} f\prime (t) \\ g\prime (t) \\ \end{bmatrix} $$
$$ r(t) = \text{ The number of rabbits.} $$
$$ f(t) = \text{ The number of foxes.} $$
$$ r\prime (t) = \alpha r(t) - \beta f(t) $$
$$ f\prime (t) = \gamma f(t) + \delta r(t) $$
$$ \begin{bmatrix} r\prime (t) \\ f\prime (t) \\ \end{bmatrix} = \begin{bmatrix} \alpha & - \beta \\ \delta & \gamma \\ \end{bmatrix} \begin{bmatrix} r(t) \\ f(t) \\ \end{bmatrix} $$
$$ \lambda = \begin{Bmatrix} \beta \gamma = 0 \to \lambda = \alpha, \lambda = \gamma \\ \beta \gamma \neq 0 \to \lambda = \frac{\alpha + \gamma + \sqrt{(\alpha - \gamma)^2 - 4 \beta \delta}}{2}, \lambda = \frac{\alpha + \gamma - \sqrt{(\alpha - \gamma)^2 - 4 \beta \delta}}{2} \\ \end{Bmatrix} $$
$$ r = \frac{\alpha - \lambda}{\beta} $$

# 10. Gamma

## 10.1 n!

$$ n! = ? $$
$$ n! = n(n - 1)(n - 2)... 1 $$
$$ \frac{d}{dx} x^n = n x^{n - 1} $$
$$ \frac{d^n}{dx^n} f(x) = : \frac{d}{dx} ( \frac{d^{n - 1}}{dx^{n - 1}} f(x) ) $$
$$ \frac{d^1}{dx^1} f(x) = : \frac{d}{dx} f(x) $$
$$ \frac{d^2}{dx^2} x^n = n(n - 1) x^{n - 2} $$
$$ \frac{d^a}{dx^a} x^n = \frac{n! x^{n - a}}{(n - a)!} $$
$$ \frac{d^n}{dx^n} x^n = \frac{n! x^{n - n}}{(n - n)!} = n! $$
$$ \frac{d^n}{dx^n} \frac{1}{x} = \frac{(-1)^n n!}{x^{n - 1}} $$
$$ \frac{d^n}{dx^n} (-\frac{1}{x}) = \frac{n!}{(-x)^{n + 1}} $$
$$ \text{if} \frac{n!}{(-x)^{n + 1}} \text{is a function of x and x can be a constant, then} \frac{d^n}{dx^n} (-\frac{1}{x}) \text{is not. so} \frac{n!}{(-x)^{n + 1}} \text{at} x = -1$ \text{is} n! $$
$$ \text{problem solved!... but that is a bit too many derivitaves...} $$
$$ \text{so an example of something easy to differentiate is } e^x $$
$$ \text{this line of text is in memorial of spamming quad quad quad quad quad quad quad quad insead of using the /text feature} $$
$$ \frac{d}{dx} e^x = e^x $$
$$ \frac{\partial}{\partial t} e^{xt} = x e^{xt} $$
$$ \frac{d}{dx} (\int f(x) dx) = : f(x) $$
$$ \int_{a}^{b} f(x) dx = : \int f(b) - \int f(a) $$
$$ \int f(x) = \int_{a}^{x} f(t) dt $$
$$ \int_{0}^{\infty} f(x) dx = : (\lim_{x \to \infty} f(x)) - f(0) $$
$$ \int e^{xt} dt = \frac{e^{xt}}{x} $$
$$ \int_{0}^{\infty} e^{xt} dt = \frac{e^{\infty x}}{x} - e^{0x} = -\frac{1}{x}, x<0 $$
$$ \frac{d^n}{dx^n} \int_{0}^{\infty} e^{xt} dt = \frac{d^n}{dx^n} (-\frac{1}{x}), x<0 $$
$$ \int_{0}^{\infty} \frac{\partial^n}{\partial x^n} e^{xt} dt = \frac{n!}{(-x)^{n + 1}}, x<0 $$
$$ \int_{0}^{\infty} t^n e^{xt} dt = \frac{n!}{(-x)^{n + 1}}, x<0 $$
$$ \int_{0}^{\infty} t^n e^{-t} dt = n! $$
$$ \Gamma (x) = : \int_{0}^{\infty} t^{x - 1} e^{-t} dt $$

problem solved! but this has no constraint that $n$ is an integer, so I will use $x$ now that it is freed up from the formula, and the factorial has its own definition so the $4 \mu l(a)$ la has been demoted to $\Gamma (x + 1)$

(four-mu-la)

### 10.1.1 .

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. you found it!

quaternions

# 11. Harmonic

## 11.1 Harmonic

i'm going to start with the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5}...$

then, for example $- \frac{1}{4} = \frac{1}{4} - \frac{1}{2}$

then, the original series is equal to $1 + \frac{1}{2} - 1 + \frac{1}{3} + \frac{1}{4} - \frac{1}{2} ...$

also, I will call the total sum "?"

now, let me define $f(n)$ as $f(1) = 1$ and $f(2) = 1 + \frac{1}{2} - 1 + \frac{1}{3} = \frac{1}{2} + \frac{1}{3}$ than $f(3)$ equals the next few terms, so $\frac{1}{3} + \frac{1}{4} + \frac{1}{5}$

in genaral, $f(n)$ = $\frac{1}{n} + \frac{1}{n + 1} + ...$ with n terms OR $\frac{1}{n} + \frac{1}{n} + ... + \frac{1}{2n - 1}$

but than $f(\infty) = ?$

$f(n,x) = \frac{1}{n} + \frac{1}{n + 1} + ... + \frac{1}{xn - 1}$ $x$ as $\frac{\text{integer}}{n}$ for integer n

$f(n,x + \frac{1}{n}) - f(n,x) = \frac{1}{xn}$

$n(f(n,x + \frac{1}{n}) - f(n,x)) = \frac{1}{x}$

as $n \to \infty$ $\frac{1}{n} \to 0$ and $x$ could be anything AND the previous expression equals $\frac{df(\infty,x)}{dx}$

but $ln \prime(x) = \frac{1}{x}$ and $ln(1) = 0$ AND $f(\infty,1) = \frac{1}{1 + \infty} = 0$

therefore $f(\infty,x) = ln(x)$

but $f(n,2) = f(n)$ and $? = f(\infty)$

so, $? = f(\infty,2) = ln(2)$

so $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5}$ FOREVER $ = ln(2)$

yay!

### 11.1.1 regular harmonic

Now I'll start with a seiries like $1 + \frac{1}{2} + \frac{1}{3} - 1 + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \frac{1}{2}$ is $f(\infty, 3)$, so ln(3)

the simpler $1 + \frac{1}{2} + \frac{1}{3} - 1 - \frac{1}{2} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \frac{1}{3} - \frac{1}{4}$, if you don't get it, its 3 positave 2 negatave.

for the same reasons as before, this equals (remember that $n \to \infty$ in the following) $\frac{1}{2n} + \frac{1}{2n + 1} + ... + \frac{1}{3n - 1}$, so $(\frac{1}{n} + \frac{1}{n + 1} + ... + \frac{1}{3n - 1}) - (\frac{1}{n} + \frac{1}{n + 1} + ... + \frac{1}{2n - 1})$, so $f(n, 3) - f(n, 2)$, so $f(\infty, 3) - f(\infty, 2)$, so $ln(3) - ln(2)$, so $ln(\frac{3}{2})$, now I will define $g(m, n)$ as m posataves, n negataves, so $g(m, n) = ln(\frac{m}{n})$

now, with the power of $g$ on our side, we can solve for $1 + \frac{1}{2} + \frac{1}{3}...$ (which by the way is the point of this chapter) as 1 posatave, 0 negataves, so $ln(1) - ln(0)$, but $ln(0)$ is undefined, so time to define it!

### 11.1.2 $ln(0)$

$ln(0) = u$ (u for undefined)

$e^u = 0$

$\frac{1}{e^{-u}} = 0$

$e^{\infty} = \infty$

$\frac{1}{\infty} = 0 \Rightarrow u = - \infty \quad \And \quad ln(1) = 0 \Rightarrow 1 + \frac{1}{2} + \frac{1}{3}... = \infty$

### 11.1.3 $\frac{1}{\infty}$

I'll just say it, I spent a long time trying to figure it out a way to prove that $1 + \frac{1}{2} + \frac{1}{3}... = \infty$, and I realized that $e^{-\infty} = 0$ and $\frac{1}{\infty} = 0$ are true by convention, and the only proofs like "if you divide $1$ by $e$ engough times, it eventually has to hit zero" and "$\frac{1}{\infty}$ 'aproaches ' zero" are nor rigoris because the word "approach" proveably doesent have rigoris definition, so you just have to trust the conclution.

### 11.1.4 conclution

$$ 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} ... = ln(2) $$
$$ \text{m posataves, n negataves} = ln(\frac{m}{n}) $$
$$ 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}... = \infty $$

# 12. Binomial

## 12.1 $(1 + x)^n$

$$ (1 + x)^0 = 1 $$
$$ (1 + x)^1 = 1 + x $$
$$ (1 + x)^2 = 1 + 2x + x^2 $$
$$ (1 + x)^3 = 1 + 3x + 3x^2 + x^3 $$
$$ (1 + x)^n = \sum\limits_{k = 0}^{n} f(n, k) x^k = \sum\limits_{k = -\infty}^{\infty} f(n, k) x^k $$
$$ f(n, 0) = 1 $$
$$ f(n, k) = 0, $$

for $k$ bigger than $n$ or smaller than $0$

$$ (1 + x)^n = (1 + x) (1 + x)^{n - 1} = (1 + x) \sum\limits_{k = -\infty}^{\infty} f(n - 1, k) x^k = \sum\limits_{k = -\infty}^{\infty} f(n - 1, k) x^k + \sum\limits_{k = -\infty}^{\infty} f(n - 1, k) x^{k + 1} = \sum\limits_{k = -\infty}^{\infty} f(n - 1, k) x^k +\sum\limits_{k = -\infty}^{\infty} f(n - 1, k - 1) x^k $$
$$ \sum\limits_{k = -\infty}^{\infty} f(n, k) x^k = \sum\limits_{k = -\infty}^{\infty} (f(n - 1, k) + f(n - 1, k - 1)) x^k $$
$$ f(n, k) = f(n - 1, k) + f(n - 1, k - 1) $$

### 12.1.1 discrete calculus

$$ f(n, k) - f(n - 1, k) = f(n - 1, k - 1) $$
$$ \Delta f(x) = : f(x + 1) - f(x) $$
$$ x^{\frac{n}{}} = : \prod\limits_{k = x - n + 1}^{x} k = x \cdot (x - 1) \cdot (x - 2) ... \text{n times} = \frac{x!}{(x - n)!} $$
$$ \Delta x^{\frac{n}{}} = \frac{(x + 1)!}{(x - n + 1)!} - \frac{x!}{(x - n)!} = (x + 1) \frac{x!}{(x - n + 1)!} - \frac{x!}{\frac{(x - n + 1)!}{x - n + 1}} = (x + 1) x^{\frac{n - 1}{}} - (x - n + 1) x^{\frac{n - 1}{}} = x^{\frac{n - 1}{}} ((x + 1) - (x - n + 1)) $$
$$ \Delta x^{\frac{n}{}} = n x^{\frac{n - 1}{}} $$
$$ \Delta f(x, k) = f(x + 1, k) - f(x, k) = f((x + 1), k) - f((x + 1) - 1, k) = f((x + 1) - 1, k - 1) $$
$$ \Delta f(x, k) = f(x, k - 1) $$
$$ \Delta \frac{x^{\frac{n}{}}}{n!} = \frac{x^{\frac{n - 1}{}}}{(n - 1)!} $$
$$ f(x, k) = \frac{x^{\frac{k}{}}}{k!} = \frac{x!}{k!(x - k)!} = \begin{pmatrix} x \\ k \\ \end{pmatrix} $$
$$ (1 + x)^n = \sum\limits_{k = 0}^{\infty} \begin{pmatrix} n \\ k \\ \end{pmatrix} x^k = \sum\limits_{k = 0}^{\infty} \frac{n! x^k}{k!(n - k)!} $$

if $x$ is closer to $0$, than this will converge faster!

## 12.2 $\pi$

$$ \pi = 2 \int_{-1}^{1} \sqrt{1 - x^2} dx = 4 \int_{0}^{1} \sqrt{1 - x^2} dx $$

but at $0$, this is $1$, and at $\frac{1}{2}$, this will converge really fast!

But

$$ \int_{0}^{\frac{1}{2}} \sqrt{1 - x^2} dx = \frac{\sqrt{3}}{8} + \frac{\pi}{12} $$

### 12.2.1 $\sqrt{3}$

$$ \sqrt{3} = \sqrt{4 - 1} = \sqrt{4(1 - \frac{1}{4})} = 2 \sqrt{1 - \frac{1}{4}} = 2 (1 - \frac{1}{4})^{\frac{1}{2}} $$

which converges really fast!

thus...

$$ \pi = 12 ((\sum\limits_{k = 0}^{\infty} \frac{\begin{pmatrix} \frac{1}{2} \\ k \\ \end{pmatrix} (-\frac{1}{4})^{k + 1}}{k + 1}) - \frac{\sum\limits_{k = 0}^{\infty} \begin{pmatrix} \frac{1}{2} \\ k \\ \end{pmatrix} (-\frac{1}{4})^k}{4}) $$

where

$$ \begin{pmatrix} \frac{1}{2} \\ k \\ \end{pmatrix} = \frac{\prod\limits_{m = \frac{1}{2} - k + 1}^{\frac{1}{2}} m}{k!} = \frac{\prod\limits_{m = 1 - k}^{0} m + \frac{1}{2}}{k!} $$

# 13. Fractional calculus

## 13.1 \(\frac{calculas}{2}\)

\[ \frac{d^{\frac{1}{2}}}{dx^{\frac{1}{2}}} f(x) = ? \]
\[ f(x) = x^n \]
\[ \frac{d}{dx} x^n = n x^{n - 1} \]
\[ \frac{d^2}{dx^2} x^n = n (n - 1) x^{n - 2} \]
\[ \vdots \]
\[ \frac{d^k}{dx^k} x^n = n \cdot (n - 1) \cdot (n - 2) \cdot ... \cdot (n - k + 1) x^{n - k} = \frac{n \cdot (n - 1) \cdot (n - 2) \cdot ... \cdot 1 x^{n - k}}{(n - k) \cdot (n - k - 1) \cdot (n - k - 2) \cdot ... \cdot 1} \]
\[ \frac{d^k}{dx^k} x^n = \frac{n! x^{n - k}}{(n - k)!} \]
\[ \frac{d^p}{dx^p} x^n = \frac{n! x^{n - p}}{(n - p)!} \]
\[ \text{where } (n - p)! = : \int_{0}^{\infty} t^{n - p} e^{-t} dt \]

### 13.1.1 \(\quad_2 \int_{0}^{x}\)

\[ \int f(x) dx = \int_{0}^{x} f(t) dt \]
\[ \int \int f(x) dx^2 = ? = \quad_2 \int_{0}^{x} f(t) dt^2 \]
\[ ? = \int_{0}^{x} \int_{0}^{y} f(t) dt dy \]
\[ \frac{d^2}{dx^2} ? = \frac{\partial^2}{\partial x^2} ? = : f(x) \]
\[ ! = \int_{0}^{x} (x - t) f(t) dt = \int_{0}^{x} g(x, t) dt \]
\[ g(x, t) = (x - t) f(t) \]
\[ \frac{\partial}{\partial x} ! = \frac{\partial}{\partial x} \int_{0}^{x} g(x, t) dt = \frac{\int_{0}^{x + dx} g(x + dx, t) dt - \int_{0}^{x} g(x, t) dt}{dx} = \frac{\int_{0}^{x} g(x + dx, t) dt - \int_{0}^{x} g(x, t) dt + g(x + dx, x + dx) dt}{dx} = \frac{\int_{0}^{x} g(x + dx, t) dt - \int_{0}^{x} g(x, t) dt}{dx} + \frac{g(x + dx, x + dx) dt}{dx} = \int_{0}^{x} \frac{g(x + dx, t) - g(x, t)}{dx} dt + \frac{(x + dx - x - dx) f(t) dt}{dx} = \int_{0}^{x} \frac{\partial}{\partial x} g(x, t) dt \]
\[ \frac{\partial}{\partial x} g(x, t) = \frac{\partial}{\partial x} (x - t) f(t) = f(t) \]
\[ \frac{\partial}{\partial x} ! = \int_{0}^{x} f(t) dt \]
\[ \frac{\partial^2}{\partial x^2} ! = f(x) \]
\[ ? = \text{ } ! \]
\[ \quad_2 \int_{0}^{x} f(t) dt^2 = \int_{0}^{x} (x - t) f(t) dt \]
\[ \quad_2 \int_{0}^{x} f(t) dt^2 - \quad_2 \int_{0}^{a} f(t) dt^2 = \int_{0}^{x} (x - t) f(t) dt - \int_{0}^{a} (x - t) f(t) dt \]
\[ \quad_2 \int_{a} f(x) dx^2 = \int_{a}^{x} (x - t) f(t) dt \]

### 13.1.2 \(\quad_3 \int_{a}^{x}\)

\[ \quad_3 \int_{a}^{x} f(t) dt^3 = \quad_2 \int_{a}^{x} \int_{a}^{t} f(s) ds dt^2 = \int_{a}^{x} (x - t) \int_{a}^{t} f(s) ds dt \]
\[ (uv) \prime = u v \prime + u \prime v \]
\[ \int (uv) \prime dx = \int u v \prime dx + \int u \prime v dx \]
\[ uv = \int u \frac{dv}{dx} dx + \int \frac{du}{dx} v dx \]
\[ uv = \int u dv + \int du v \]
\[ \int u dv = uv - \int du v \]
\[ u = \int_{a}^{t} f(s) ds \]
\[ dv = (x - t) dt \]
\[ du = f(t) dt \]
\[ v = - \frac{1}{2} (x - t)^2 \]
\[ \quad_3 \int_{a}^{x} f(t) dt^3 = \int_{t=a}^{x} u dv = [uv]_ {t=a}^{x} - \int_{a}^{x} - \frac{1}{2} (x - t)^2 f(t) dt = [- \frac{1}{2} (x - t)^2 \int_{a}^{t} f(s) ds]_ {t=a}^{x} + \int_{a}^{x} - \frac{1}{2} (x - t)^2 f(t) dt = (- \frac{1}{2} (x - a)^2 \int_{a}^{a} f(s) ds) - (- \frac{1}{2} (x - x)^2 \int_{a}^{x} f(s) ds) + \int_{a}^{x} \frac{1}{2} (x - t)^2 f(t) dt \]
\[ \quad_3 \int_{a} f(x) dx^3 = \frac{1}{2} \int_{a}^{x} (x - t)^2 f(t) dt \]

### 13.1.3 \(\quad_p \int_{a}^{x}\)

\[ \vdots \]
\[ \quad_n \int_{a}^{x} f(t) dt = \int_{a}^{x} g(x, t, n) f(t) dt \]
\[ \quad_{n + 1} \int_{a}^{x} f(t) dt^{n + 1} = \quad_n \int_{a}^{x} \int_{a}^{t} f(s) ds dt^n = \int_{a}^{x} g(x, t, n) \int_{a}^{t} f(s) ds dt \]
\[ u = \int_{a}^{t} f(s) ds \]
\[ dv = g(x, t, n) dt \]
\[ du = f(t) dt \]
\[ v \quad ? = ? \quad - g(x, t, n + 1) \]
\[ \quad_{n + 1} \int_{a}^{x} f(t) dt^{n + 1} = \int_{t=a}^{x} u dv = [uv]_ {t=a}^{x} - \int_{a}^{x} - g(x, t, n + 1) f(t) dt = [- g(x, t, n + 1) \int_{a}^{t} f(s) ds]_ {t=a}^{x} + \int_{a}^{x} g(x, t, n + 1) f(t) dt = (- g(x, a, n + 1) \int_{a}^{a} f(s) ds) - (- g(x, x, n + 1) \int_{a}^{x} f(s) ds) + \int_{a}^{x} g(x, t, n + 1) f(t) dt = \int_{a}^{x} g(x, t, n + 1) f(t) dt + g(x, x, n + 1) \int_{a}^{x} f(s) ds = \int_{a}^{x} g(x, t, n + 1) f(t) dt \]
\[ g(x, x, n + 1) = 0 \text{ AND } \frac{d}{dt} g(x, t, n + 1) = - g(x, t, n) \Rightarrow v = - g(x, t, n + 1) \]
\[ \frac{1}{n!} (x - x)^n = 0 \]
\[ \frac{d}{dt} \frac{1}{n!} (x - t)^n = - \frac{1}{n} n (x - t)^{n - 1} = - \frac{1}{(n - 1)!} (x - t)^{n - 1} \]
\[ g(x, t, 2) = x - t \]
\[ \frac{1}{(2 - 1)!} (x - t)^{2 - 1} = x - t \]
\[ g(x, t, n) = \frac{1}{(n - 1)!} (x - t)^{n - 1} \]
\[ \quad_n \int_{a} f(x) dx^n = \frac{1}{(n - 1)!} \int_{a}^{x} (x - t)^{n - 1} f(t) dt \]
\[ \quad_n \int_{a} f(x) dx^n = \frac{1}{\Gamma (n)} \int_{a}^{x} (x - t)^{n - 1} f(t) dt \]

(this is Gamma)

\[ \quad_\frac{1}{2} \int_{a} f(x) dx^{\frac{1}{2}} = : \frac{1}{\Gamma (\frac{1}{2})} \int_{a}^{x} (x - t)^{n - 1} f(t) dt = \frac{1}{\sqrt{\pi}} \int_{a}^{x} (x - t)^{n - 1} f(t) dt \]

### 13.1.4 differintegral (not derivigral)

\[ \frac{d^{\frac{1}{2}}}{dx^{\frac{1}{2}}} f(x) = : \frac{d}{dx} \frac{1}{\sqrt{\pi}} \int_{a}^{x} (x - t)^{n - 1} f(t) dt \]
\[ \frac{d^p}{dx^p} f(x) = : \begin{Bmatrix} p = 0 : f(x) \\ p \text{ is an integer and greater than } 0 : \frac{d^p}{dx^p} f(x) \\ p \text{ is an integer and less than } 0 : \frac{1}{(|p| - 1)!} \int_{a}^{x} (x - t)^{|p| - 1} f(t) dt \\ p < 0 : \frac{1}{\Gamma (|p|)} \int_{a}^{x} (x - t)^{|p| - 1} f(t) dt \\ p > 0 : \frac{d^k}{dx^k} (\frac{1}{\Gamma (\alpha)} \int_{a}^{x} (x - t)^{\alpha - 1} f(t) dt) \\ \text{where } p + \alpha = \text{integer } k \\ \end{Bmatrix} \]

and here's a graph!

> 

the calculator could not take derivitaves of fractional integrals, so I used the limit definition of the derivitave and sadly, that is why you can only go from \(-1\) to \(1\)

# 14. Fibbonacci

## 14.1 fibbonacci

\[ F_n = F_{n - 1} + F_{n - 2} \]
\[ F_1 = 1 \]
\[ F_2 = 1 \]
\[ x^2 = x + 1 \]
\[ x = \begin{Bmatrix} \frac{1}{2} + \frac{\sqrt{5}}{2} = \phi \\ \frac{1}{2} - \frac{\sqrt{5}}{2} = \psi \\ \end{Bmatrix} \]
\[ (-\frac{1}{x})^2 = \frac{1}{x^2} \]
\[ 1 = \frac{1}{x} + \frac{1}{x^2} \]
\[ 1 - \frac{1}{x} = \frac{1}{x^2} = (-\frac{1}{x})^2 \]
\[ (-\frac{1}{x})^2 = (-\frac{1}{x}) + 1 \]
\[ x = 1 + \frac{1}{x} \]
\[ -\frac{1}{x} ? = ? x = 1 + \frac{1}{x} \]
\[ -1 - \frac{1}{x} = -x ? = ? \frac{1}{x} \]
\[ -x^2 ? = ? 1 \]
\[ x ? = ? i, -i \]
\[ \phi \ne i \]
\[ \phi \ne -i \]
\[ \psi \ne i \]
\[ \psi \ne -i \]
\[ \text{Thus...} \]
\[ \phi^{-1} = -\psi \]
\[ \psi^{-1} = -\phi \]
\[ \psi^{n - 1} = -\phi \psi^n \]
\[ \phi^{n - 1} = -\psi \phi^n \]
\[ \text{And don't forget that!} \]
\[ \phi^2 = \phi + 1 \]
\[ \phi^3 = \phi^2 + \phi = 2 \phi + 1 \]
\[ \phi^4 = 2 \phi^2 + \phi = 3 \phi + 2 \]
\[ \vdots \]
\[ \phi^n = c_{n, n} \phi + c_{n, n - 1} \]
\[ \phi^{n + 1} = c_{n + 1, n + 1} \phi + c_{n + 1, n} = \phi^n \phi = c_{n, n} \phi^2 + c_{n, n - 1} \phi = c_{n, n} \phi + c_{n, n} + c_{n, n - 1} \phi \]
\[ c_{n + 1, n} = c_{n, n} \]
\[ c_{n + 2, n} = c_{n + 1, n} = c_{n, n} \]
\[ \vdots \]
\[ c_{n + k, n} = c_{n, n} \]
\[ C_n = : c_{n, n} \]
\[ \phi^n = C_n \phi + C_{n - 1} \]
\[ \phi^{n + 1} = C_{n + 1} \phi + C_n = \phi^n \phi = C_n \phi^2 + C_{n - 1} \phi = C_n \phi + C_n + C_{n - 1} \phi \]
\[ C_{n + 1} = C_n + C_{n - 1} \]
\[ C_n = C_{n - 1} + C_{n - 2} \]
\[ \phi^2 = \phi + 1 = \phi C_2 + C_1 \]
\[ C_1 = 1 \]
\[ C_2 = 1 \]
\[ C_n = F_n \]
\[ \phi^n = F_n \phi + F_{n - 1} \]
\[ \text{Yay, now I can solve for $F_n$ if it weren't for the second fibbonacci term, so how can I solve that?} \]
\[ \text{Well, this is only because $\phi^2 = \phi + 1$, but same goes for $\psi$, so...} \]
\[ \psi^n = F_n \psi + F_{n - 1} \]
\[ \text{And subtracting, we get...} \]

$$ \phi^n - \psi^n = F_n \phi + F_{n - 1} - F_n \psi - F_{n - 1} = (\phi - \psi) F_n = (\frac{1}{2} + \frac{\sqrt{5}}{2} - \frac{1}{2} + \frac{\sqrt{5}}{2}) F_n = \sqrt{5} F_n $$
$$ F_n = \frac{\phi^n - \psi^n}{\sqrt{5}} = \frac{\phi^n - \psi^n}{\phi - \psi} $$
$$ F_n \approx \frac{\phi^n}{\sqrt{5}} $$

### 14.1.1 luca numbers

$$ L_n = : F_{n + 1} + F_{n - 1} = L_{n - 1} + L_{n - 2} $$
$$ L_1 = 1 $$
$$ L_2 = 3 $$
$$ L_0 = 2 \text{ And } F_0 = 0 \text{ btw.} $$
$$ L_n = \frac{\phi^{n + 1} - \psi^{n + 1}}{\phi - \psi} + \frac{\phi^{n - 1} - \psi^{n - 1}}{\phi - \psi} = \frac{\phi^{n + 1} - \psi^{n + 1} + \phi^{n - 1} - \psi^{n - 1}}{\phi - \psi} $$
$$ \text{I hope that you remembered!} $$
$$ L_n = \frac{\phi^n \phi - \psi^n \psi - \phi^n \psi + \psi^n \phi}{\phi - \psi} = \frac{\phi^n (\phi - \psi) + \psi^n (\phi - \psi)}{\phi - \psi} $$
$$ L_n = \phi^n + \psi^n $$
$$ L_n \approx \phi^n $$

### 14.1.2 fibbonacci numbers?

$$ L_{n + 1} + L_{n - 1} = \phi^{n + 1} + \psi^{n + 1} + \phi^{n - 1} + \psi^{n - 1} = \phi^n \phi + \psi^n \psi - \phi^n \psi - \psi^n \phi = \phi^n (\phi - \psi) - \psi^n (\phi - \psi) = \phi^n \sqrt{5} - \psi^n \sqrt{5} = \frac{\phi^n - \psi^n}{\sqrt{5}} 5 $$
$$ L_{n + 1} + L_{n - 1} = 5 F_n $$
$$ F_n = \frac{L_{n + 1} + L_{n - 1}}{5} $$
$$ L_n \phi + L_{n - 1} = (\phi^n + \psi^n) \phi + \phi^{n - 1} + \psi^{n - 1} = \phi^{n + 1} + \psi^n \phi + \phi^{n - 1} - \psi^n \phi = \phi^{n + 1} + \phi^{n - 1} = \phi^n (\phi + \phi^{-1}) $$
$$ \phi^n \sqrt{5} = L_n \phi + L_{n - 1} = \phi^{n + 1} + \phi^{n - 1} $$
$$ \phi^n = \frac{L_n \phi + L_{n - 1}}{\sqrt{5}} = \frac{\phi^{n + 1} + \phi^{n - 1}}{\sqrt{5}} $$
$$ \phi^n = \phi^{n - 1} + \phi^{n - 2} \text{ btw} $$

### 14.1.3 everything up 'til now

$$ F_1 = 1 $$
$$ F_2 = 1 $$
$$ F_0 = 0 $$
$$ \phi^1 = \phi $$
$$ \phi^2 = \phi + 1 $$
$$ \phi^0 = 1 $$
$$ L_1 = 1 $$
$$ L_2 = 3 $$
$$ L_0 = 2 $$

.

$$ F_n = F_{n - 1} + F_{n - 2} $$
$$ \phi^n = \phi^{n - 1} + \phi^{n - 2} $$
$$ L_n = L_{n - 1} + L_{n - 2} $$

.

$$ F_n = \frac{\phi^n - \psi^n}{\sqrt{5}} $$
$$ \phi^n = F_n \phi + F_{n - 1} = \frac{L_n \phi + L_{n - 1}}{\sqrt{5}} $$
$$ L_n = \phi^n + \psi^n $$

.

$$ F_n = \frac{L_{n + 1} + L_{n - 1}}{5} $$

$$ \phi^n = \frac{\phi^{n + 1} + \phi^{n - 1}}{\sqrt{5}} $$
$$ L_n = F_{n + 1} + F_{n - 1} $$

.

$$ F_n \approx \frac{\phi^n}{\sqrt{5}} $$
$$ \phi^n \approx \frac{F_{n + 1}}{F_n} $$
$$ L_n \approx \phi^n $$

### 14.1.4 one more thing

$$ \text{how would one combine } \phi^n = F_n \phi + F_{n - 1} \text{ and } \phi^n = \frac{L_n \phi + L_{n - 1}}{\sqrt{5}}? $$
$$ \text{for one, rearrange the second one} $$
$$ \phi^n = F_n \phi + F_{n - 1} $$
$$ \phi^n \sqrt{5} = L_n \phi + L_{n - 1} $$
$$ F_{n - 1} = F_n - F_{n - 2} $$
$$ L_{n - 1} = F_n + F_{n - 2} $$
$$ \phi^n = F_n \phi + F_n - F_{n - 2} $$
$$ \phi^n \sqrt{5} = L_n \phi + F_n + F_{n - 2} $$
$$ (1 + \sqrt{5}) \phi^n = 2 \phi \phi^n = (F_n + L_n) \phi + F_n - F_{n - 2} + F_n + F_{n - 2} = (F_n + L_n) \phi + 2F_n $$
$$ \phi \phi^n = \frac{1}{2} (F_n + L_n) \phi + F_n $$
$$ \phi^n = \frac{1}{2} (F_n + L_n) + F_n \phi^{-1} = \frac{1}{2} F_n + \frac{1}{2} L_n - F_n \psi = \frac{1}{2} F_n + \frac{1}{2} L_n - \frac{1}{2} F_n + \frac{\sqrt{5}}{2} F_n $$
$$ \phi^n = \frac{1}{2} L_n + \frac{\sqrt{5}}{2} F_n \approx \frac{1}{2} \phi^n + \frac{\sqrt{5}}{2} \frac{\phi^n}{\sqrt{5}} = \phi^n $$

## 14.2 fibbonacci 2

$$ S_n = ? $$
$$ S_0 = a $$
$$ S_1 = b $$
$$ S_n = S_{n - 1} + S_{n - 2} $$
$$ F_{-1} = 1 \text{ Btw.} $$
$$ S_2 = a + b $$
$$ S_3 = a + b + b = a + 2b $$
$$ S_4 = a + 2b + a + b = 2a + 3b $$
$$ S_5 = 2a + 3b + a + 2b = 3a + 5b $$

Yes, I know that this text is not centered and is in a different font. I'm tired of proving things that the corrasponding youtube narrator just says "I'll leave this as an exersize for the veiwer", for example I used lines \(58\) thru \(88\) just to prove that the numbers that you were seeing were the fibbonacci numbers, I got too lazy to prove this next one, so I hate to say it, but I will leave proving the following statement as an exersize for the viewer:

$$ S_n = F_n b + F_{n - 1} a = \frac{(\phi^n - \psi^n) b + (\phi^{n - 1} - \psi^{n - 1}) a}{\phi - \psi} = \frac{\phi^n b - \psi^n b - \psi \phi^n a + \phi \psi^n a}{\phi - \psi} $$
$$ S_n = \frac{\phi^n (b - \psi a) - \psi^n (b - \phi a)}{\phi - \psi} $$

# 15. Probability

## 15.1 to cheat, or not to cheat?

let's start with a game! The game is that you flip a coin 10 times and then you add the total number of heads and you want the most heads total, this game is preatty popular in some hypothetical town where there is a rumor going around that some pepole are using (somehow) using a weighted coin that has a $75$% chance of landing heads, the notation is $\frac{3}{4}$ chance, because what else (other than $100$) are you going to take a percent of? Anyways, this town has anominated you to be the detectave and test everyone with a test has all three of the properties:

1, false accusation: accuse less than $5$% of fair players

2, true accusation: accuse more than $80$% of cheaters

3, make it short: do as little coin flips as possible because we don't have all day

this is usualy where I solve the puzzle, but now's the chance to solve it yourself using the tools in the next chapter

### 15.1.1 solution

#### binomial coeficient

do you remember that time when I proved the binomial therom ? Well, I can count the number of heads and the number of tails and the total number of flips is always the same. For example, no matter if you get $1$ heads and $4$ tails, $3$ heads and $2$ tails, whatever it is, it shuld add up to $5$ flips total. The same happens with the probability, because if there always has to be $5$ flips, then there could be $0$ to $5$ heads and $5$ minus that tails, so if you add up the probability of each thing happening, then you should have to get $1$ or $100$%

even if I proved that:

$$ (1 + x)^n = \sum\limits_{k = 0}^{\infty} \begin{pmatrix} n \\ k \\ \end{pmatrix} x^k = \sum\limits_{k = 0}^{\infty} \frac{n! x^k}{k!(n - k)!} $$

you can use a simalar argument that:

$$ (x + y)^n = \sum\limits_{k = 0}^{\infty} \begin{pmatrix} n \\ k \\ \end{pmatrix} x^k y^{n - k} = \sum\limits_{k = 0}^{\infty} \frac{n! x^k y^{n - k}}{k!(n - k)!} $$

so if I made $x$ into the probability $p$ of landing heads, made $y$ into $(1 - p)$, and made $n$ into the number of flips, than on the left, I would get $1^n$, which is just $1$, but on the right, knowing that multaplying probabilities is the probability that they happen in parellel (if independent!), I would get the sum of the probability of any number number of heads times the probability of getting tails $5$ minus the total number of heads times (which is the total number of tails) times each orientation e.g.heads, tails, heads tails, heads is the same as heads, heads, heads, tails, tails. So this is taking the probability of getting some number of heads and tails, adding them up, and saying that it is $100$%, going one step back, each term is the probability of getting some number of heads which I will call $k$, I will also swap $5$ with $n$.

This seems preaty useful and what I did was took apart the binomial therom and the probability is the probability of getting $k$ heads and $n - k$ tails and each oreintation. I will write this probability below saying that there are $n$ flips, $k$ heads, and probability $p$ of landing heads.

$$ P(n, k, p) = \begin{pmatrix} n \\ k \\ \end{pmatrix} p^k (1 - p)^{n - k} $$

#### binomial approxamation (ish)

Realisticly, the test wouldn't test a specific number of heads or tails, so it would be something more like $7$ or more heads out of $10$ flips, you could say that it is the probability of $7$ heads OR $8$ heads OR $9$ heads OR $10$ heads, but for one, it is hard to define an "OR", for example if there are two coins $a$ and $b$, one has probability $a$ of landing heads, one has probability $b$ of landing heads, the probability of $a$ AND $b$ landing heads is simply $ab$, but the probability of $a$ OR \

(b\) landing heads (or both!) is \(1 - (1 - a) (1 - b)\) and that is preaty easy to prove, anyways my point still stands! And I should not use that definiton, because it is computationally intensive, hard to read, and most of all, it uses an "I'll leave this as an exersize for the veiwer". Anyways, I am tired of writing text and just want to prove this new formula.

So, now that you know how to minipulate probabilities, I won't use something like the binomial therom and I will instead just build it from the ground up, so the factor of \(p^k\) sounds about right, but if I don't use the \((1 - p)^{n - k}\) term, than it wouldn't necessarily mean that the rest are tails, but what about the \(n\) choose \(k\) term? You might be asking (actually probably not, because I forgot to tell you what that oreintation term is called a looooooooooong time ago) well, same as before, just replace the tails with question marks! Actually, as I am writing this, I realised that those question marks need an oreintation too, exept sometimes they don't if they are all the same then they don't and it is really confusing. Actually, as I was writing that, I realized, no actuall I couldn't find a counter counter point, but what I did realise was that the \(\frac{n!}{k!(n - k)!}\) was the thing that was going to change.

there are sooooo many counter counter counter counter counter points, and I know that the solution is something something \(1 - (1 - p)^n\), so I will just go off of that. \(1 - (1 - p)^n\) Is a term that means the probability of \(p\) OR \(p\) OR \(p\) OR \(p\) OR \(p \dots\) \(n\) times, so the probability of \(1\) or more heads out of \(n\) total flips, I just made this up, but to add \(k\) into the equation, replace \(p\) with \(p^k\), so now it is the probability of \(p^k\) or more heads (if this is wrong, than I will replace it later), anyways its time to do the thing where I say "Anyways, I am tired of writing text and just want to prove this new formula."

\[ 1 - (1 - p^k)^n \]

## 15.1.2

i't been \(2\) months since I worked on this page. i'll exept that (hold on, I need to find it somewhere in this page)

\[ 1 - \prod\limits_{i = k}^{n} (1 - \begin{pmatrix} n \\ k \\ \end{pmatrix} p^k (1 - p)^{n - k}) \]

# 16. Complex II

## 16.1 the roots of unity

$$ z = r e^{i \theta} $$
$$ \tau = : 2 \pi $$
$$ e^{i \tau} = 1 $$
$$ e^{k i \tau} = 1 $$
$$ z = r e^{i \theta + k i \tau} $$
$$ z^n = 1 $$
$$ (r e^{i \theta} e^{k i \tau})^n = 1 $$
$$ r^n {e^{i \theta}}^n {e^{k i \tau}}^n = 1 $$
$$ r^n e^{i \theta n} e^{k i \tau n} = 1 $$
$$ r^n = 1 $$
$$ r = 1 $$
$$ e^{i \theta n} e^{k i \tau n} = 1 $$
$$ e^{i \theta n} = e^{m i \tau} $$
$$ (z^n)^m = 1^m = 1 $$
$$ i \theta n = m i \tau $$
$$ \theta n = \tau m $$
$$ \theta = \tau \frac{m}{n} $$
$$ rou_{m, n} = e^{i \tau \frac{m}{n}} $$

The "rou" means root of unity, because unity means $1$, and root because $z$ is the $n$'th root of $1$. but all of the roots of unity can be constructed with what is called the principal value $rou_{1, n}$ (actually no, the principal value is $rou_{0, n}$, so just $1$), but that is:

$$ e^{i \frac{\tau}{n}} $$

This is because the more common $e^{i \tau \frac{m}{n}}$ is just $(e^{i \frac{\tau}{n}})^m$, so the following statement is:

$$ rou_{m, n} = (rou_{1, n})^m $$
$$ rou_n = : rou_{1, n} $$
$$ rou_{m, n} = rou_n^m $$

this also looks kind of like notation, but it is just a power.

# 17. The strand puzzle

## 17.1 house numbers

First things first, I have to copy the strand puzzle word for word, having the same punctuation, same capitalization, and same timing for pressing enter, and here it is

"There is a man who lives on a long street, numbered on

his side one, two, three, and so on, and that all the

numbers on one side of him added up exactly the same

as all the numbers on the other side of him.

.

He said he knew there was more than

fifty houses on that side of the street, but not so many

as five hundred."

just to clarify, there are no houses on the other side of the street, and the house numbers don't skip any. To solve the puzzle, you find the number of his house, and the total number of houses on the street / number of the last house on the street, and that one is in between $50$ and $500$. For future reference, I will call those unknowns $x$ and $y$ respectively. When someone gave ramanujan (I'm a big fan of him by the way), he (supposedly) solved it in under $10$ seconds.

### 17.1.1 calculation

also, because it is against MY rules to use something that I haven't derived (which is why the polynomial page has more than $10$ lines), and why most of the contents of this website are derivations), first I will derive the formula for the sums of integers using discrete calculus

#### the formula for the sums of integers (I couldn't find a better name)

$$\sum\limits_{k = 0}^{n} k = f(n)$$
$$\Delta f(n) = \sum\limits_{k = 0}^{n + 1} (k) - \sum\limits_{k = 0}^{n} (k) = n + 1 + \sum\limits_{k = 0}^{n} (k) - \sum\limits_{k = 0}^{n} (k)$$
$$\Delta f(n) = n + 1$$
$$f(n - 1) + n = f(n)$$
$$f(n - 1) = f(n) - n$$
$$f(1) = 1$$
$$f(1 - 1) = f(0) = f(1) - 1$$
$$f(0) = 0$$
$$\Delta^2 g(n) = : \Delta (\Delta g(n))$$
$$\Delta^2 f(n) = \Delta (n + 1) = n + 1 + 1 - n - 1 = 1$$
$$\Delta f(0) = 0 + 1 = 1$$
$$\Delta^2 f(0) = 1$$
$$\text{for the rest of this page, until I say "ao!", the things about } f(n) \text{ and } g(n) \text{ work for any } f \text{ and } g$$
$$\Delta (f(n) + g(n)) = f(n + 1) + g(n + 1) - f(n) - g(n) = f(n + 1) - f(n) + g(n + 1) - g(n) = \Delta f(n) + \Delta g(n)$$
$$f(n + 1) = f(n) + \Delta f(n)$$
$$\Delta (c f(n)) \text{(in respect to n)} = c f(n + 1) - c f(n) = c (f(n) + \Delta f(n)) - c f(n) = c f(n) + c \Delta f(n) - c f(n) = c \Delta f(n)$$

**putting it all together**

$$ \Delta \begin{pmatrix} n \\ k \\ \end{pmatrix} \text{(in respect to n)} = \begin{pmatrix} n \\ k - 1 \\ \end{pmatrix} $$
$$ \begin{pmatrix} n \\ k \\ \end{pmatrix} = \frac{n^{\frac{k}{}}}{k!} $$
$$ \Delta f(n) = 5 \begin{pmatrix} n \\ 0 \\ \end{pmatrix} + 1 \begin{pmatrix} n \\ 1 \\ \end{pmatrix} + 4 \begin{pmatrix} n \\ 2 \\ \end{pmatrix} + 2 \begin{pmatrix} n \\ 3 \\ \end{pmatrix} $$
$$ f(n) = f(0) + 5 \begin{pmatrix} n \\ 1 \\ \end{pmatrix} + 1 \begin{pmatrix} n \\ 2 \\ \end{pmatrix} + 4 \begin{pmatrix} n \\ 3 \\ \end{pmatrix} + 2 \begin{pmatrix} n \\ 4 \\ \end{pmatrix} = f(0) \begin{pmatrix} n \\ 0 \\ \end{pmatrix} + 5 \begin{pmatrix} n \\ 1 \\ \end{pmatrix} + 1 \begin{pmatrix} n \\ 2 \\ \end{pmatrix} + 4 \begin{pmatrix} n \\ 3 \\ \end{pmatrix} + 2 \begin{pmatrix} n \\ 4 \\ \end{pmatrix} $$
$$ \text{ao!} $$
$$ \Delta^2 f(n) = \begin{pmatrix} n \\ 0 \\ \end{pmatrix} $$
$$ \Delta f(0) = 1 $$
$$ \Delta f(n) = \begin{pmatrix} n \\ 0 \\ \end{pmatrix} + \begin{pmatrix} n \\ 1 \\ \end{pmatrix} $$
$$ f(0) = 0 $$
$$ f(n) = \begin{pmatrix} n \\ 1 \\ \end{pmatrix} + \begin{pmatrix} n \\ 2 \\ \end{pmatrix} $$
$$ \begin{pmatrix} n \\ 0 \\ \end{pmatrix} = 1 $$
$$ \begin{pmatrix} n \\ 1 \\ \end{pmatrix} = n $$
$$ \begin{pmatrix} n \\ 2 \\ \end{pmatrix} = \frac{n (n - 1)}{2} $$
$$ f(n) = \frac{n (n - 1)}{2} + n = \frac{n (n - 1)}{2} + \frac{2n}{2} = \frac{(n - 1) n + (2) n}{2} = \frac{n (n - 1 + 2)}{2} $$
$$ \sum\limits_{k = 0}^{n} k = \frac{n (n + 1)}{2} $$

## 17.1.2 back to the strand puzzle or whatever I was doing because I don't remember

now I can find formulas for the sums of house numbers and say that they are equal. The first starts at $1$ and stops at $x - 1$, so the formula is:

$$ \frac{(x - 1) x}{2} $$

Now, for the second formula, there are two methods. If I say "ao!", then forget the last $2$ lines, if i say "oa!", than remember them. With that being said...

one method is to pull out an $x$ at each of the $y - x$ steps, which amounts to an addded $x(y - x)$, so it is:

$$ \frac{(y - x) (y - x + 1)}{2} + x(y - x) = \frac{1}{2}((y - x) (y - x + 1) + 2x(y - x)) = \frac{1}{2}(y^2 - xy + y - xy + x^2 - x + 2xy - 2x^2) = \frac{1}{2}(y^2 + y - x^2 - x) = \frac{y (y + 1)}{2} - \frac{x (x + 1)}{2} $$

ao!

one method is to realize that this is the difference between the first $y$ houses and the first $x$ houses, so it is:

$$ \frac{y (y + 1)}{2} - \frac{x (x + 1)}{2} $$

oa!

you can probably tell that the first was made by me, but time for some algebra!

$$ \frac{(x - 1) x}{2} = \frac{y (y + 1)}{2} - \frac{x (x + 1)}{2} $$
$$ (x - 1) x = y (y + 1) - x (x + 1) $$
$$ x^2 - x = y^2 + y - x^2 - x $$
$$ 2x^2 = y^2 + y $$

Bit more!

$$ 2 (4x^2) = 4y^2 + 4y $$
$$ 2 (2x)^2 = 4y^2 + 4y + 1 - 1 = (4y^2 + 4y + 1) - 1 = (2y + 1)^2 - 1 $$
$$ (2y + 1)^2 - 2 (2x)^2 - 1 = 0 $$
$$ (2y + 1)^2 - 2 (2x)^2 = 1 $$
$$ X = : 2x $$
$$ Y = : 2y + 1 $$
$$ Y^2 - 2X^2 = 1 $$

Which you might (but probably don't) know as the...

## 17.1.3 pell equation

$$ Y^2 - 2X^2 \approx 0 $$
$$ Y^2 \approx 2X^2 $$
$$ \frac{Y^2}{X^2} \approx 2 $$
$$ (\frac{Y}{X})^2 \approx 2 $$
$$ \frac{Y}{X} \approx \sqrt{2} $$

But that is besides the point. The pell equation can be rearanged into $Y = \sqrt{2X^2 + 1}$, then plug in values for $X$ until $Y$ is an integer. Doing this, (and remembering to divide by $2$ and take the floor), I'll list the solutions that we get.

$$ 6 \quad 8 $$
$$ 35 \quad 49 $$

Just barely misses!

$$ 204 \quad 288 $$

There it is!

## 17.1.4 in genaral

the real soluion would be a formula for the $n$'th solution to the strand puzzle droppng the second rule, it is a bit tedious to check using this method and someone probably would have used the formula and this wouldn't work. I usually don't tell you the video that this page is built on, but this time, you would notice that it takes a completely different method. So yes, I built the rest of this page from the ground up.

Something that is easy to notice but hard to prove is that the bigger the $X$ and $Y$, the better the approxamation of $\sqrt{2}$, so I will conjecture the following: Any ratio $\frac{Y}{X}$ for $X$ and $Y$ solving the pell equation*, is closer to $\sqrt{2}$ than any other fraction with denominator less than or equal to $X$.

*The numbers $X$ and $Y$ could solve the alternitave $Y^2 - 2X^2 = - 1$ that alternates with the pell equation as $X$ and $Y$ get bigger, so instead of the pell equation, from now on, it is more like $|Y^2 - 2X^2| = 1$.

By the way, the conjecture was an "I'll leave this as an exersize for the veiwer", and I made up the proof as I went.

Proof (by contradiction): Let's say that there is a fraction $\frac{b}{a}$ that is a better approxamation of $\sqrt{2}$ than $\frac{Y}{X}$ with $a$
strictly *less than $X$. $X$ and $Y$ solve $|Y^2 - 2X^2| = 1$, and $X$ is the next biggest solution than $a$. Well, what is $|b^2 - 2a^2|$? By the thing that is easy to notice but hard to prove, even if $|b^2 - 2a^2| = 1$, it still dosen't work, so $|b^2 - 2a^2| \neq 1$, and by the fact that $\sqrt{2}$ is irrational** (I'll prove that later), $|b^2 - 2a^2| \neq 0$, and by the fact that $a$ and $b$ are integers, $|b^2 - 2a^2| = an$ $integer$. So $|b^2 - 2a^2|$ is an integer, does not equal $0$, and does not equal $1$. But the closer to $1$, the better the approxamation (unless the numbers are is bigger). Thus, the proof is complete!

*If $a = X$, than $|b^2 - 2a^2| = |b^2 - 2X^2|$. If $b = Y$, than I can ignore that case because $\frac{Y}{X}$ cannot be a better approxamation of $\sqrt{2}$ than $\frac{Y}{X}$. So if $b < Y$, $b^2 < Y^2$ * , so $|b^2 - 2X^2|$ also is not $1$.

( ) If $a$, $b$, $X$, and $Y$ are all negatave, than $b^2 > Y^2$. But for one, if they are both negatave, then they are not in reduced fractions. And for another, if one is negatave, than the whole thing is negatave, and $\sqrt{2}$ is not. Also if $b > Y$ (yes, it was only a constraint on $a$ and not $b$), than it still wouldn't work.

** Warning! I will swap $a$ and $b$ in this proof: Note that an even number is two times an integer, if the square of a number is even, than the number that is being squared is even, and a ratio of two even numbers is not in reduced form. You can derive these because I am too busy writing this proof down. Lets say that a fraction $\frac{a}{b}$ in lowest terms (that will be important in the contradiction part of proof by contradiction) that equals $\sqrt{2}$. Going back, this means that $a^2 = 2b^2$. But $b^2$ is an integer, so $a^2$ is even, so $a$ is even. if $a$ is even, than lets say that $a = 2m$ for some integer $m$, $a^2 = 4m^2$, so $4m^2 = 2b^2$, so $2m^2 = b^2$. But $m^2$ is an integer, so $b^2$ is even, so $b$ is even, so $a$ and $b$ are both even, but if they are both even, (and if you remembered), this is a contradiction!

Side note! With this proof, you can prove the thing that is easy to notice but hard to prove, but this would be a circular argument, because we need it to be true for it to be true. To this day, I haven't found a proof of this fact. But if I do, then you know the drill by now, I'll write the proof here and put this in quotation marks.

Now, the proof is finally complete! Where was I again?

### 17.1.5 Time to keep going

$$ Y^2 - 2X^2 = 1 $$
$$ \hat{Y} = 2X + Y $$
$$ \hat{X} = X + Y $$
$$ \hat{Y}^2 - 2 \hat{X}^2 = (2X + Y)^2 - 2(X + Y)^2 = 4X^2 + 4XY + Y^2 - 2X^2 - 4XY - 2Y^2 = 2X^2 - Y^2 = -(Y^2 - 2X^2) $$
$$ \hat{Y}^2 - 2 \hat{X}^2 = - 1 $$
$$ Y_N = 2 \hat{X} + \hat{Y} $$
$$ \text{(N for next)} $$
$$ X_N = \hat{X} + \hat{Y} $$

But the calculations that I did earlier still work, so

$$ Y_N^2 - 2X_N^2 = 1 $$

So like I said, solutions for the pell equation and inverse pell equation alternate.

Okay, this actially happend: I was soving the strand puzzle and got up to this point. I met someone new, and naturally, I opened with how I got here and where I was with the strand puzzle. She said "can you prove that this is the next solution to the pell equation", and I said "I'm too tired fot that". My point is that you shold just run with it.

From now on, $X$ will be replaced with $X_n$, $Y$ will be replaced with $Y_n$, $\hat{X}$ will be replaced with $X_{n + 1}$, $\hat{Y}$ will be replaced with $Y_{n + 1}$, $X_N$ will be replaced with $X_{n + 2}$, and $Y_N$ will be replaced with $Y_{n + 2}$. this is because it will look more like a formula on $n$ when you solve for $Y_n$ as appose to $Y$. Also these can be solutions to the inverse pell equation. For the pell equation, I want to have a $2n$, so it has to be the inverse pell equation first, an easy answer is

$$ X_1 = 1 $$
$$ Y_1 = 1 $$

and the first solution to the strand puzzle is $1 \quad 1$, so $2 \quad 3$ falls out of that, you can verify this solution yourself (you might recognize $\frac{17}{12}$ from the mathologer video, it translates to $6 \quad 8$ considered as the first real solution, but we'll get there), so

$$ X_2 = 2 $$
$$ Y_2 = 3 $$

and because of the recurance...

$$ X_{n + 1} = X_n + Y_n $$
$$ Y_{n + 1} = 2X_n + Y_n $$
$$ X_{n + 2} = X_{n + 1} + Y_{n + 1} = X_n + Y_n + 2X_n + Y_n = 3X_n + 2Y_2 = 2(X_n + Y_n) + X_n = 2X_{n + 1} + X_n $$
$$ Y_{n + 2} = 2X_{n + 1} + Y_{n + 1} = X_n + X_n + Y_n + Y_n + 2X_n + Y_n = 4X_n + 3Y_n = 2(2X_n + Y_n) + Y_n = 2Y_{n + 1} + Y_n $$
$$ X_n = 2X_{n - 1} + X_{n - 2} $$
$$ Y_n = 2Y_{n - 1} + Y_{n - 2} $$

### 17.1.6 The final dash in the silaspe marithon (yes, that is a referance to mathologer marithon)

For now, I will just copy some text from the fibbonacci page, and minipulate it.

$$ z^2 = z + 1 $$
$$ z = \begin{Bmatrix} 1 + \sqrt{2} = c \\ 1 - \sqrt{2} = d \\ \end{Bmatrix} $$

$$ c^2 = 2c + 1 $$
$$ c^3 = 2c^2 + c = 5c + 2 $$
$$ c^4 = 5c^2 + 2c = 12c + 5 $$
$$ \vdots $$
$$ c^n = c_{n, n} c + c_{n, n - 1} $$
$$ c^{n + 1} = c_{n + 1, n + 1}c + c_{n + 1, n} = c^n c = c_{n, n} c^2 + c_{n, n - 1} c = 2c_{n, n}c + c_{n, n} + c_{n, n - 1} c $$
$$ c_{n + 1, n} = c_{n, n} $$
$$ c_{n + 2, n} = c_{n + 1, n} = c_{n, n} $$
$$ \vdots $$
$$ c_{n + k, n} = c_{n, n} $$
$$ C_n = : c_{n, n} $$
$$ c^n = C_n c + C_{n - 1} $$
$$ c^{n + 1} = C_{n + 1}c + C_n = c^n c = C_n c^2 + C_{n - 1} c = 2C_n c + C_n + C_{n - 1} c $$
$$ C_{n + 1} = C_n + C_{n - 1} $$
$$ C_n = 2C_{n - 1} + C_{n - 2} $$
$$ c^2 = 2c + 1 = c C_2 + C_1 $$
$$ C_1 = 1 $$
$$ C_2 = 2 $$
$$ C_n = X_n $$
$$ c^n = X_n c + X_{n - 1} $$
$$ \text{Yay, now I could solve for $X_n$ if it weren't for the second term, so how can I solve that?} $$
$$ \text{Well, this is only because $c^2 = 2c + 1$, but same goes for $d$, so...} $$
$$ d^n = X_n d + X_{n - 1} $$
$$ \text{and subtracting, we get...} $$
$$ c^n - d^n = X_n c + X_{n - 1} - X_n d - X_{n - 1} = (c - d) X_n = (1 + \sqrt{2} - 1 + \sqrt{2}) X_n = 2\sqrt{2} X_n $$
$$ X_n = \frac{c^n - d^n}{2\sqrt{2}} = \frac{c^n - d^n}{c - d} $$
$$ \text{On a different note, rearanging the reccurance, we get} $$
$$ X_{n - 2} = X_n - 2X_{n - 1} $$
$$ Y_{n - 2} = Y_n - 2Y_{n - 1} $$
$$ X_2 = 2 = 2X_1 + X_0 = 2 + X_0 $$
$$ Y_2 = 3 = 2Y_1 + Y_0 = 2 + Y_0 $$
$$ X_0 = 0 $$
$$ Y_0 = 1 $$
$$ \text{but that is besides the point, what I want is a formula for } Y \text{. Skipping through a lot of trial and error, you might try this:} $$
$$ X_n = 2X_{n - 1} + X_{n - 2} $$
$$ X_{n - 1} = 2X_{n - 2} + X_{n - 3} $$
$$ X_n + X_{n - 1} = 2X_{n - 1} + X_{n - 2} + 2X_{n - 2} + X_{n - 3} $$
$$ (X_{(n)} + X_{(n) - 1}) = 2(X_{(n - 1)} + X_{(n - 1) - 1}) + (X_{(n - 2)} + X_{(n - 2) - 1}) $$
$$ X_1 + X_{1 - 1} = X_1 + X_0 = 1 + 0 $$
$$ X_1 + X_{1 - 1} = 1 $$
$$ X_2 + X_{2 - 1} = X_2 + X_1 = 2 + 1 $$
$$ X_2 + X_{2 - 1} = 3 $$
$$ \text{Sound familiar? If not, I'll write down the formlas that we have just derived along with the formulas for } Y. $$
$$ (X_{(n)} + X_{(n) - 1}) = 2(X_{(n - 1)} + X_{(n - 1) - 1}) + (X_{(n - 2)} + X_{(n - 2) - 1}) $$
$$ X_1 + X_{1 - 1} = 1 $$
$$ X_2 + X_{2 - 1} = 3 $$
$$ \text{Now for } Y. $$
$$ Y_n = 2Y_{n - 1} + Y_{n - 2} $$
$$ Y_1 = 1 $$
$$ Y_2 = 3 $$
$$ \text{Now, it should click.} $$
$$ Y_n = X_n + X_{n - 1} $$
$$ Y_n = \frac{c^n - d^n}{2\sqrt{2}} + \frac{c^{n - 1} - d^{n - 1}}{2\sqrt{2}} = \frac{c^n - d^n + c^{n - 1} - d^{n - 1}}{2\sqrt{2}} = \frac{c^n - d^n + \frac{1}{c} c^n - \frac{1}{d} d^n}{2\sqrt{2}} $$

$$ cd = (1 + \sqrt{2})(1 - \sqrt{2}) = 1 - \sqrt{2} + \sqrt{2} - 2 = -1 $$
$$ Y_n = -\frac{d^n - c^n + \frac{1}{d} d^n - \frac{1}{c} c^n}{2\sqrt{2}} = \frac{cd d^n - cd c^n + cd \frac{1}{d} d^n - cd \frac{1}{c} c^n}{2\sqrt{2}} = \frac{cd d^n - cd c^n + cd^n - dc^n}{2\sqrt{2}} = \frac{cd^n (d + 1) - dc^n (c + 1)}{2\sqrt{2}} $$
$$ c + 1 = 1 + \sqrt{2} + 1 = \sqrt{2}(1 + \sqrt{2}) = c\sqrt{2} $$
$$ d + 1 = 1 - \sqrt{2} + 1 = \sqrt{2}(\sqrt{2} - 1) = -d\sqrt{2} $$
$$ Y_n = \frac{-cdd^n\sqrt{2} - cdc^n\sqrt{2}}{2\sqrt{2}} = \frac{-cdc^n - cdd^n}{2} $$
$$ -cd = 1 $$
$$ Y_n = \frac{c^n + d^n}{2} $$

### 17.1.7 The final step in the silaspe marithon

You might recognize these two formulas from the mathologer video thit I showed you earlier. Around half of the page after and including the "in genaral" chapter was improv, including things like "the proof". The only things that weren't improv were the result for $X_n$, the proof was using*, the other thing was the reccurance relation, I used a reccurance relation that went backwards, but it still counts. You can use these formulas (that I'll write down) for solutions for $x_n$ and $y_n$ (I bet that you forgot about them). I forgot to tell you that $x_n$ and $y_n$ are the $n$'th solutions to the strand puzzle respecively. Another thing to notice is that on the github repo, this is the $395$'th line, this is already the longest page so far, but from now on, if I reach $400$ lines, I'll write down "$400$ lines!!".

*Bunnymatics, a study of the population of immortal bunies. They can be a child, or an adult. Each day, the child bunnies turn into adult bunnies, and the adult bunnies asexually reproduce and then clone themselves (probably an adult thing). Day one, there is one adult bunny. if you give the bunnies price of $1$ for child bunnies and price $c$ for adult bunnies, one more day is multiplying the price by $c$

$400$ lines!!!!!!

$$ X_n = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}} $$
$$ Y_n = \frac{(1 + \sqrt{2})^n + (1 - \sqrt{2})^n}{2} $$
$$ X_n \approx 2x_n $$
$$ Y_n \approx 2y_n + 1 $$
$$ x_n = \frac{X_{2n}}{2} $$
$$ y_n = \frac{Y_{2n}}{2} - \frac{1}{2} $$
$$ x_n = \frac{(1 + \sqrt{2})^{2n} - (1 - \sqrt{2})^{2n}}{4\sqrt{2}} $$
$$ y_n = \frac{(1 + \sqrt{2})^n + (1 - \sqrt{2})^n}{4} - \frac{1}{2} $$

# 18. The infamous "arctan puzzle"

### 18.0.1 First, some history

Before this website was even created, I was on a vacation to europe for family, and I solved at last $2$ puzzles. After that, I said that I solved the strand puzzle and the arctan puzzle. First, I solved the strand puzzle in praha, but I only solved fir $x_n$ and not $y_n$, the time that I truly solved the strand puzzle was while I was writing it down on the page. I solved the strand puzzle first by the way. I know what you might be thinking,

## 18.1 what even is the arctan puzzle?

$$ z = a + bi = re^{i\theta} = r(cos(\theta) + i \text{ } sin(\theta)) = rcos(\theta) + irsin(\theta) $$
$$ a = rcos(\theta) $$
$$ b = rsin(\theta) $$

Good, you can convert from polar to cartesian coordinates, but what about back from cartesian to polar? ($r$ Is the distance from the origin at $0$, and $\theta$ is the angle counerclockwise from the positave $x$ axis. I'm saying this, because either I didn't say it earlier, or as a reminder) You would probably use the pythagorian therom to prove this, but this website dosen't heve illustrations, so I will prove the pythagorian therom without geometry. First, take the complex conjegate.

$$ a - bi = rcos(\theta) - irsin(\theta) $$
$$ cos(\theta) = cos(-\theta) $$
$$ -sin(\theta) = sin(-\theta) $$
$$ a - bi = rcos(-\theta) + irsin(-\theta) $$
$$ a - bi = re^{-i\theta} $$
$$ \text{Side note!} $$
$$ \text{ccong} (e^{i\theta}) = e^{-i\theta} = \frac{1}{e^{i\theta}} $$
$$ \text{End of side note.} $$
$$ (a + bi)(a - bi) = a^2 - abi + bia + b^2 = a^2 + b^2 = re^{i\theta} re^{-i\theta} = r^2 $$
$$ \text{Side note!.. Again.} $$
$$ r = |z| \text{ The distance from } 0 \text{, get it?} $$
$$ z \text{ ccong} (z) = |z|^2 $$
$$ \text{ccong} (z) = \frac{|z|^2}{z} $$
$$ \frac{1}{z} = \frac{|z|^2}{\text{ccong} (z)} $$
$$ \text{End of side note... Again.} $$
$$ a^2 + b^2 = r^2 $$

Okay, that is the pythagonian therom done, bou I was looking for $r$.

$$ r \text{ (Incert is greater than or equal to symbol here.) } 0 $$
$$ r = \sqrt{a^2 + b^2} $$

That's $r$, but what about $\theta$?

$$ \frac{b}{a} = \frac{rsin(\theta)}{rcos(\theta)} = \frac{sin(\theta)}{cos(\theta)} = tan(\theta) $$
$$ \text{Good, } r \text{ dosen't change } \theta! $$
$$ \theta = arctan(\frac{b}{a}) = ? $$

There's the arctan. By the way, this was when I used $h$ instead of $dx$. If $arctan = tan^{-1}$, than I tried to find the derivitves of the inverse for the taylor seiries. I accedentally proved the chain rule with the proof that is now on the calculus page as a bonus. I thought "different method for finding $\theta$. First, normalise $z$."

$$ ln(z) = ln(e^{i\theta}) = i\theta $$
$$ i^2 = -1 $$
$$ (i) (-i) = 1 $$
$$ \frac{1}{i} = -i $$

$$ \theta = -i \text{ } ln(z) $$
$$ \text{But what about } ln(z) \text{?} $$
$$ \text{You might remember the formula } \frac{a^{dx} - 1}{dx} \text{ for } ln(a) \text{, so let's try that!} $$
$$ \theta = -i \frac{z^{dx} - 1}{dx} $$
$$ \text{But then, what about }z^{dx} \text{?} $$
$$ \theta = -i \text{ } \frac{((1 + i dx)^{\frac{\theta}{dx}})^{dx} - 1}{dx} = \frac{1}{i} \text{ } \frac{(1 + i dx)^\theta - 1}{dx} = \text{ } \frac{(1 + i\theta dx) - 1}{i dx} $$
$$ \theta = \theta $$

So, all that we have proved is that $\theta = \theta$. Yay!

Ok, so that's the end of that train of thought. I googled "derivitave of arctan" for the result. When I found the awnser $\frac{1}{1 + x^2}$, I diden't find an awnser why. So, of course I asked ChatGPT, the proof went something like this:

Actually, I forgot it and this is my best guess, with $\frac{}{f}$ as the inverse of $f$:

### 18.1.1 derivitave of arctan

$$ f(\frac{}{f} (x)) = : x $$
$$ \frac{d}{dx} f(\frac{}{{}^f} (x)) = f\prime (\frac{}{f} (x)) \frac{}{f}\prime (x) = 1 $$
$$ \frac{}{f}\prime (x) = \frac{1}{f\prime (\frac{}{f} (x))} $$
$$ f(x) = tan(x) $$
$$ \frac{}{f} = arctan(x) $$
$$ arctan\prime (x) = \frac{1}{tan\prime (arctan(x))} $$
$$ \text{Oh! I forgot to tell you about (well, for one, it's pi day today) the other trigonometric functions other than } \text{sin} \text{ pronounced "sine", and } cos \text{ pronounced "cosine" or "cos", but I prefer cosine, theres } tan = \frac{sin}{cos} \text{ pronounced "tan" or "tangent", which I use interchangibly in real life, and it's inverse } arctan \text{ pronounced execly how it is spelled, but theres also } sec = \frac{1}{cos} \text{ pronounced "secant" or "sec", but I prefer secant, but theres also } csc = \frac{1}{sin} \text{ counterintuitivly, but it's pronounced "cosec" or "cosecant", but I prefer cosec. You can also square any of these for a } 2 \text{ superscript in front of the function instead of the parenthasies} $$
$$ tan\prime (x) = (\frac{sin(x)}{cos(x)}) \prime = \frac{sin\prime (x) cos(x) - sin(x) cos\prime (x)}{cos^2(x)} = \frac{cos(x) cos(x) + sin(x) sin(x)}{cos^2(x)} $$
$$ \text{Okay, we have } cos^2 + sin^2 \text{. Eccept, I still have to derive that.} $$
$$ z = e^{i\theta} = r(cos(\theta) + i \text{ } sin(\theta)) = rcos(\theta) + irsin(\theta) $$
$$ r = 1 $$
$$ z = cos(\theta) + isin(\theta) = a + bi $$
$$ a = cos(\theta) $$
$$ b = sin(\theta) $$
$$ r = \sqrt{a^2 + b^2} $$
$$ cos^2 (\theta) + sin^2 (\theta) = 1 $$
$$ \text{Okay, time to keep going.} $$
$$ tan\prime (x) = \frac{1}{cos^2(x)} $$
$$ tan\prime (x) = sec^2 (x) $$
$$ tan^2 (x) = \frac{sin^2 (x)}{cos^2 (x)} = \frac{sin^2 (x) + cos^2 (x) - cos^2 (x)}{cos^2 (x)} = \frac{sin^2 (x) + cos^2 (x)}{cos^2 (x)} - \frac{cos^2 (x)}{cos^2 (x)} $$
$$ tan^2 (x) = sec^2 (x) - 1 $$
$$ arctan\prime (x) = \frac{1}{sec^2 (arctan(x))} = \frac{1}{1 + sec^2 (arctan(x)) - 1} = \frac{1}{1 + tan^2 (arctan(x))} = \frac{1}{1 + tan(arctan(x))tan(arctan(x))} $$
$$ arctan\prime (x) = \frac{1}{1 + x^2} $$

### 18.1.2 Many weeks later...

I asked ChatGPT for awnsers, and it solved it immediety. This is why I called the arctan puzzle the "tricky puzzle that took $1$ ChatGPT search to solve". Warning: This counts as a taylor seiries (404 page not found) for $arctan(x)$ in the interval $-1 < x < 1$ (or equal to $1$)!)

$$ arctan(0) = 0 $$

$$ arctan(x) = \int_{0}^{x} \frac{1}{1 + t^2} dt $$

#### Quick tangent (no pun intended, plus this is more about arctan than tan)

$$ \sum\limits_{n=0}^{\infty}x^n = ? $$
$$ \text{By the convention in the harmonic page below, if } -1 < x < 1 \text{, than the powers of } x \text{ approach } 0 $$

.

$$ ? x = x \sum\limits_{n=0}^{\infty}x^n = \sum\limits_{n=0}^{\infty}x^{n + 1} = \sum\limits_{n=1}^{\infty + 1}x^n = \sum\limits_{n=0}^{\infty}x^n - x^0 + x^{\infty + 1} $$
$$ x^0 = 1 $$
$$ x^{\infty + 1} \to 0 $$
$$ ? x = ? - 1 $$
$$ ? (x - 1) = -1 $$
$$ ? = \frac{-1}{x - 1} $$
$$ \sum\limits_{n=0}^{\infty}x^n = \frac{1}{1 - x} $$

#### back on track

$$ \frac{1}{1 + x^2} = \frac{1}{1 - (-x^2)} = \sum\limits_{n=0}^{\infty} (-x^2)^n = \sum\limits_{n=0}^{\infty} (-1)^n x^{2n} $$
$$ arctan(x) = \int_{0}^{x} \sum\limits_{n=0}^{\infty} (-1)^n t^{2n} dt = \sum\limits_{n=0}^{\infty} \int_{0}^{x} (-1)^n t^{2n} dt $$
$$ \int_{0}^{x} (-1)^n t^{2n} dt = (-1)^n \frac{x^{2n + 1}}{2n + 1} $$
$$ arctan(x) = \sum\limits_{n=0}^{\infty} (-1)^n \frac{x^{2n + 1}}{2n + 1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} \dots $$

# 19. Modular arithmetic

## 19.1 Like the title suggests, what would happen if you restart counting after $10$?

You (supposedly) started counting when you were very young, so starting with counting to the interers, we will rebuild arithmatic from the ground up.

$$ 1 $$
$$ 2 $$
$$ 3 $$
$$ 4 $$
$$ 5 $$
$$ 6 $$
$$ 7 $$
$$ 8 $$
$$ 9 $$
$$ 10 $$
$$ 1 $$
$$ 2 $$

Okay, this seems about right, but the whole point (kinda) is for there to be one digit, so instead of $10$, what about $0$?

$$ 1 $$
$$ 2 $$
$$ 3 $$
$$ 4 $$
$$ 5 $$
$$ 6 $$
$$ 7 $$
$$ 8 $$
$$ 9 $$
$$ 0 $$
$$ 1 $$
$$ 2 $$

That's better, just use the last digit of a number. But you might be asking: What about the equals sign? In this case, the equals sign tells if two numbers have the same last digit, I'll use the O-equals sign for this.

$$ n = 10 $$
$$ k \circeq k + n $$
$$ k \circeq k + 2n $$
$$ k \circeq k + 3n $$
$$ \vdots $$
$$ k \circeq k - n $$
$$ k \circeq k - 2n $$
$$ k \circeq k - 3n $$
$$ \vdots $$
$$ \text{Remember, these are only integers so far!} $$
$$ a \circeq b \to M(a) = M(b) $$
$$ a \circeq M(a) $$

A function like $M$ would be useful for converting back and forth, but what should it be? What about $M(k)$ equals the $k$'th term in, I'll call it the modular counting. Also, instead of the objectave of finding what numbers are equal to, you try to find it's $M$. Also, there are only $10$ numbers in modular arithmetic, because

$$ -1 < M(k) < n. $$

$$ k = k_d n + M(k) $$
$$ k \equiv M(k) \text{, So } M(M(k)) = M(k) \text{.} $$
$$ M(a + b) = ? $$
$$ a = a_d n + M(a) $$
$$ b = b_d n + M(b) $$
$$ M(a + b) = M(a_d n + M(a) + b_d n + M(b)) = M((a_d + b_d) n + M(a) + M(b)) $$
$$ M(a + b) = M(M(a) + M(b)) $$
$$ M(ab) = ? $$
$$ M(ab) = M((a_d n + M(a))(b_d n + M(b))) = M(a_d n b_d n + a_d n M(b) + M(a) b_d n + M(a) M(b)) = M((a_d b_d n + a_d M(b) + b_d M(a)) n + M(a) M(b)) $$
$$ M(ab) = M(M(a)M(b)) $$
$$ M(a - b) = M(M(a) + M(-b)) $$
$$ M(-b) \equiv -b \equiv -M(b) $$
$$ M(a - b) = M(M(a) - M(b))) $$

### 19.1.1 Non-integers

$$ M(\frac{a}{b}) = M(M(a)M(\frac{1}{b})) $$

Remember, these are only integers so far, this should be the part where you learn about non-integers, but modular arithmetic actually has only $10$ numbers in it. But first, we need to take a leap of faith, the multiplicitave inverse.

$$ 3 \cdot 7 = 21 \equiv 1 $$
$$ \text{Thus} $$
$$ 7 \equiv \frac{1}{3} $$
$$ \text{and} $$
$$ 3 \equiv \frac{1}{7} $$
$$ \frac{5}{7} = 5 \cdot 3 = 15 \equiv 5. $$

But there is a problem, and it's not $\frac{1}{10}$, but here is the problem with $\frac{1}{10}$ anyways.

$$ 10 \equiv 0 $$
$$ \frac{1}{10} ? \equiv ? \frac{1}{0} $$
$$ \frac{1}{10} \equiv ? $$
$$ ? \cdot 10 \equiv 1 $$
$$ \text{any number} \cdot 10 \equiv 0 $$
$$ ? \equiv \frac{1}{0} $$
$$ \frac{1}{10} \equiv \frac{1}{0} $$
$$ \text{The problem is that } 1 \text{ divided by a non-} 0 \text{ number is undefined, say } 2. $$
$$ 2 \equiv / \equiv 0 $$
$$ \frac{1}{2} ? \equiv / \equiv ? \frac{1}{0} $$
$$ \frac{1}{2} \equiv ? $$
$$ 1 \cdot 2 \equiv 2 \ne 1 $$
$$ ? \ne 1 $$
$$ 2 \cdot 2 \equiv 4 \ne 1 $$
$$ ? \ne 2 $$
$$ 3 \cdot 2 \equiv 6 \ne 1 $$
$$ ? \ne 3 $$
$$ \vdots $$
$$ 5 \cdot 2 \equiv 0 \text{ again} $$
$$ \vdots $$
$$ 10 \cdot 2 \equiv 0 \ne 1 $$
$$ ? \ne 3 $$
$$ \frac{1}{2} \equiv \frac{1}{0} $$

So the promlem is that you can't divide by some non-0 numbers like I said. This happens when you are dividing by a factor of the base of the modulas (the base of the modulas is the term where you restart couning first) and the numerator does not also have a factor of it. Said simply

$$ \frac{a}{b} \equiv \frac{1}{0} \text{ precisely when } n \text{ divides } b \text{, and } a \text{ does not divide } b. $$
$$ \text{The solution, of coarse, is to swich the base from } 10 \text{ to } 5 $$
$$ n = 5. $$
$$ \text{Side note! } 2 \text{ and } 3 \text{ are } i \text{ and } -i \text{ respectavely. Actually I don't know which is } i \text{ and which is } -i \text{, but here's proof!} $$
$$ i^2 = -1 $$
$$ (-i)^2 = -1 $$
$$ 2^2 = 4 = 5 - 1 \equiv -1 $$
$$ 3^2 = 9 = 2 \cdot 5 - 1 \equiv -1 $$
$$ i \cdot -i = 1 $$
$$ 2 \cdot 3 = 6 \equiv 1 $$

($200$ lines btw)

$$ i + (-i) = 0 $$
$$ -(i) = -i $$
$$ -(-i) = i $$
$$ 2 + 3 = 5 = 5 \cdot 1 + 0 \equiv 0 $$
$$ -2 = 5 \cdot (-1) + 3 \equiv 3 $$
$$ -3 = 5 \cdot (-1) + 2 \equiv 2 $$

So, because $0$ $1$ $i$ $-i$ and $-1$ would multiplty to get another one of $0$ $1$ $i$ $-i$ and $-1$. This means that if you want to multiply numbers mod $5$, just use the beautiful angle addition of the complex plane. Let's try to fnd the mod $7$ equivilant using last monday's work. That took all day. First, the times tables mod $7$

$$ \begin{bmatrix} x \quad 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 0 \quad 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 \quad 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 2 \quad 0 & 2 & 4 & 6 & 1 & 3 & 5 \\ 3 \quad 0 & 3 & 6 & 2 & 5 & 1 & 4 \\ 4 \quad 0 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 \quad 0 & 5 & 3 & 1 & 6 & 4 & 2 \\ 6 \quad 0 & 6 & 5 & 4 & 3 & 2 & 1 \\ \end{bmatrix} $$

So, iI would want numbers $2$ - $5$ to corraspond with letters $a$ - $d$. And no letter for $6$, because it is already equal to $-1$ mod $7$. So the puzzle is to solve for complex numbers $a$ $b$ $c$ and $d$ with the following times table:

$$ \begin{bmatrix} x \quad 0 & 1 & a & b & c & d & -1 \\ 0 \quad 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 \quad 0 & 1 & a & b & c & d & -1 \\ a \quad 0 & a & c & -1 & 1 & b & d \\ b \quad 0 & b & -1 & a & d & 1 & c \\ c \quad 0 & c & 1 & d & a & -1 & b \\ d \quad 0 & d & b & 1 & -1 & c & a \\ -1 \quad 0 & -1 & d & c & b & a & 1 \\ \end{bmatrix} $$
$$ j = \pm i $$
$$ k = -j = \mp i $$
$$ \text{So, mod } 5 $$
$$ 2 = j $$
$$ 3 = k $$
$$ \text{back to mod } 7 \text{ with a more flattened out times table without multiplication by } 0 \text{ or } 1 \text{, because that is preatty self explanitory.} $$
$$ aa = c $$
$$ ab = -1 $$
$$ ac = 1 $$
$$ ad = b $$
$$ a (-1) = d $$
$$ bb = a $$
$$ bc = d $$
$$ bd = 1 $$
$$ b (-1) = c $$
$$ cc = a $$
$$ cd = -1 $$
$$ c (-1) = d $$
$$ dd = c $$
$$ d (-1) = a $$
$$ (-1) (-1) = 1 $$

I'll let you think about it.

SPOILERS BELOW

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## 19.1.2 awnser

You might have gussed that all of these have absolute value $1$, but you might not have known that these are the [roots of unity(https://silaspe.github.io/maths/complex_2.html). But remember to use that $j$ thing. And yes, as oppose to $e^{i\theta} = cos(\theta) + i \text{ } sin(\theta)$,

$$ e^{j\theta} = cos(\theta) + j sin(\theta). $$

Also, $300$ lines by the way.

You might have also noticed that these numbers can circle around in two different ways. (I found the $d$ one before the $b$ one by the way)

$$ 1b = b $$
$$ bb = a $$
$$ ab = -1 $$
$$ (-1) b = c $$
$$ cb = d $$
$$ db = 1. $$

and also

$$ 1d = d $$
$$ dd = c $$
$$ cd = -1 $$
$$ (-1) d = a $$
$$ ad = b $$
$$ bd = 1. $$

Here's a diagram of both in the complex plane:

And for $d$:

And also, if two numbers on the complex plane have absolute value $1$, and they multiply to $1$, then they are complex congegates or as you might know as $ccong$, so the correct picture translating back from letters $a$ - $d$ and $-1$ to numbers $2$ - $5$ and $6$ would be the $b$ - picture with up being $j$ and down being $k$. And here it is:

Isn't it beautiful? I can just add the angles and multiply the magnitudes of any two numbers to get their product mod $7$. $6$ is on $-1$ which makes sense because $6 = 7 - 1$. Also, numers on opposite sides add to $7$, and "opposite side" in complex numbers means "negatave", so that is also just so perfect to me to. Also, if two numbers are complex conjugates on the unit circle, then they are inverses of each other, and I've never seen that in a real desmos graph before! Also, that was a bit of a lie,

because I'm actually editing on the repo, so the graph isn't loading, and I would have to click on the tab with the graph to see it. Alternatively, I could just hit the "Commit changes" button on the top-right and wait for \(2\) minutes to see it on the website. Yeah, I'll do that.

I will add only this line of text to my website, because it would be sad to add anything else to my website on easter.

I will add only this line of text to my website, because it would be sad to add anything else to my website on April \(1\)'st. Well, no it wouldn't be, but no one would belive something added on April \(1\)'st. So I'll take the day off.

Okay, you know the drill. Time to copy paste and tweak the preveous line, just give me a sec... Wait! April \(2\)'nd is not a holiday. But that means that I have to work today. I think I liked the quadruple weekend more.

And, would you look at that. I forgot to work at all yesterday, and I have to take the day off for family. Like I said many times: I'll imbed the desmos graphs tomorrow, or maybe today if I am lucky

At this point on April \(4\)'th, I'll just stop working on this page and write down below the day that I imbed the desmos graphs.

April \(6\)'th, I deleted the thing I did the day before, and that's why it dosen't look different.

## 19.1.3 how to make your own diagram for modular arithmetic in base \(p\) in the desmos graphing calculator

You don't actually need to write out the full times table, you just have to find the smallest primitave root \(r\) of the base (a primitave root of a base is a number where it's power's would result in all different numbers (exept \(0\)) in the modular base. e.g. \(2\) in mod \(5\), \(2^0 ⊜ 1\), \(2^1 ⊜ 2\), \(2^2 ⊜ 4\), \(2^3 ⊜ 3\), \(2^4 ⊜ 1\). One way to find one is to google it, or use brute force) and then put it a \(p - 1\)'th around the unit circle from \(1\), which is just the point (\(1\), \(0\)). Anyways, first boot up the desmos graphing calculator, then I realized that the colors of the functions you draw go red, blue, green, purple, black, and back to red. First, type \(x\) and check if it is a good color for your circle, if it is, remove the letter \(x\) and copy this "x^{2}+y^{2}=1" and paste it into desmos, if not, press the button on the top-right of the box that you are writing in, which I will call the \(x\) button, and repeat until you either find a good color, or choose the least worst color available. In my case, black. Oh, and paste in `p=1` to make a slider called \(p\) and set it to the base that you chose. then do the same thing for the dots, but click on the emptyness underneith the box before pressing the \(x\) button. Once you find a good dot color, paste in `\left(0,0\right)`. Then, this is the tricky part, click on the emptyness underneith the box, then press the \(x\) button, repeat \(5\) times to get the same color, this will be known as "repeat until same color". then copy and paste in `\left(1,0\right)`, repeat until same color, than paste in `\left(\cos\left(\frac{2\pi}{p - 1}\right),\sin\left(\frac{2\pi}{p - 1}\right)\right)`, repeat until same color, than paste in `\left(\cos\left(2\frac{2\pi}{p - 1}\right),\sin\left(2\frac{2\pi}{p - 1}\right)\right)`, repeat until same color, than paste in `\left(\cos\left(3\frac{2\pi}{p - 1}\right),\sin\left(3\frac{2\pi}{p - 1}\right)\right)`... For this reason, you should probably choose \(p\) to be small like \(5\) or \(7\), and I probably should've told you that earlier. Then the best part, press "label" on each dot and click on the big underscore to start naming, the dot in the center is named " \(0\) ". the one directly to the right and not up or down is named " \(1\) ". The name of the one that directly counterclockwise to the previous is found by multiplying the previous's value by \(r\) and subtracting \(p\) until you get \(0\) or a number less than \(p\). And yeah, that's really all there is to it, I'm sorry for those who decided to read through this wall of text (which is the biggest one yet) and didn't just skip to here, but for those pepole who did read this whole thing, you should now have a beautiful diagram for modular arithmatic.

Say goodbye to the modular arithmetic page and say hello to a new page, It's name is the name of the first chapter on the brainstorm page without a question mark... Or just scroll down the dropdown menu or home page.

And with that, the page has ended.

# 20. Geometric algebra

### 20.0.1 why can't you/you can't multiply two vectors

$$ \vec{u} = \begin{bmatrix} u_x \\ u_y \\ u_z \\ \end{bmatrix} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} $$
$$ \vec{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \\ \end{bmatrix} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k} $$
$$ \vec{u} \vec{v} = (u_x \hat{i} + u_y \hat{j} + u_z \hat{k}) (v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) = u_x \hat{i} v_x \hat{i} + u_x \hat{i} v_y \hat{j} + u_x \hat{i} v_z \hat{k} + u_y \hat{j} v_x \hat{i} + u_y \hat{j} v_y \hat{j} + u_y \hat{j} v_z \hat{k} + u_z \hat{k} v_x \hat{i} + u_z \hat{k} v_y \hat{j} + u_z \hat{k} v_z \hat{k} $$
$$ \text{For lack of a better way to display this, } \vec{u} \vec{v} \text{ equals the sum of the things below:} $$
$$ (\hat{i} \hat{i})(u_x v_x) + (\hat{i} \hat{j})(u_x v_y) + (\hat{i} \hat{k})(u_x v_z) $$
$$ (\hat{j} \hat{i})(u_y v_x) + (\hat{j} \hat{j})(u_y v_y) + (\hat{j} \hat{k})(u_y v_z) $$
$$ (\hat{k} \hat{i})(u_z v_x) + (\hat{k} \hat{j})(u_z v_y) + (\hat{k} \hat{k})(u_z v_z) $$

## 20.1 geometric algebra

WARNING! This page requires knowing linear algebra, mostly just vectors, and the first part of A Swift Introduction to Geometric Algebra (literally, that was the name) to know what a $k$-vector is. (Also, it's where these ideas come from (at least in this chapter and the complex numbers chapter coming soon... And the Maxwell's equation chapter at the end).) Here is the definition for the product of basis vectors: The prodct of a basis vector $e_i$ and it self is $1$, and the product of two basis vectors $e_i$ and $e_j$ equals $-e_j e_i (i \ne j)$. This means that you can do this at any point in the product of basis vectors (this should make sense). By the way, $U = \hat{i} \hat{j} \hat{k}$

$$ \hat{i} = \hat{x} $$
$$ \hat{j} = \hat{y} $$
$$ \hat{k} = \hat{z} $$
$$ \text{Now is about as good of a time as any to simplify the product.} $$
$$ \vec{u} \vec{v} = \begin{pmatrix} (\hat{x} \hat{x})(u_x v_x) + (\hat{x} \hat{y})(u_x v_y) + (\hat{x} \hat{z})(u_x v_z) + \\ (\hat{y} \hat{x})(u_y v_x) + (\hat{y} \hat{y})(u_y v_y) + (\hat{y} \hat{z})(u_y v_z) + \\ (\hat{z} \hat{x})(u_z v_x) + (\hat{z} \hat{y})(u_z v_y) + (\hat{z} \hat{z})(u_z v_z) \\ \end{pmatrix} = \begin{pmatrix} (\hat{x} \hat{x})(u_x v_x) + (\hat{x} \hat{y})(u_x v_y) - (\hat{z} \hat{x})(u_x v_z) - \\ (\hat{x} \hat{y})(u_y v_x) + (\hat{y} \hat{y})(u_y v_y) + (\hat{y} \hat{z})(u_y v_z) + \\ (\hat{z} \hat{x})(u_z v_x) - (\hat{y} \hat{z})(u_z v_y) + (\hat{z} \hat{z})(u_z v_z) \\ \end{pmatrix} = \begin{pmatrix} u_x v_x + (\hat{x} \hat{y})(u_x v_y) - (\hat{z} \hat{x})(u_x v_z) - \\ (\hat{x} \hat{y})(u_y v_x) + u_y v_y + (\hat{y} \hat{z})(u_y v_z) + \\ (\hat{z} \hat{x})(u_z v_x) - (\hat{y} \hat{z})(u_z v_y) + u_z v_z \\ \end{pmatrix} = u_x v_x + u_y v_y + u_z v_z + \begin{pmatrix} (\hat{x} \hat{y})(u_x v_y - u_y v_x) + \\ (\hat{y} \hat{z})(u_y v_z - u_z v_y) + \\ (\hat{z} \hat{x})(u_z v_x - u_x v_z) \\ \end{pmatrix} = \vec{u} \cdot \vec{v} + \begin{pmatrix} (\hat{x} \hat{y})(u_x v_y - u_y v_x) + \\ (\hat{y} \hat{z})(u_y v_z - u_z v_y) + \\ (\hat{z} \hat{x})(u_z v_x - u_x v_z) \\ \end{pmatrix} $$
$$ \text{But now, I need to turn a bivector into a vector. What about that } U \text{ thing?} $$

Puzzle time! Prove that $U \vec{v} = \vec{v} U$.

$$ U^2 = \hat{x} \hat{y} \hat{z} \hat{x} \hat{y} \hat{z} = -\hat{x} \hat{y} \hat{z} \hat{x} \hat{z} \hat{y} = \hat{x} \hat{y} \hat{z} \hat{z} \hat{x} \hat{y} = \hat{x} \hat{y} \hat{x} \hat{y} = -\hat{x} \hat{x} \hat{y} \hat{y} $$
$$ U^2 = -1 $$

$U$ is usually called $i$ for this reason.

$$ U^3 = \hat{x} \hat{y} \hat{z} \hat{x} \hat{y} \hat{z} \hat{x} \hat{y} \hat{z} = -\hat{x} \hat{y} \hat{x} \hat{z} \hat{y} \hat{z} \hat{x} \hat{y} \hat{z} = \hat{x} \hat{x} \hat{y} \hat{z} \hat{y} \hat{z} \hat{x} \hat{y} \hat{z} = \hat{y} \hat{z} \hat{y} \hat{z} \hat{x} \hat{y} \hat{z} = -\hat{y} \hat{y} \hat{z} \hat{z} \hat{x} \hat{y} \hat{z} = -\hat{x} \hat{y} \hat{z} = -U = \hat{x} \hat{z} \hat{y} = -\hat{z} \hat{x} \hat{y} = \hat{z} \hat{y} \hat{x} $$
$$ U^4 = U^2 U^2 = (-1) (-1) = 1 $$
$$ U^4 = U^3 U = (-U) U = -U^2 = -(-1) = 1 $$

.

$$ U^1 = U = \hat{x} \hat{y} \hat{z} $$
$$ U^2 = 1 $$
$$ U^3 = -U = \hat{z} \hat{y} \hat{x} = -\hat{x} \hat{y} \hat{z} $$
$$ U^4 = (-U) U = 1 $$

.

$$ \hat{x} \hat{y} = \hat{x} \hat{y} \hat{z} \hat{z} = U \hat{z} $$
$$ \hat{y} \hat{z} = \hat{y} \hat{z} \hat{x} \hat{x} = -\hat{y} \hat{x} \hat{z} \hat{x} = \hat{x} \hat{y} \hat{z} \hat{x} = U \hat{x} $$
$$ \hat{z} \hat{x} = \hat{z} \hat{x} \hat{y} \hat{y} = -\hat{z} \hat{y} \hat{x} \hat{y} = -(-U) \hat{y} = U \hat{y} $$
$$ \text{Yes! Now I can finaly solve the puzzle.} $$
$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \begin{pmatrix} (U \hat{z})(u_x v_y - u_y v_x) + \\ (U \hat{x})(u_y v_z - u_z v_y) + \\ (U \hat{y})(u_z v_x - u_x v_z) \\ \end{pmatrix} = \vec{u} \cdot \vec{v} + U \begin{pmatrix} \hat{x} (u_y v_z - u_z v_y) + \\ \hat{y} (u_z v_x - u_x v_z) + \\ \hat{z} (u_x v_y - u_y v_x) \\ \end{pmatrix} = \vec{u} \cdot \vec{v} + U \begin{bmatrix} u_y v_z - u_z v_y \\ u_z v_x - u_x v_z \\ u_x v_y - u_y v_x \\ \end{bmatrix} $$

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$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + U \text{ } \vec{u} \times \vec{v} $$
$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \hat{x} \hat{y} \hat{z} \text{ } \vec{u} \times \vec{v} $$
$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \vec{u} \times \vec{v} \text{ } i $$

Also, the cross product only works in $3d$ while this $U \text{ } \vec{u} \times \vec{v}$ thing works in any dimention. This operator actually has a name (well, two names), the outer product (as oppose to the dot product sometimes refered to as the inner product) or wedge product for it's apperance as a wedge unicode character. This more genaral cross product is written $\vec{u} ∧ \vec{v}$. Also, I have to interrupt this for...

$100$ lines! But

$$ \vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \vec{u} ∧ \vec{v}. $$

### 20.1.1 $\frac{1}{\vec{v}}$ (and $\vec{v}^2$)

$$ \text{Lets say that we are in dimension } d \text{. First, basis vectors} $$
$$ e_1 = \hat{i} = \hat{x} $$
$$ e_2 = \hat{j} = \hat{y} $$
$$ e_3 = \hat{k} = \hat{z} $$
$$ e_4 = \hat{l} = \hat{w} $$
$$ \vdots $$
$$ \vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_d\\ \end{bmatrix} = \sum\limits_{n = 1}^{d} u_n e_n $$
$$ \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_d \\ \end{bmatrix} = \sum\limits_{n = 1}^{d} v_n e_n $$
$$ \vec{u} \vec{v} = \begin{bmatrix} (\hat{x} \hat{x}) u_1 v_1 & (\hat{x} \hat{y}) u_1 v_2 & (\hat{x} \hat{z}) u_1 v_3 & (\hat{x} \hat{w}) u_1 v_4 & \dots & (\hat{x} e_d) u_1 v_d \\ (\hat{y} \hat{x}) u_2 v_1 & (\hat{y} \hat{y}) u_2 v_2 & (\hat{y} \hat{z}) u_2 v_3 & (\hat{y} \hat{w}) u_2 v_4 & \dots & (\hat{y} e_d) u_2 v_d \\ (\hat{z} \hat{x}) u_3 v_1 & (\hat{z} \hat{y}) u_3 v_2 & (\hat{z} \hat{z}) u_3 v_3 & (\hat{z} \hat{w}) u_3 v_4 & \dots & (\hat{z} e_d) u_3 v_d \\ (\hat{w} \hat{x}) u_4 v_1 & (\hat{w} \hat{y}) u_4 v_2 & (\hat{w} \hat{z}) u_4 v_3 & (\hat{w} \hat{w}) u_4 v_4 & \dots & (\hat{w} e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ (e_d \hat{x}) u_d v_1 & (e_d \hat{y}) u_d v_2 & (e_d \hat{z}) u_d v_3 & (e_d \hat{w}) u_d v_4 & \dots & (e_d e_d) u_d v_d \\ \end{bmatrix} = \begin{bmatrix} u_1 v_1 & (\hat{x} \hat{y}) u_1 v_2 & (\hat{x} \hat{z}) u_1 v_3 & (\hat{x} \hat{w}) u_1 v_4 & \dots & (\hat{x} e_d) u_1 v_d \\ (\hat{y} \hat{x}) u_2 v_1 & u_2 v_2 & (\hat{y} \hat{z}) u_2 v_3 & (\hat{y} \hat{w}) u_2 v_4 & \dots & (\hat{y} e_d) u_2 v_d \\ (\hat{z} \hat{x}) u_3 v_1 & (\hat{z} \hat{y}) u_3 v_2 & u_3 v_3 & (\hat{z} \hat{w}) u_3 v_4 & \dots & (\hat{z} e_d) u_3 v_d \\ (\hat{w} \hat{x}) u_4 v_1 & (\hat{w} \hat{y}) u_4 v_2 & (\hat{w} \hat{z}) u_4 v_3 & u_4 v_4 & \dots & (\hat{w} e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ (e_d \hat{x}) u_d v_1 & (e_d \hat{y}) u_d v_2 & (e_d \hat{z}) u_d v_3 & (e_d \hat{w}) u_d v_4 & \dots & u_d v_d \\ \end{bmatrix} $$
$$ \vec{u} \vec{v} = (u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 + \dots + u_d v_d) + \begin{bmatrix} & (\hat{x} \hat{y}) u_1 v_2 & (\hat{x} \hat{z}) u_1 v_3 & (\hat{x} \hat{w}) u_1 v_4 & \dots & (\hat{x} e_d) u_1 v_d \\ -(\hat{x} \hat{y}) u_2 v_1 & & (\hat{y} \hat{z}) u_2 v_3 & (\hat{y} \hat{w}) u_2 v_4 & \dots & (\hat{y} e_d) u_2 v_d \\ -(\hat{x} \hat{z}) u_3 v_1 & -(\hat{y} \hat{z}) \end{bmatrix}$$

u_3 v_2 & & (\hat{z} \hat{w}) u_3 v_4 & \dots & (\hat{z} e_d) u_3 v_d \\ -(\hat{x} \hat{w}) u_4 v_1 & -(\hat{y} \hat{w}) u_4 v_2 & -(\hat{z} \hat{w}) u_4 v_3 & & \dots & (\hat{w} e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ -(\hat{x} e_d) u_d v_1 & -(\hat{y} e_d) u_d v_2 & -(\hat{z} e_d) u_d v_3 & -(\hat{w} e_d) u_d v_4 & \dots & \\ \end{bmatrix} \]

\[ \text{It was really frustrating to write a plus in between, so I just gave up. Just assume that the results above and below are summed together (I kept swiching between "Just assume that the result above is summed together" and "Just assume that the results above and below are summed together") (you can assume that every matrix on this page is a sum). Here's an idea! Square the vector.} \]

\[ \vec{v} \vec{v} = (v_1 v_1 + v_2 v_2 + v_3 v_3 + v_4 v_4 + \dots + v_d v_d) + \begin{bmatrix} & (\hat{x} \hat{y}) v_1 v_2 & (\hat{x} \hat{z}) v_1 v_3 & (\hat{x} \hat{w}) v_1 v_4 & \dots & (\hat{x} e_d) v_1 v_d \\ -(\hat{x} \hat{y}) v_2 v_1 & & (\hat{y} \hat{z}) v_2 v_3 & (\hat{y} \hat{w}) v_2 v_4 & \dots & (\hat{y} e_d) v_2 v_d \\ -(\hat{x} \hat{z}) v_3 v_1 & -(\hat{y} \hat{z}) v_3 v_2 & & (\hat{z} \hat{w}) v_3 v_4 & \dots & (\hat{z} e_d) v_3 v_d \\ -(\hat{x} \hat{w}) v_4 v_1 & -(\hat{y} \hat{w}) v_4 v_2 & -(\hat{z} \hat{w}) v_4 v_3 & & \dots & (\hat{w} e_d) v_4 v_d \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ -(\hat{x} e_d) v_d v_1 & -(\hat{y} e_d) v_d v_2 & -(\hat{z} e_d) v_d v_3 & -(\hat{w} e_d) v_d v_4 & \dots & \\ \end{bmatrix} \]

(incert filler text here)

\[ \text{By the way, }
\hat{\hat{\hat{

$$ \frac{1}{\vec{v}} = \frac{\vec{v}}{|| \vec{v} ||^2}. $$
$$ \text{By the way, this means that the inverse of a vector is a vector, } \frac{\vec{u}}{\vec{v}} = \frac{\vec{u} \vec{v}}{|| \vec{v} ||^2} \text{. The inverse of a } 2d \text{ is it's circle inverse, the inverse of a } 3d \text{ vector is it's sphere inverse, the inverse of a } 4d \text{ vector is it's hypersphere inverse, and so on.} $$

**fun fact!**

As you know, $\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 + \dots + u_d v_d$. But what about $\vec{v} \cdot \vec{u}$? And what about the wedge product? (By the way, this is about the vector product or sometimes called the geometric product)

$$ \vec{v} \cdot \vec{u} = v_1 u_1 + v_2 u_2 + v_3 u_3 + v_4 u_4 + \dots + v_d u_d = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 + \dots + u_d v_d $$
$$ \vec{v} \cdot \vec{u} = \vec{u} \cdot \vec{v} $$
$$ \vec{v} \vec{u} = \vec{u} \cdot \vec{v} + \vec{v} ∧ \vec{u} $$
$$ \vec{u} ∧ \vec{v} = \begin{bmatrix} & (\hat{x} \hat{y}) u_1 v_2 & (\hat{x} \hat{z}) u_1 v_3 & (\hat{x} \hat{w}) u_1 v_4 & \dots & (\hat{x} e_d) u_1 v_d \\ -(\hat{x} \hat{y}) u_2 v_1 & & (\hat{y} \hat{z}) u_2 v_3 & (\hat{y} \hat{w}) u_2 v_4 & \dots & (\hat{y} e_d) u_2 v_d \\ -(\hat{x} \hat{z}) u_3 v_1 & -(\hat{y} \hat{z}) u_3 v_2 & & (\hat{z} \hat{w}) u_3 v_4 & \dots & (\hat{z} e_d) u_3 v_d \\ -(\hat{x} \hat{w}) u_4 v_1 & -(\hat{y} \hat{w}) u_4 v_2 & -(\hat{z} \hat{w}) u_4 v_3 & & \dots & (\hat{w} e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ -(\hat{x} e_d) u_d v_1 & -(\hat{y} e_d) u_d v_2 & -(\hat{z} e_d) u_d v_3 & -(\hat{w} e_d) u_d v_4 & \dots & \\ \end{bmatrix} $$
$$ \vec{v} ∧ \vec{u} = \begin{bmatrix} & (\hat{x} \hat{y}) v_1 u_2 & (\hat{x} \hat{z}) v_1 u_3 & (\hat{x} \hat{w}) v_1 u_4 & \dots & (\hat{x} e_d) v_1 u_d \\ -(\hat{x} \hat{y}) v_2 u_1 & & (\hat{y} \hat{z}) v_2 u_3 & (\hat{y} \hat{w}) v_2 u_4 & \dots & (\hat{y} e_d) v_2 u_d \\ -(\hat{x} \hat{z}) v_3 u_1 & -(\hat{y} \hat{z}) v_3 u_2 & & (\hat{z} \hat{w}) v_3 u_4 & \dots & (\hat{z} e_d) v_3 u_d \\ -(\hat{x} \hat{w}) v_4 u_1 & -(\hat{y} \hat{w}) v_4 u_2 & -(\hat{z} \hat{w}) v_4 u_3 & & \dots & (\hat{w} e_d) v_4 u_d \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ -(\hat{x} e_d) v_d u_1 & -(\hat{y} e_d) v_d u_2 & -(\hat{z} e_d) v_d u_3 & -(\hat{w} e_d) v_d u_4 & \dots & \\ \end{bmatrix} = -\begin{bmatrix} & -(\hat{x} \hat{y}) v_1 u_2 & -(\hat{x} \hat{z}) v_1 u_3 & -(\hat{x} \hat{w}) v_1 u_4 & \dots & -(\hat{x} e_d) v_1 u_d \\ (\hat{x} \hat{y}) v_2 u_1 & & -(\hat{y} \hat{z}) v_2 u_3 & -(\hat{y} \hat{w}) v_2 u_4 & \dots & -(\hat{y} e_d) v_2 u_d \\ (\hat{x} \hat{z}) v_3 u_1 & (\hat{y} \hat{z}) v_3 u_2 & & -(\hat{z} \hat{w}) v_3 u_4 & \dots & -(\hat{z} e_d) v_3 u_d \\ (\hat{x} \hat{w}) v_4 u_1 & (\hat{y} \hat{w}) v_4 u_2 & (\hat{z} \hat{w}) v_4 u_3 & & \dots & -(\hat{w} e_d) v_4 u_d \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ (\hat{x} e_d) v_d u_1 & (\hat{y} e_d) v_d u_2 & (\hat{z} e_d) v_d u_3 & (\hat{w} e_d) v_d u_4 & \dots & \\ \end{bmatrix} = -\begin{bmatrix} & (\hat{x} \hat{y}) u_1 v_2 & (\hat{x} \hat{z}) u_1 v_3 & (\hat{x} \hat{w}) u_1 v_4 & \dots & (\hat{x} e_d) u_1 v_d \\ -(\hat{x} \hat{y}) u_2 v_1 & & (\hat{y} \hat{z}) u_2 v_3 & (\hat{y} \hat{w}) u_2 v_4 & \dots & (\hat{y} e_d) u_2 v_d \\ -(\hat{x} \hat{z}) u_3 v_1 & -(\hat{y} \hat{z}) u_3 v_2 & & (\hat{z} \hat{z}) u_3 v_4 & \dots & (\hat{z} e_d) u_3 v_d \\ -(\hat{x} w) u_4 v_1 & -(\hat{y} w) u_4 v_2 & -(\hat{z} w) u_4 v_3 & & \dots & (w e_d) u_4 v_d \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ -(\hat{x} e_d) u_d v_1 & -(\hat{y} e_d) u_d v_2 & -(\hat{z} e_d) u_d v_3 & -(w e_d) u_d v_4 & \dots & \\ \end{bmatrix} $$
$$ \vec{v} ∧ \vec{u} = - \vec{u} ∧ \vec{v} $$

Also, halfway through (well, more like mostly through), if I added one more $\hat{w}$, it would be "unabe to render expression". By the way, $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ lines.

$$ \vec{v} \vec{u} = \vec{u} \cdot \vec{v} - \vec{u} ∧ \vec{v} $$

## 20.1.2 complex numbers

$$ \text{complex numbers (surprisingly in geomatric algebra) come from } 2d \text{ geomatric algebra. So time to re-derive that! (And use } i \text{ this time instead of } U \text{.) (I thought that } i \text{ should have a hat, but that wouldn't work.) Here's the definitions and link to the complex numbers page:} $$

.

$$ \hat{i} = \hat{x} \text{ (This is so that } \hat{i} \text{ isn't confused with } i \text{).} $$
$$ \hat{j} = \hat{y} $$
$$ \hat{x} \hat{y} = i $$
$$ \hat{x}^2 = 1 $$
$$ \hat{y}^2 = 1 $$
$$ \hat{x} \hat{y} = -\hat{y} \hat{x} \text{(} = i \text{)} $$

\[ \text{And with those, you can derive} \]
\[ i^2 = -1 \]
\[ \text{this is why it was called } i. \]

.

\[ \text{Also, a bivector in } 2 \text{ dimensions has one degree of freedom (just like how a vector behaves in } 1d \text{) (by the way, a } k \text{-vector in } n \text{ dimensions has } \begin{pmatrix} n \\ k \\ \end{pmatrix} \text{ DoF's (degrees of freedom)), so I'll call it, I dunno, a pseudoscalar. (} i \text{ Is always the unit pseudoscalar no matter the dimension.)} \]
\[ \vec{u} = \begin{bmatrix} u_x \\ u_y \\ \end{bmatrix} = u_x \hat{x} + u_y \hat{y} \]
\[ \vec{v} = \begin{bmatrix} v_x \\ v_y \\ \end{bmatrix} = v_x \hat{x} + v_y \hat{y} \]
\[ \vec{u} \vec{v} = \begin{pmatrix} (\hat{x} \hat{x}) u_x v_x & (\hat{x} \hat{y}) u_x v_y \\ (\hat{y} \hat{x}) u_y v_x & (\hat{y} \hat{y}) u_y v_y \\ \end{pmatrix} = (u_x v_x + u_y v_y) + (u_x v_y - u_y v_x)i \]

\(300\) Lines!

\[ \vec{u} ∧ \vec{v} = i \cdot Det \begin{bmatrix} u_x & u_y \\ v_x & v_y \\ \end{bmatrix} \]

Here's a determinat video and linear algebra playlist for context (make sure to skip the cross product video) (also, that determinant is probably the only matrix on this page. I'll come back here when I finish the page and put the awnser here: ). ( \(\text{ }\) : ) \(\text{ }\) Btw.)

\[ \text{Text over time! (As oppose to voice over, it's a font swich.) What's something that veiwers of this website would want from geomatric algebra? I think I have an idea: rotate a vector by the angle between two other vectors. I'l do it in } 20 \text{ minutes.} \]
\[ \text{It's the next day.} \]
\[ \vec{v} i = (v_x \hat{x} + v_y \hat{y})i = v_x \hat{x} i + v_y \hat{y} i = v_x \hat{x} \hat{x} \hat{y} + v_y \hat{y} \hat{x} \hat{y} = v_x \hat{y} - v_y \hat{x} \hat{y} \hat{y} = v_x \hat{y} - v_y \hat{x} \]

```
def complexify(v_x, v_y):
  return v_x + v_y * j
```

\[ \text{I just realized that this isn't a code repo(sitory). It was my instinct to write it in code form.} \]
\[ \text{complexify} (v_x \hat{x} + v_y \hat{y}) = v_x + v_y i \]

(incert filler text here)

\[ \text{complexify} (\vec{v} i) = i \text{ complexify} (\vec{v}) \]
\[ i \vec{v} = i(v_x \hat{x} + v_y \hat{y}) = i v_x x + i v_y \hat{y} \]
\[ \text{But } v_x \text{ and } v_y \text{ are scalars, so} \]
\[ i \vec{v} = v_x i \hat{x} + v_y i \hat{y} = v_x \hat{x} \hat{y} \hat{x} + v_y \hat{x} \hat{y} \hat{y} = -v_x \hat{y} \hat{x} \hat{x} + v_y \hat{x} = -v_x \hat{y} + v_y \hat{x} \]
\[ \text{complexify} (i \vec{v}) = -i \text{ complexify} (\vec{v}). \]
\[ \text{Next: vector times a complex number.} \]
\[ \text{vectorize} (a + bi) = a \hat{x} + b \hat{y} \]
\[ \text{complexify} (\text{vectorize} (z)) = z \]
\[ \text{vectorize} (\text{complexify} (\vec{v})) = \vec{v} \]
\[ (a \hat{x} + b \hat{y})(c + di) = a \hat{x} c + a \hat{x} di + b \hat{y} c + b \hat{y} di = \begin{pmatrix} ac (\hat{x}) & ad (\hat{x} i) \\ bc (\hat{y}) & bd (\hat{y} i) \\ \end{pmatrix} = \begin{pmatrix} ac (\hat{x}) & ad (\hat{x} \hat{x} \hat{y}) \\ bc (\hat{y}) & bd (\hat{y} \hat{x} \hat{y}) \\ \end{pmatrix} = \begin{pmatrix} ac (\hat{x}) & ad (\hat{y}) \\ bc (\hat{y}) & -bd (\hat{x} \hat{y} \hat{y}) \\ \end{pmatrix} = \begin{pmatrix} ac (\hat{x}) & ad (\hat{y}) \\ bc (\hat{y}) & -bd (\hat{x}) \\ \end{pmatrix} = (ac - bd) \hat{x} + (ad + bc) \hat{y} \]
\[ z = a + bi \]
\[ w = c + di \]
\[ \text{vectorize} (z) w = \text{vectorize} (zw) \]
\[ \vec{u} \text{ complexify} (\vec{v}) = \text{vectorize} (\text{complexify} (\vec{u}) \text{ complexify} (v)) \]
\[ \vec{v} z = \text{vectorize} (\text{complexify} (\vec{v}) z) \]

Wait a minute! (Which was the catch phrase of sugey (pronounced soo-gey) in the quick introduction that I mentioned earlier.) If multiplying a vector by a complex number acts like comlex number multiplication (rotating and scaling) (exempt it returns a

vector, not a complex number), and a vector times a vector is a complex number, that what rotation is that? (Well, the rotation and scaling of a complex number scales by the magnitude and rotates by the angle counterclockwise from the positave \(x\) axis.)

\[ \text{By the way, } \vec{v} \vec{u} = \text{ccong} (\vec{u} \vec{v}) \]
\[ |a + bi| = \sqrt{a^2 + b^2} \]
\[ a + bi = \vec{u} \vec{v} \]
\[ a = \vec{u} \cdot \vec{v} = u_x v_x + u_y v_y \]
\[ bi = \vec{u} ∧ \vec{v} = (u_x v_y - u_y v_x)i \text{ (This also means that } b = u_x v_y - u_y v_x = \vec{u} \times \vec{v} \text{).} \]
\[ || \vec{u} \vec{v} || = \sqrt{(\vec{u} \cdot \vec{v})^2 + (\vec{u} \times \vec{v})^2} = \sqrt{(u_x v_x + u_y v_y)^2 + (u_x v_y - u_y v_x)^2} = \sqrt{u_x v_x u_x v_x + u_x v_x u_y v_y + u_y v_y u_x v_x + u_y v_y u_y v_y + u_x v_y u_x v_y - u_x v_y u_y v_x - u_y v_x u_x v_y + u_y v_x u_y v_x} = \sqrt{u_x^2 v_x^2 + u_x^2 v_y^2 + u_y^2 v_x^2 + u_y^2 v_y^2} = \sqrt{(u_x^2 + u_y^2)(v_x^2 + v_y^2)} = \sqrt{u_x^2 + u_y^2} \sqrt{v_x^2 + v_y^2} \]
\[ || \vec{u} || = \sqrt{u_x^2 + u_y^2} \]
\[ || \vec{v} || = \sqrt{v_x^2 + v_y^2} \]
\[ || \vec{u} \vec{v} || = || \vec{u} || \text{ } || \vec{v} || \]
\[ \text{Next: the part that I have no idea how to prove!} \]
\[ \vec{u} \vec{v} = || \vec{u} || \text{ } || \vec{v} || e^{i \theta} \]
\[ \theta = \text{ The angle counterclockwise from } \vec{u} \text{ towards } \vec{v}. \]
\[ \text{By the way, with this can derive two new equations! (But you probably know the first.)} \]
\[ \vec{u} \cdot \vec{v} = || \vec{u} || \text{ } || \vec{v} || cos(\theta) \]
\[ \vec{u} ∧ \vec{v} = || \vec{u} || \text{ } || \vec{v} ||sin(\theta) i \]

But what I wanted was that \(\theta!\) But I know that it is (almost) impossible to find \(\theta\) when all you know is \(e^{i \theta}\). Wait a minute! If a complex number of the form \(e^{i \theta}\) is just a rotation, and if multiplying a vector by a complex number acts like they were both complex numbers (exempt it returns a vector), than you can rotate a vecor \(\vec{w}\) by the angle between two unit vectors (or just ones with inverse magnitude, but normalizing them is easier) without having to even know about complex numbers!

\[ \hat{u} = \frac{\vec{u}}{|| \vec{u} ||} \]
\[ \hat{v} = \frac{\vec{v}}{|| \vec{v} ||} \]
\[ \hat{u} \hat{v} = e^{i \theta} \]
\[ \text{So, the rotation is} \]
\[ \vec{w} \hat{u} \hat{v}. \]

(Mic drop.) Also, that was on \(400\) lines exactly.

\[ \text{One more thing before maxwell's equation and rotors.} \]
\[ (c + di)(a \hat{x} + b \hat{y}) = ca \hat{x} + cb \hat{y} + dia \hat{x} + dib \hat{y} = \begin{pmatrix} ca (\hat{x}) & cb (\hat{y}) \\ da (i \hat{x}) & db (i \hat{y}) \\ \end{pmatrix} = \begin{pmatrix} ca (\hat{x}) & cb (\hat{y}) \\ da (\hat{x} \hat{y} \hat{x}) & db (\hat{x} \hat{y} \hat{y}) \\ \end{pmatrix} = \begin{pmatrix} ca (\hat{x}) & cb (\hat{y}) \\ -da (\hat{y} \hat{x} \hat{x}) & db (\hat{x}) \\ \end{pmatrix} = \begin{pmatrix} ca (\hat{x}) & cb (\hat{y}) \\ -da (\hat{y}) & db (\hat{x}) \\ \end{pmatrix} = (ca + db) \hat{x} + (cb - da) \hat{y} = (a \hat{x} + b \hat{y})(c - di) \]
\[ z \vec{v} = \vec{v} \text{ ccong} (z) \]
\[ \text{But if you remember, } \vec{v} \vec{u} = \text{ccong} (\vec{u} \vec{v}) \text{ and } \vec{u} \vec{v} \text{ is a complex number, so...} \]
\[ \vec{u} \vec{v} \vec{w} = \vec{w} \text{ ccong} (\vec{u} \vec{v}) = \vec{w} \vec{v} \vec{u} \]
\[ \text{One more thing.} \]
\[ \theta (\vec{u}, \vec{v}) = \text{ The angle counterclockwise from } \vec{u} \text{ towards } \vec{v}. \]
\[ \vec{w} \vec{u} \vec{v} = \vec{w} \text{ Scaled by the length of } \vec{u} \text{ and } \vec{v} \text{, and rotated by } \theta (\vec{u}, \vec{v}). \]

## 20.1.3 Rotors

A rotor is a way to rotate a vector by any angle \(\theta\) in any plane (Yes, plane. I think it makes the same if not more sense to rotate in a plane. I also think that this is simalar to the ∧ v.s. \(\times\) Product.) in any dimention. Let's start with the simplest possible rotor I can think of, in \(3d\), \(90\)°, \(\hat{x}\) - \(\hat{y}\) plane. (I'll call \(\vec{v}\) rotated by the name of \(\vec{v} \prime\))

\[ \vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z} \]
\[ \vec{v} \prime = v_x \hat{y} - v_y \hat{x} + v_z \hat{z} \]
\[ \vec{v} \prime = \vec{v} \hat{x} \hat{y} ? \]
\[ \vec{v} \hat{x} \hat{y} = (v_x \hat{x} + v_y \hat{y} + v_z \hat{z}) \hat{x} \hat{y} = v_x \hat{x} \hat{x} \hat{y} + v_y \hat{y} \hat{x} \hat{y} + v_z \hat{z} \hat{x} \hat{y} = v_x \hat{y} - v_y \hat{x} \hat{y} \hat{y} - v_z \hat{x} \hat{z} \hat{y} = (v_x) \hat{y} - (v_y) \hat{x} + (v_z) \hat{x} \hat{y} \hat{z} \]
\[ \text{What was that thing he said at this point? Oh right, it was: Well, it almost worked, the } x \text{ and the } y \text{ coordinates got rotated correctly, but the } z \text{ coordinate got turned into this trivector, let's try something else.} \]
\[ \text{Do you remember that time when I proved that } z \vec{v} = \vec{v} \text{ ccong} (z) \text{, that means that I can now prove this: } \vec{v} z = \text{ccong} (z) \vec{v} \text{. I'll assume that that means } \hat{y} \hat{x} \text{, I'll try that!} \]
\[ \hat{y} \hat{x} \vec{v} = \hat{y} \hat{x} (v_x \hat{x} + v_y \hat{y} + v_z \hat{z}) = \hat{y} \hat{x} v_x \hat{x} + \hat{y} \hat{x} v_y \hat{y} + \hat{y} \hat{x} v_z \hat{z} = v_x \hat{y} \hat{x} \hat{x} + v_y \hat{y} \hat{x} \hat{y} + v_z \hat{y} \hat{x} \hat{z} = v_x \hat{y} - v_y \hat{x} \hat{y} \hat{y} - v_z xyz = (v_x) y - (v_y) x - (v_z) \hat{x} \hat{y} \hat{z} \]
\[ \text{Once again, the } x \text{ and the } y \text{ coordinates got rotated correctly, but the } z \text{ coordinate got turned into the inverse trivector this time, maybe doing both at once would cancel out?} \]
\[ \hat{y} \hat{x} \vec{v} \hat{x} \hat{y} = \hat{y} \hat{x} (v_x \hat{x} + v_y \hat{y} + v_z \hat{z}) \hat{x} \hat{y} = \hat{y} \hat{x} v_x \hat{x} \hat{x} \hat{y} + \hat{y} \hat{x} v_y \hat{y} \hat{x} \hat{y} + \hat{y} \hat{x} v_z \hat{z} \hat{x} \hat{y} = v_x \hat{y} \hat{x} \hat{y} + v_y \hat{y} \hat{x} \hat{y} \hat{x} \hat{y} + \hat{y} \hat{x} v_z \hat{z} \hat{x} \hat{y} = - v_x \hat{x} \hat{y} \hat{y} - v_y \hat{y} \hat{y} \hat{x} \hat{x} \hat{y} + \hat{y} \hat{x} v_z \hat{z} \hat{x} \hat{y} = - v_x \hat{x} - v_y \hat{y} + \hat{y} \hat{x} v_z \hat{z} \hat{x} \hat{y} \]
\[ \hat{y} \hat{x} v_z \hat{z} \hat{x} \hat{y} = v_z \hat{y} \hat{x} \hat{z} \hat{x} \hat{y} = -v_z \hat{y} \hat{x} \hat{x} \hat{z} \hat{y} = -v_z \hat{y} \hat{z} \hat{y} = v_z \hat{y} \hat{y} \hat{z} \]
\[ \hat{y} \hat{x} v_z \hat{z} \hat{x} \hat{y} = v_z \hat{z} \]
\[ \hat{y} \hat{x} \vec{v} \hat{x} \hat{y} = - v_x \hat{x} - v_y \hat{y} + v_z \hat{z} \]
\[ \text{ The } z \text{ coordinate is correct! Success! But the } x \text{ and } y \text{ got messed up, of coarse they did, we rotated twice. (Once for } \hat{y} \hat{x} \text{ on the left, and once for } \hat{x} \hat{y} \text{ on the right.) let's add back in that angle } \theta \text{ to prove that we did rotate twice.} \]

Do you remember euler's identity, \(e^{i \theta} = cos(\theta) + isin(\theta)\) right? Well, this actually for any \(i\) whose square is \(-1\), the thing is that \((\hat{x} \hat{y})^2 = \hat{x} \hat{y} \hat{x} \hat{y} = -\hat{x} \hat{x} \hat{y} \hat{y} = -1\), so

\[ e^{\hat{x} \hat{y} \theta} = cos(\theta) + \hat{x} \hat{y} \text{ } sin(\theta). \]
\[ \text{And just to make sure, let's test for } \theta = 90° \]
\[ cos(90°) + \hat{x} \hat{y} \text{ } sin(90°) = 0 + \hat{x} \hat{y} \text{ } 1 = \hat{x} \hat{y}. \]
\[ \text{Also, } cos(\theta) = c \text{ and } sin(\theta) = s \]
\[ \text{So, replace } \hat{x} \hat{y} \text{ and } \hat{y} \hat{x} \text{ with } c + \hat{x} \hat{y} \text{ } s \text{ and } cos(-\theta) + \hat{x} \hat{y} \text{ } sin(-\theta) \text{ respectively.} \]
\[ \text{(The second one can simplify to } c - \hat{x} \hat{y} \text{ } s \text{).} \]
\[ \vec{v} \prime = (v_x c - v_y s) \hat{x} + (v_y c + v_x s) \hat{y} + v_z \hat{z} \]
\[ (c - \hat{x} \hat{y} \text{ } s) \text{ } \vec{v} \text{ } (c + \hat{x} \hat{y} \text{ } s) = (c - \hat{x} \hat{y} \text{ } s) (v_x \hat{x} + v_y \hat{y} + v_z \hat{z}) (c + \hat{x} \hat{y} \text{ } s) \]
\[ (c - \hat{x} \hat{y} \text{ } s) \text{ } \vec{v} \text{ } (c + \hat{x} \hat{y} \text{ } s) = c v_x \hat{x} \text{ } c + c v_x \hat{x} \hat{x} \hat{y} \text{ } s - \hat{x} \hat{y} \text{ } s v_x \hat{x} \text{ } c - \hat{x} \hat{y} \text{ } s v_x \hat{x} \hat{x} \hat{y} \text{ } s + c v_y \hat{y} \text{ } c + c v_y \hat{y} \hat{x} \hat{y} \text{ } s - \hat{x} \hat{y} \text{ } s v_y \hat{y} \text{ } c - \hat{x} \hat{y} \text{ } s v_y \hat{y} \hat{x} \hat{y} \text{ } s + c v_z \hat{z} \text{ } c + c v_z \hat{z} \hat{x} \hat{y} \text{ } s - \hat{x} \hat{y} \text{ } s v_z \hat{z} \text{ } c - \hat{x} \hat{y} \text{ } s v_z \hat{z} \hat{x} \hat{y} \text{ } s = (v_x cc) \hat{x} + (v_x cs) \hat{x} \hat{x} \hat{y} - (v_x sc) \hat{x} \hat{y} \hat{x} - (v_x ss) \hat{x} \hat{y} \hat{x} \hat{x} \hat{y} + (v_y cc) \hat{y} + (v_y cs) \hat{y} \hat{x} \hat{y} - (v_y sc) \hat{x} \hat{y} \hat{y} - (v_y ss) \hat{x} \hat{y} \hat{y} \hat{x} \hat{y} + (v_z cc) \hat{z} + (v_z cs) \hat{z} \hat{x} \hat{y} - (v_z sc) \hat{x} \hat{y} \hat{z} - (v_z sin(\theta) sin(\theta)) \hat{x} \hat{y} \hat{z} \hat{x} \hat{y} \]
\[ \text{I had to split it in } 2 \text{, it reached the limit of LaTeX symbols per line.} \]
\[ \hat{x} \hat{x} \hat{y} = \hat{y} \]
\[ \hat{x} \hat{y} \hat{x} = -\hat{y} \hat{x} \hat{x} = -\hat{y} \]
\[ \hat{x} \hat{y} \hat{x} \hat{x} \hat{y} = \hat{x} \hat{y} \hat{y} = \hat{x} \]
\[ \hat{y} \hat{x} \hat{y} = -\hat{y} \hat{y} \hat{x} = -\hat{x} \]
\[ \hat{x} \hat{y} \hat{y} = \hat{x} \]
\[ \hat{x} \hat{y} \hat{y} \hat{x} \hat{y} = \hat{x} \hat{x} \hat{y} = \hat{y} \]

$$ \hat{z} \hat{x} \hat{y} = -\hat{x} \hat{z} \hat{y} = \hat{x} \hat{y} \hat{z} $$
$$ \hat{x} \hat{y} \hat{z} \hat{x} \hat{y} = -\hat{x} \hat{y} \hat{x} \hat{z} \hat{y} = \hat{x} \hat{x} \hat{y} \hat{z} \hat{y} = \hat{y} \hat{z} \hat{y} = -\hat{z} \hat{y} \hat{y} = -\hat{z} $$
$$ (c - \hat{x} \hat{y} \text{ } s) \text{ } \vec{v} \text{ } (c + \hat{x} \hat{y} \text{ } s) = (v_x cc) \hat{x} + (v_x cs) \hat{y} + (v_x sc) \hat{y} - (v_x ss) \hat{x} + (v_y cc) \hat{y} - (v_y cs) \hat{x} - (v_y sc) \hat{x} - (v_y ss) \hat{y} + (v_z cc) \hat{z} + (v_z cs) \hat{x} \hat{y} \hat{z} - (v_z sc) \hat{x} \hat{y} \hat{z} + (v_z ss) \hat{z} $$
$$ (v_x cc) \hat{x} + (v_x cs) \hat{y} + (v_x sc) \hat{y} - (v_x ss) \hat{x} = (v_x cc - v_x ss) \hat{x} + (v_x cs + v_x sc) \hat{y} = (v_x(cos(\theta) cos(\theta) - sin(\theta) sin(\theta))) \hat{x} + (v_x(cos(\theta) sin(\theta) + sin(\theta) cos(\theta))) \hat{y} $$

And then, with some trigonometry identities...

$$ cos(\theta + \theta) = cos(\theta) cos(\theta) - sin(\theta) sin(\theta) $$

$500$ Lines!

$$ sin(\theta + \theta) = cos(\theta) sin(\theta) + sin(\theta) cos(\theta) $$
$$ (v_x cc) \hat{x} + (v_x cs) \hat{y} + (v_x sc) \hat{y} - (v_x ss) \hat{x} = (v_x cos(2 \theta)) \hat{x} + (v_x sin(2 \theta)) \hat{y} $$
$$ (v_y cc) \hat{y} - (v_y cs) \hat{x} - (v_y sc) \hat{x} - (v_y ss) \hat{y} = (v_y cc - v_y ss) \hat{y} - (v_y cs + v_y sc) \hat{x} = (v_y(cos(\theta) cos(\theta) - sin(\theta) sin(\theta))) \hat{y} - (v_y(cos(\theta) sin(\theta) + sin(\theta) cos(\theta))) \hat{x} $$
$$ (v_y cc) \hat{y} - (v_y cs) \hat{x} - (v_y sc) \hat{x} - (v_y ss) \hat{y} = (v_y cos(2 \theta)) \hat{y} - (v_y sin(2 \theta)) \hat{x} $$
$$ (v_x cc) \hat{x} + (v_x cs) \hat{y} + (v_x sc) \hat{y} - (v_x ss) \hat{x} + (v_y cc) \hat{y} - (v_y cs) \hat{x} - (v_y sc) \hat{x} - (v_y ss) \hat{y} = (v_x cos(2 \theta)) \hat{x} + (v_x sin(2 \theta)) \hat{y} + (v_y cos(2 \theta)) \hat{y} - (v_y sin(2 \theta)) \hat{x} = (v_x cos(2 \theta) - v_y sin(2 \theta)) \hat{x} + (v_y cos(2 \theta) + v_x sin(2 \theta)) \hat{y} $$
$$ (v_z cc) \hat{z} + (v_z cs) \hat{x} \hat{y} \hat{z} - (v_z sc) \hat{x} \hat{y} \hat{z} + (v_z ss) \hat{z} = (v_z cc) \hat{z} + (v_z ss) \hat{z} = (v_z(cc + ss)) \hat{z} $$

And then, with some more trigonometry identities...

$$ (v_z cc) \hat{z} + (v_z cs) \hat{x} \hat{y} \hat{z} - (v_z sc) \hat{x} \hat{y} \hat{z} + (v_z ss) \hat{z} = (v_z) \hat{z} $$
$$ \text{Also, } cos(2 \theta) = C \text{ and } sin(2 \theta) = S $$
$$ \vec{v} \prime \prime = (v_x C - v_y S) \hat{x} + (v_y C + v_x S) \hat{y} + v_z \hat{z} $$
$$ e^{-\hat{x} \hat{y} \theta} \vec{v} e^{\hat{x} \hat{y} \theta} = \vec{v} \prime \prime $$
$$ \text{Yessssssss

# 21. Set theory

## 21.1 What even is a set?

A set is a well defined collection of objects, a set could contain the two shoes on you feet, or the \(5\) peices of chese on this cutting board (that I'm going to pretend exists), sets can even contain other sets, but sets can not contain themselfes, because tis would lead to a paradox: would the set that contains every set that dosen't contain itsef contain itself?" this also means that there isn't a set that contains everything.

But the thing is, using some symbols, you can describe almost all of math. These symbols can just be pronounced as words, and it would make a sentence, such as " \(¬ \exists (x): |x| < 0\) " as " there does not exist \(x\) such that the absolute value of \(x\) is strictly less than \(0\) ". Time to rapidfire through each one's pronunciation and meaning.

### 21.1.1 meanings of things

\[ ∀ \text{ Is pronounced "for any" or "for all" (but I prefer "for any") and means what it says. It than has an open parentheses, a thing (or sometimes multiple things separated by a comma) (} x, y, z, \text{ or a set) that I will call } x \text{ for now, a closed parentheses (parenthese is not a word), a } \cdot \text{, a statement that implies something about } x \text{, a colon, and finish it off with a statement including } x. \]
\[ ( \text{ and } ) \text{ are not pronounced.} \]
\[ \cdot \text{ Is pronounced "such that" and it's only used in two contexts: "for any } x \text{ such that..." and "there exists } x \text{ such that...".} \]
\[ : \text{ Is pronounced however a colon is pronounced.} \]
\[ , \text{ Is pronounced however a comma is pronounced.} \]
\[ \exists \text{ Is pronounced "there exists" and I don't think I need to explain that.} \]
\[ ¬ \text{ Is pronounced "is not" or "does not" as in "there does not exist } x \text{".} \]
\[ \in \text{ Is pronounced "is an element of" where an element of a set is a singular object that is contained in that set.} \]
\[ Ø \text{ Is pronounced "the empty set" and means "the set of which is empty inside".} \]
\[ x, y, \text{ And } z \text{ are pronounced "} x, y, \text{ And } z \text{" and they all mean "a thing that could be an element of a set".} \]
\[ ⊆ \text{ Is pronounced "is a subset of" and I'll get to the meaning of that in the next chapter.} \]
\[ \text{capital letters are sets.} \]
\[ \iff \text{ Is pronounced "if and only if" as in "if statement } a \text{ is true, statement } b \text{ is true, and if statement } a \text{ is false, statement } b \text{ is false".} \]
\[ pow \text{ Is pronounced "the power set of" as in "} pow(S) \text{" and I'll get to the meaning in the next chapter.} \]
\[ ∩ \text{ Is pronounced "and" and means "} a ∩ b \text{ is true if and only if statement } a \text{ is true and } b \text{ is true", it can also mean the intersection of two sets, in that case, it is pronounced "intersectioned with", but I'll get to it's formal meaning in the next next next next chapter..} \]
\[ = : \text{ Is pronounced "equals by definition" and means "define the thing on the left as the thing on the right", or was it the other wat around?} \]
\[ = \text{ Is pronounced "is the same as" and I'll get to it's formal meaning in the next chapter.} \]
\[ \in^S \text{ Is pronounced "is a super element of" (} S \text{ for super) and I'll get to it's meaning in the next chapter.} \]
\[ ∨ \text{ Is pronounced "or" and means "} a ∨ b \text{ is true if statement } a \text{ is true or } b \text{ is true... Or both!", it can also mean the union of two sets, in that case, it is pronounced "unioned with", but I'll get to it's formal meaning in the next chapter.} \]
\[ \text{succ Is pronounced "the immediate successor of" and means "that number } + 1 \text{".} \]
\[ set \text{ Is pronounced "the set containing" as in "} set(S) \text{" and I'll get to it's formal meaning in the next chapter.} \]
\[ → \text{ Is technically called the if then sign, but it is pronounced "implies" and means "statement } a → b \text{ is true if statement } a \text{ being true implies statement } b \text{ is true", so } a → b \text{ is true if statement } a \text{ is true and statement } b \text{ is true, or if statement } a \text{ is false and statement } b \text{ is false, or if statement } a \text{ is true and statement } b \text{ is false, but not if statement } a \text{ is false and statement } b \text{ is true. Also, if there was an element sign two spaces behind, pronounce it "being an element of" as oppose to "is an element of".} \]
\[ \text{I don't think that I need to explain the } < \text{sign.} \]

### 21.1.2 definitions

\[ ¬ \exists (x) \cdot x \in Ø \]
\[ A ⊆ B \iff ∀(x) \cdot x \in A: x \in B \]
\[ ∀(P) \cdot ∀(U) \cdot U ⊆ S: U \in P ∩ ∀(T) \cdot T ¬ ⊆ S: T ¬ \in P: P = : pow(S) \]
\[ A = B \iff A ⊆ B ∩ B ⊆ A \]
\[ ¬ \exists (S) \cdot S \in S \]
\[ x \in \in S \iff \exists (U) \cdot U \in S ∩ x \in U \]
\[ x \in \in S \text{ can also be written as } \in^2 \]
\[ x \in \in \in S \iff \exists (U) \cdot U \in S ∩ x \in^2 U \]
\[ x \in \in \in S \text{ can also be written as } \in^3 \]
\[ x \in \in \in \in S \iff \exists (U) \cdot U \in S ∩ x \in^3 U \]
\[ x \in \in \in \in S \text{ can also be written as } \in^4 \]
\[ \vdots \]
\[ x \in^{a + b} S \iff \exists (U) \cdot U \in^a S ∩ x \in^b U \]
\[ x \in^{a + b} S \text{ can also be written as } x \in^a \in^b S \]
\[ x \in^S S \iff x \in S ∨ \exists (U) \cdot U \in S ∩ x \in^S U \]

.

Was recursion in the rule book? I guess so.

Recursion? Get it?

### 21.1.3 numbers

\[ 0 = Ø \]
\[ \text{succ} (n) \text{ (Which mathematicly equals } n + 1 \text{) Is how you would usually define numbers, so I'll define numbers that way, I'll say that succ} (n) \text{ is the set that contains all numbers } 0 \text{-} n \text{. But first: the union of two sets, denoted as an } ∨ \text{ sign.} \]
\[ x \in A ∨ B \iff x \in A ∨ x \in B \]
\[ ∀(S) \cdot E \in S ∩ ∀(T) \cdot T ¬= E: T ¬ \in S: S = : set(E) \]
\[ \text{Around } 100 \text{ lines?? (I might add another definition, but at the time of typing this, this is on } 95 \text{ lines.)} \]
\[ \text{succ} (n) = : set(n) ∨ n \]

### 21.1.4 back to definitions

\[ x \in A ∩ B \iff x \in A ∩ x \in B \]
\[ \text{Here's another definition of the subset: } ∀(A, B) \cdot ¬ \exists (x) \cdot x \in A ¬ → x \in B: A ⊆ B. \]
\[ \text{And another one! } A ⊆ B \iff ¬ \exists (x) \cdot x \in A ∩ x ¬ \in B \]
\[ n_1 < n_2 \iff n_1 \in n_2 \]
\[ \text{WARNING! the next statement is the axiom of choice, kinda controversial.} \]
\[ ∀(S) \cdot S ¬= Ø: \exists (x) \cdot x \in S \]

### 21.1.5 group theory

A group (call it \(G\)) is a certain type of set, including an addition like thing represented with a \(+\) sign (this addition like thing could also be multiplication), let's start with the set of numbers \(0\) - \(4\) under modular addition. To be a group, it has to follow \(4\) different rules.

\[ 1 \text{, Closure} \]
\[ ∀(a, b) \cdot a \in G ∩ b \in G: a + b \in G \]

Because it is modular, this holds true for modular addition.

\[ 2 \text{, Associativity} \]
\[ ∀(a, b, c) \cdot a \in G ∩ b \in G ∩ c \in G: (a + b) + c = a + (b + c) \]

Because addition is associative, this holds true for modular addition.

$$ 3 \text{, Identity} $$
$$ \exists (e) \cdot e \in G \cap \forall(a) \cdot a \in G: a + e = a \cap \forall(b) \cdot b \in G: e + b = b $$

Because of $0$, this holds true for modular addition.

$$ 4 \text{, Inverses} $$
$$ \forall(a) \cdot a \in G: \exists (b) \cdot b \in G \cap a + b = e \cap b + a = e $$

Because of negataves and them looping back arround, this holds true for modular addition. (Also, modular multiplication almost works, but it fails at this step because there is no $\frac{1}{0}$.)

Thus, the set of numbers $0$ - $4$ under modular addition is a group.

## 21.1.6 set theory proofs?

No.

OOOO

# 23. Code repo

This page is not on the website (yet), because it is entirely pseudo-code and python code.

### 23.0.1 asterisk operator (pseudo)

```
define Asterisk(x,y,n)
  (coment) if n is 1, it is adding x and y
  (coment) if n is 2, it is multiplying x and y
  (coment) if n is 3, it is it is taking x to the power of y
  (coment) and so on.

  if n = 1
    return x + y

  (coment) because I said that earlier.

  else if y = 1
    return x

  (coment) x times 1 = x, x to the power of 1 = x, ect

  else
    return Asterisk(x,Asterisk(x,y - 1,n),n - 1)

  (coment) this makes sense if you think about it.
```

### 23.0.2 asterisk operator (python)

```
def Asterisk(a, b, n):
  if n == 1:
    return a + b
  elif n == 2:
    return a*b
  elif b == 1:
    return a
  else:
    return Asterisk(a,Asterisk(a,b - 1,n),n - 1)
```

### 23.0.3 alternitave asterisk operator (python)

```
def (a, b, n):
  if n == 1:
    return a + b
  if n == 2:
    return a*b
  k = a
  for i in range(b - 1):
    k = Asterisk(a, k, n - 1)
```

### 23.0.4 the exact digits of square roots (pseudo)

404 Page not found.

### 23.0.5 the exact digits of square roots (python) (I have been working on this since this page was made)

```
def Root(x, y, n):
  f = 0
  while f**x <= y:
    f += 1
  f -= 1
  a = f
  l = 0
  for i in range(n):
    a *= 10
    l += 1
    while a**x <= y*(10**(l*x)):
      a += 1
    a -= 1
  a -= f*(10**l)
  if n > 1:
    return str(f) + '.' + str(a)
```

```
  else:
    return str(f)
```

## 23.0.6 the exact digits of logarithms (pseudo)

```
404 Page not found.
```

## 23.0.7 the exact digits of logarithms (python)

```
def Log(x, y, n):
  f = 0
  while x**f <= y:
    f += 1
  f -= 1
  a = f
  l = 0
  for i in range(n):
    a *= 10
    l += 1
    while x**a <= y**(10**(l)):
      a += 1
    a -= 1
  a -= f*(10**l)
  if n > 1:
    return str(f) + '.' + str(a)
  else:
    return str(f)
```

**as a cherry on top, the exact digits of the roots of polynomials (pseudo)**

```
def Roots(p(x), StartValue, n):
  f = StartValue
  while p(f) <= y:
    f += 1
  f -= 1
  a = f
  l = 0
  for i in range(n):
    a *= 10
    l += 1
    while p(a*(10**(-l) <= p(y):
(coment) you will have to expand and simplify the equation so that it is a differance of integers
      a += 1
    a -= 1
  a -= f*(10**l)
  if n > 1:
    return str(f) + '.' + str(a)
  else:
    return str(f)
```

## 23.0.8 set theory numbers

```
def SetTheoryNumbers(n):
  string = "{"
  k = n - 1
  while k >= 0:
    string += SetTheoryNumbers(k)
    k -= 1
  string += "}"
  return string
```

## 23.0.9 alternitave set theory numbers

```
def AlmostSetTheoryNumbers(n):
  if n == 0:
    return ""
  return SetTheoryNumbers(n - 1) + AlmostSetTheoryNumbers(n - 1)
```

```
def SetTheoryNumbers(n):
  return "{" + AlmostSetTheoryNumbers(n) + "}"
```

## 23.0.10 base names

```
def SmallestPrimeDivisor(n):
  k = 2
  while n % k != 0:
    k += 1
  return k
```

```
def AlmostBase(n):
  if n == 2:
    return "bi"
  if SmallestPrimeDivisor(n) == n:
    return "un" + AlmostBase(n - 1) + "sna"
  string = ""
  k = n
  while k != 1:
    spdk = SmallestPrimeDivisor(k)
    string += AlmostBase(spdk)
    k /= spdk
  return string
```

```
def base(n):
  if n < 2:
    return "n must be a positave integer!"
  return AlmostBase(n) + "nary"
```

## 23.0.11 base names (first \(n\) values)

```
def SmallestPrimeDivisor(n):
  k = 2
  while n % k != 0:
    k += 1
  return k
```

```
for n in range(101):
  L = [None, None]
  if n == 2:
    L += ["bi"]
    print("binary")
  if SmallestPrimeDivisor(n) == n:
    L.append("un" + L[n - 1] + "sna")
    print("un" + AlmostBase(n - 1) + "snanary")
  string = ""
  k = n
  while k != 1:
    spdk = SmallestPrimeDivisor(k)
    string += L[spdk]
    k /= spdk
  L.append(string)
  print(string + "nary")
```

## 23.0.12 the exact digits of reciprocals (in any base) (using my home made method)

```
def frac(n, b):
  if n == 1:
    return "this code can not compute one over one!"
  if b == 1:
    return "the base must not equal one!"
  if n < b:
    nb = n
  else:
    nb = b
  for K in range(nb - 1):
    k = K + 2
    if n % k == 0:
      if b % k == 0:
        return "this code will not work if you give it two numbers that are coprime"
  rem = b % n
  string = "0. repeating " + str(b // n)
  for i in range(n - 2):
    rem *= b
    string += " " + str(rem // n)
    rem %= n
    if rem == 1:
      return string
```

also, this is the \(400\)'th commit to this branch.

## 23.0.13 virus (DO NOT RUN THIS CODE)

```
def crash(exponent):
  result = "1"
```

```
  for i in range(exponent):
    result += result
  return result
```

```
print(crash(crash(crash(crash(crash(crash(crash(crash(crash(crash(10000))))))))))
```

## 23.0.14 virus #2 (it goes without saying at this point)

```
result = "1"
while true:
  result += result
  print result
```

## 23.0.15 group theory (I've kinda been doing a lot of group theory today and yesterday)

```
def GroupTheory(TheNumberOfFaces, TransformationNumberOne, TransformationNumberOnesIndex, TransformationNumberTwo, TransformationNumberTwosIndex):
  if (TheNumberOfFaces < 2) or (TransformationNumberOne != "f" and TransformationNumberOne != "r") or (TransformationNumberOnesIndex < 0 or
TransformationNumberOnesIndex >= TheNumberOfFaces) or (TransformationNumberTwo != "f" and TransformationNumberTwo != "r") or (TransformationNumberTwosIndex <
0 or TransformationNumberTwosIndex >= TheNumberOfFaces):
    return "that isn't group theory!"
  if TransformationNumberOne == "f":
    if TransformationNumberTwo == "f":
      return "r" + str((TransformationNumberTwosIndex - TransformationNumberOnesIndex) % TheNumberOfFaces)
    return "f" + str((TransformationNumberOnesIndex - TransformationNumberTwosIndex) % TheNumberOfFaces)
  if TransformationNumberTwo == "f":
    return "f" + str((TransformationNumberTwosIndex - TransformationNumberOnesIndex) % TheNumberOfFaces)
  return "r" + str((TransformationNumberOnesIndex + TransformationNumberTwosIndex) % TheNumberOfFaces)
```

## 23.0.16 binary names

```
404 Page not found.
```

## 23.0.17 alternitave numbers

```
def SmallestPrimeDivisor(n):
  k = 2
  while n % k != 0:
    k += 1
  return k
```

```
def AlmostNumber(n, threshold):
  if n <= threshold and n > 0:
    isPrime = "false"
    return str(n)
  if SmallestPrimeDivisor(n) == n:
    isPrime = "true"
    return "((" + AlmostNumber(n - 1) + ")+1)"
  else:
    isPrime = "false"
  string = ""
  k = n
  while k != 1:
    spdk = SmallestPrimeDivisor(k)
    string += AlmostBase(spdk)
    k /= spdk
  return string
```

```
def number(n, threshold):
  if isPrime == "false":
    return AlmostNumber(n, threshold)
  result = ""
  for i in range(1, len(AlmostNumber(n, threshold)) - 1):
    result += AlmostNumber(n, threshold)[i]
  return result
```

## 23.0.18 magic sequences

### magic sequences

```
def MagiSeq(n):
  MagicSequence = [1]
  MagicNext = 1
  for i in range(n - 1):
    MagicNext *= 2
```

```
    MagicNext %= n
    MagicSequence += [MagicNext]
    if MagicNext in MagicSequence[:-1]:
      return MagicSequence
```

## 23.0.19 The Apocalyptic Numbers

### The Apocalyptic Numbers

```
for i in range(1000):
  power = str(2 ** i)
  for j in range(len(power)):
    if power[j] == "6":
      if len(power) >= j + 2:
        if power[j + 1] == "6":
          if power[j + 2] == "6":
            if power[j + 3] != "6":
              result = ""
              for k in range(j):
                result += power[k]
              result += "  666  "
              for k in range(j + 3, len(power)):
                result += power[k]
              if i != 157:
                print()
              print(f"{i}, 2^{i} = {result}")
```

## 23.0.20 nsuemtb etrh etohreyo#r#y

```
def SmallestPrimeDivisor(n):
  k = 2
  while n % k != 0:
    k += 1
  return k
```

```
N = 100
```

```
for n in range(N):
  L = ["", "{}"]
  if SmallestPrimeDivisor(n) == n:
    L.append("{" + L[n - 1] + "}")
    print("{" + L[n - 1] + "}")
  string = "{"
  k = n
  while k != 1:
    spdk = SmallestPrimeDivisor(k)
    string += L[spdk]
    k /= spdk
  L.append(string + "}")
  print(string + "}")
```

# 24. The best way to count

First, the hindu-arabic numeral system: it is a way to write numbers. So, for example, six would be wrtten as " $6$ ". The arabic numerals are just: $0, 1, 2, 3, 4, 5, 6, 7, 8,$ and $9$. As opposed to the hindu-arabic numeral system which uses mostly decimal. For example, five hundred and sixty-seven in decimal is "five six seven", and then, if you use the arabic numerals to go from decimal to the hindu-arabic numeral system, you get " $567$ ". WARNING! (Kinda.) $567$ is NOT a number, it is just a popular way to write the corrasponding number.

The hindu-arabic numerals are a positional numeral system, which means that the value of each digit is affected by the position. Positional numeral system must've been a pretty revolutionary idea, at least, in comaprison to non-positional numeral systems, such as the roman numeral system. Positional numeral systems probably got into the top $10$ most revolutionay mathematical ideas, 'cause you can write all infinately many numbers using just $10$ digits ($10$ arabic numerals), without getting out of hand quickly (just see $100,000$ vs $MMMM

# 25. Geometric algebra 2

$$ \text{Where we last left off, we were at Maxwell's equations.} $$
$$ \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} $$
$$ \vec{\nabla} \times \vec{B} = \mu_0 (\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) $$
$$ \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} $$
$$ \vec{\nabla} \cdot \vec{B} = 0 $$
$$ \text{That's it!} $$

#### Credits

$$ \text{A.P. (For privacy reasons, I won't say his full name.) He helped sometimes.} $$
$$ \text{Sugey (that's just his username) who made a great introduction to geometric algebra.} $$
$$ \text{Silaspe (me) did evereything else.} $$

#### JK

It would be easier to go from Maxwell's equation (singular) to Maxwell's equations (plural) than to go the other way around. Here it is... After I combine some stuff. (By the way, $i = \hat{x} \hat{y} \hat{z}$ because it is $3d$.)

$$ \text{First, the differentials. There are two, to combine into the spacetime gradient, the partial derivitave with respect to time (} \frac{\partial}{\partial t} \text{), and the gradient with respect to space (} \vec{\nabla} \text{), this spacetime gradient will be called } \nabla \text{. (As opposed to } \vec{\nabla} \text{.)} $$
$$ \nabla = \frac{\partial}{\partial t} + \vec{\nabla} $$
$$ \text{Actually, no. } $$
$$ \nabla = \frac{1}{c} \frac{\partial}{\partial t} + \vec{\nabla} $$
$$ \text{Doing this may seem familiar if you've worked with relativity enough.} $$
$$ \text{Next: combine the sources that create the electric and magnetic fields (to create the source that creates the electromagnetic field), there are two (again), the charge density (} \rho \text{), and the current (} \vec{J} \text{). (by the way, this source will be caled } J \text{, as opposed to } \vec{J} \text{)To combine these, just add them!} $$
$$ J = \rho + \vec{J} $$
$$ \text{Actually, subtract them.} $$
$$ J = \rho - \vec{J} $$
$$ \text{Don't worry about the minus sign.} $$
$$ \text{Actually, wrong again, but this can be fixd by adding a factor of } c \text{ to the charge density.} $$
$$ J = c \rho - \vec{J} $$
$$ \text{Again, this is seen a lot in realitivity, so it's nothing new in physics.} $$
$$ \text{It also gives an interesting new interpretation of charge density as a current that is moving through time and not space.} $$
$$ \text{Finaly, we need to combine the electric and magnetic fields(} \vec{E} \text{ and } \vec{B} \text{) into one electromagnetic field (} F \text{). But unlike before, we can NOT just add them, the issue this time is that they are both vectors, so they're components will mix. But now's the time to kill two birds with one stone, you see, the magnetic feild is traditionally defined with a cross product and NOT a wedge product, but instead of re-defining the magnetic feild, remember how the wedge product is just } i \text{ times the cross product? Using that instead, we get this:} $$
$$ F = \vec{E} + i \vec{B} $$
$$ \text{Actually, wrong again, but this can be fixd by adding a factor of } c \text{ to the magnetic feild.} $$
$$ F = \vec{E} + ic \vec{B} $$
$$ \text{Some pepole actually call } c \vec{B} \text{ the magnetic feild, so this factor of } c \text{ is not too strangely placed} $$
$$ \text{are you ready to see Maxwell's equation? (Not to equations (plural), but equation (singular)).} $$
$$ \nabla F = J $$
$$ \text{That's it, this one simple equation describes all electromagnetic phenomina.} $$
$$ \text{Actually, I will admit, I cheated a bit} $$
$$ \nabla F = \frac{J}{c \epsilon_0} $$
$$ \text{I said it that way so that it would look better.} $$

$$ \text{To prove this foumula, we need} $$

**Maxwell's translation (I just came up with that name.)**